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We here at CertyIQ eventually got enough of the industry's greedy exam paid for. Our team of IT professionals comes with years of experience in the IT industry Prior to training CertIQ we worked in test areas where we observed the horrors of the paywall exam preparation system.

The misuse of the preparation system has left our team disillusioned. And for that reason, we decided it was time to make a difference. We had to make In this way, CertyIQ was created to provide quality materials without stealing from everyday people who are trying to make a living.

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We have developed a very scalable solution using which we are able to solve 400+ doubts every single day with an average rating of 4.8 out of 5.

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John

October 19, 2022

★★★★★

Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

Dana

September 04, 2022

★★★★★

Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago

★★★★★

Customer support is realy fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022

★★★★★

Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study.
Thank You certyiq team!

Henry Rome

2 months ago

★★★★★

These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago

★★★★★

Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's.Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today - 3 September with 942 score.

Microsoft

(AI-103)

Developing AI Apps and Agents on Azure

Total: **117 Questions**

Link: https://certyiq.com/papers/microsoft/ai-103

Question: 1 CertyIQ

HOTSPOT -
Case Study -
This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.
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To display the first question in this case study, select the “Next” button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the “Question” button to return to the question.
Overview -
Company Information -
Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.
Existing Environment -
Identity Environment -
Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.
Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.
The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement. monitor, and secure AI applications.
Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.
Generative Environment -
Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.
Project1 -
Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.
Agent1 has the following configurations:
Agent1 uses a base model deployment.
A safety evaluation pipeline is NOT enabled.
Tool invocation approval workflows are NOT enabled.
Conversation memory constraints are NOT configured.
Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
Project1 is deployed to an Azure region located in the European Union (EU).
Agent1Dev Team will use Project1 to optimize and maintain Agent1.
Project2 -
Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
Development of the solution is incomplete.
Data Environment -
Contoso stores product-related information in Azure resources that support AI applications.
The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.
Problem Statements -
Contoso identifies the following issues:
Agent1 has only general knowledge of the Contoso products.
A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.
Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.

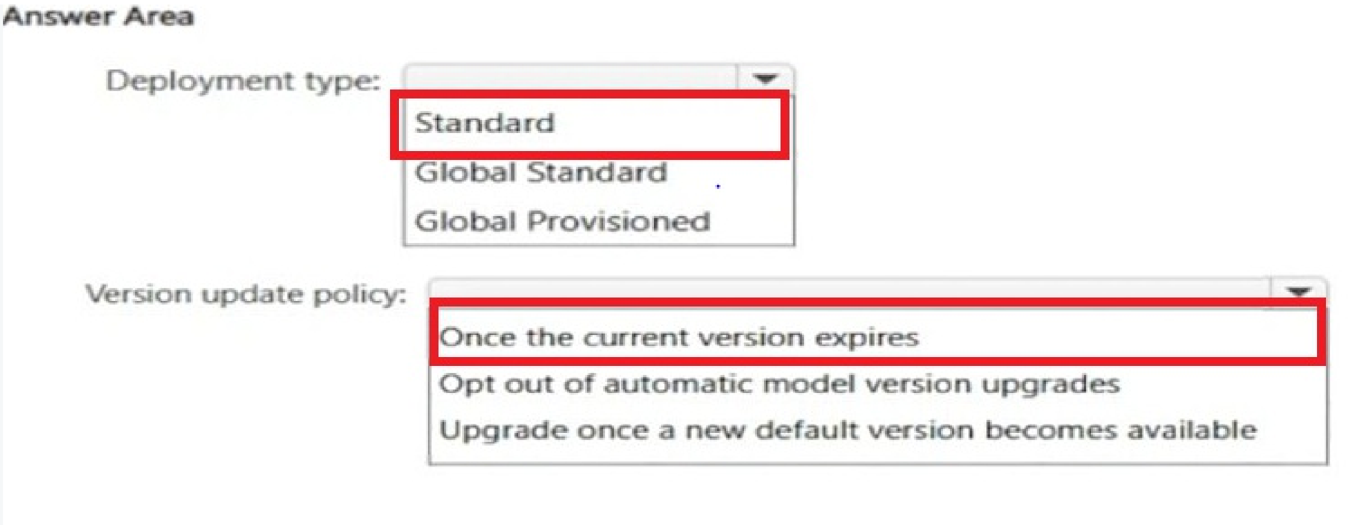
The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.
Requirements -
Planned Changes -
Contoso plans to implement the following changes:
Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.
Complete the development of the video creation solution.
Technical Requirements -
Contoso identifies the following technical requirements:
The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.
The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.
Responses generated by using the product sheet information must be relevant, complete, and accurate.
Agent1 must be able to use the product sheets to answer natural language questions about product details.
The model version used by Agent1 must remain consistent to ensure stable responses.
The data processed by the model must remain within the EU.
Security and Compliance Requirements
Contoso identifies the following security and compliance requirements:
API keys must NOT be used to access Foundry-deployed models.
Access to the Azure resources must follow the principle of least privilege.
The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.
Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.
Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements -
Contoso identifies the following business requirements:
Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
Agent1 must answer questions only about the products sold by Contoso.
You need to configure the model deployment for Agent1 to meet the technical requirements.
What should you configure? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Deployment type:
- Standard
- Global Standard
- Global Provisioned

Version update policy:
- Once the current version expires
- Opt out of automatic model version upgrades
- Upgrade once a new default version becomes available

Answer:



Explanation:

Deployment Type: Selected as **Standard** because the workload requires dynamic scaling for variable traffic without reserved throughput capacity (eliminating Provisioned options), while adhering to strict regional data residency guidelines (e.g., data remaining within a specific zone or region like the EU).

Version Update Policy: Selected as **Once the current version expires** because the model version must remain consistent to guarantee stable, predictable responses over time, rather than automatically upgrading as soon as a new default version becomes available.

Why the other answer are incorrect:

Global Standard: Incorrect because "Global" routing sends traffic dynamically to any region worldwide with available capacity. This violates strict **data residency/compliance** rules if your data must remain within a specific geographic boundary (like the EU or US).

Global Provisioned: Incorrect because "Provisioned" requires buying reserved Throughput Units (PTUs). This incurs a high, fixed cost regardless of usage, making it wrong for workloads with **variable, low-volume traffic** where minimizing costs is a priority.

Opt out of automatic model version upgrades: Incorrect because this policy is deprecated or not recommended for long-term consistency. When a model version reaches its official retirement date, it will force-upgrade anyway, meaning you cannot permanently opt out of upgrades.

Upgrade once a new default version becomes available: Incorrect because this causes the model to update automatically as soon as Microsoft releases a new default. This can **break application code** or change prompt behaviors unexpectedly, violating requirements for strict consistency.

Question: 2 CertyIQ

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Agent1 must answer questions only about the products sold by Contoso.
You need to configure Agent1 to meet the security and compliance requirements.
What should you use?

A. self-harm content filtering
B. prompt shields
C. Personally identifiable information (PII) Detection
D. violence content filtering

Answer: B

Explanation:

Technical Justification for Correct Answer: B. Prompt Shields

Why B (Prompt Shields) is the best option:

Security and Compliance Requirement Alignment: The primary reason for choosing Prompt Shields is its direct alignment with multiple security and compliance requirements stated by Contoso. Specifically, it addresses the need to protect Agent1 from revealing customer information (even when erroneously exposed) and from potential malicious instructions hidden within images in product sheets.
Protection Against Malicious Inputs: Prompt Shields are designed to filter out or shield against potentially harmful or unwanted inputs (including those that might be embedded in images), ensuring Agent1's responses remain safe and compliant.
Relevance to General Security Posture: Given the constraints around not using API keys and the emphasis on least privilege access, leveraging Prompt Shields further fortifies the security posture of Agent1 without introducing additional access vulnerabilities.

Why Other Options are Less Suitable:

A. Self-Harm Content Filtering:
Relevance: While important, self-harm content filtering does not directly address the specified security and compliance requirements related to protecting against malicious instructions or inadvertently exposed

customer data.
Scope: The primary concern here is not the content generated by Agent1 in response to self-harm queries but ensuring the security of the input process.
C. Personally Identifiable Information (PII) Detection:
Relevance: Although PII Detection is crucial for data privacy, the question's focus is on protecting Agent1 from malicious inputs and ensuring it doesn't reveal customer information **proactively**. PII Detection is more about identifying than preventing exposure.
Proactivity: The requirement implies a need for a proactive protection mechanism rather than a detection capability.
D. Violence Content Filtering:
Relevance: Similar to self-harm content filtering, this does not directly address the protection against malicious instructions or the accidental exposure of customer information as highlighted in the problem statement.
Scope: The concern is broader than just filtering violent content; it's about securing the input pipeline.

Conclusion:Given the specific security and compliance requirements outlined by Contoso, particularly the need to protect against malicious inputs and prevent the exposure of customer information, **Prompt Shields (B)** is the most appropriate choice. It directly addresses the identified risks without introducing additional vulnerabilities, aligning with the principle of least privilege and enhancing the overall security posture of Agent1.

References

Microsoft Documentation: security and compliance for Azure AI Services
Microsoft Learn: Protecting AI Models from Malicious Inputs

Question: 3

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You are planning a Microsoft Foundry project named Project1 that will contain multiple agents. Each agent will access the same Azure AI Search resource.
You need to recommend a solution to centrally manage the Azure AI Search credentials within Project1. The solution must be implemented across all the agents.
What should you recommend?

A. Enable role-based access control (RBAC) for the Azure AI Search resource.
B. Disable key-based access control on the Azure AI Search resource.
C. Add a connection to the Azure AI Search resource.
D. Create a managed private endpoint that connects to the Azure AI Search resource.

Answer: C

Explanation:

Technical Justification for Recommending Option C

For centrally managing Azure AI Search credentials across multiple agents in Project1, **Option C: Add a connection to the Azure AI Search resource** is the most suitable choice. Here's why:

Option C is the best because adding a connection to the Azure AI Search resource within the Microsoft Foundry project (Project1) allows for a centralized management approach. This connection can be securely shared or referenced across all agents within the project, ensuring that credentials are not hardcoded or duplicated across agents. Instead, agents can leverage this single, managed connection, enhancing security and simplifying credential updates.

Why Other Options are Less Suitable:

Option A: Enable role-based access control (RBAC) for the Azure AI Search resource
While RBAC is crucial for controlling access, it does not directly address the central management of credentials for accessing the Azure AI Search resource. RBAC defines what actions can be performed but does not consolidate credential management across multiple agents.
Suitability for Central Credential Management: Low
Option B: Disable key-based access control on the Azure AI Search resource
Disabling key-based access would likely increase security risks by potentially forcing the use of less secure methods or complicating access management. This does not contribute to centralizing credential management.
Suitability for Central Credential Management: Very Low (Counterproductive)
Option D: Create a managed private endpoint that connects to the Azure AI Search resource
While a managed private endpoint enhances security by providing a private, secure connection to the Azure AI Search resource, it focuses on network security rather than the central management of credentials for access by multiple agents.
Suitability for Central Credential Management: Low

Recommendation Summary: Given the need for central management of Azure AI Search credentials across multiple agents in Project1, **Option C** is the most direct and effective solution, facilitating secure, centralized credential management.

References:

For further understanding of managing connections and security in Azure and Microsoft Foundry, refer to:

1. **Microsoft Documentation - Manage connections in Microsoft Foundry**: https://learn.microsoft.com/en-us/microsoft-365/apps/foundry/manage-connections?view=o365-worldwide
2. **Azure Documentation - Secure your Azure AI Search with Azure RBAC**: https://learn.microsoft.com/en-us/azure/search/security-azure-rbac

Question: 4 CertyIQ

HOTSPOT -
Your company is piloting a customer support agent in a Microsoft Foundry project name Project1. Project1 is connected to an existing Application Insights resource, and the company's support team reviews runs in the Traces tab.
The Foundry Agent Service is configured to perform the following actions:
Retrieve the Application Insights connection string by calling project_client.telemetry.get_application_insights_connection_string().
Call configure_azure_monitor(connection_string=...) to enable telemetry.
A separate LangChain service is configured to use OpenTelemetry and has the following configurations:
Uses AzureAIOpenTelemetryTracer(connection_string=..., enable_content_recording=False)
Passes the tracer by using config= "callbacks":[azure_tracer]
Company policy has the following requirements:
Telemetry from LangChain and OpenTelemetry must be distinguishable within the same Application Insights resource.
Secrets and credentials must NOT be stored in prompts, tool arguments, or span attributes.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The LangChain service will appear in Traces without configuring a tracer.	○	○
Setting different `OTEL_SERVICE_NAME` values separates the services in Application Insights.	○	○
When using `enable_content_recording=False`, prompts and tool data will be captured in the telemetry.	○	○

Answer:

Answer Area

Statements	Yes	No
The LangChain service will appear in Traces without configuring a tracer.	○	$\boxed{○}$
Setting different `OTEL_SERVICE_NAME` values separates the services in Application Insights.	$\boxed{○}$	○
When using `enable_content_recording=False`, prompts and tool data will be captured in the telemetry.	○	$\boxed{○}$

Explanation:

Statement 1: No

OpenTelemetry tracing requires explicit configuration to start capturing data. Without setting up an active OpenTelemetry exporter or attaching a dedicated tracer (such as AzureAIOpenTelemetryTracer or calling configure_azure_monitor), no trace spans or events will be captured or forwarded to Azure Monitor.

Statement 2: Yes

The OTEL_SERVICE_NAME environment variable determines the logical name of your application service in OpenTelemetry telemetry data. Configuring distinct values for different services allows Azure Application Insights to map, filter, and isolate them into distinct components within the Application Map view.

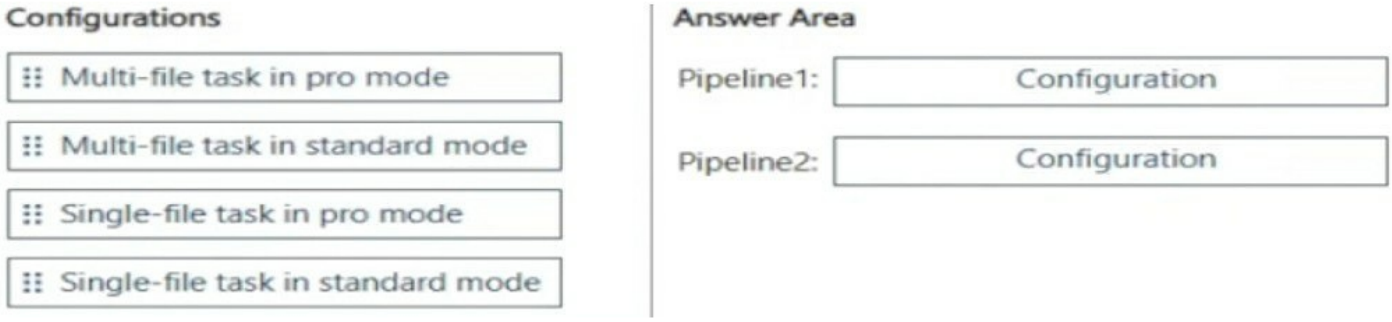
Statement 3: No

Setting enable_content_recording=False explicitly tells the tracer to **exclude** the actual text content of prompts, completions, and tool data payloads from the telemetry attributes. This protects user privacy and prevents sensitive personal data or secrets from being stored inside logs.

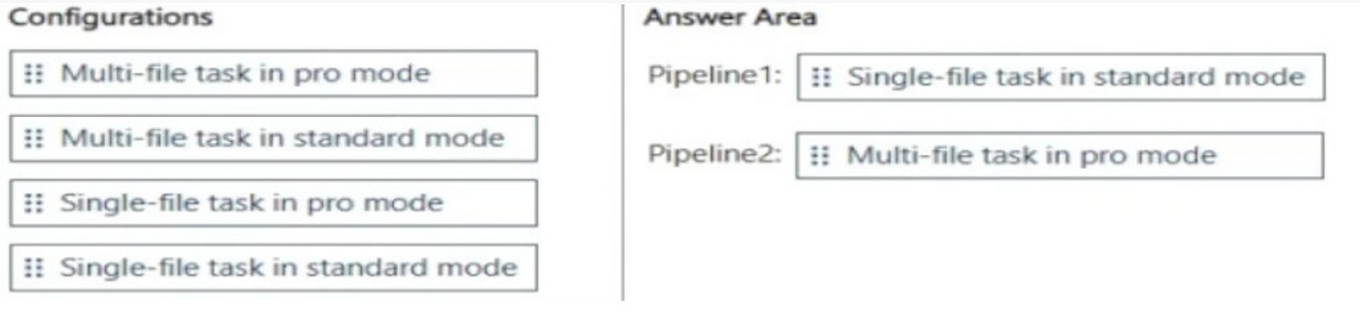
Question: 5 CertyIQ

DRAG DROP -
You have a Microsoft Foundry project that processes procurement documents submitted by suppliers.
You need to implement two pipelines by using Azure Content Understanding in Foundry Tools. The solution must meet the following requirements:
Include a pipeline named Pipeline1 that supports cost-effective, high-volume processing of standalone PDF

invoices.
Include a pipeline named Pipeline2 that supports cross-document validation by using multi-step reasoning and reference data.
How should you configure each pipeline? To answer, drag the appropriate configurations to the correct pipelines. Each configuration may be used once, more than once, of not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

Pipeline1 (Single-file task in standard mode):

This is correct for workflows processing independent, standalone documents one by one. Standard mode handles individual file parsing efficiently. It minimizes computational costs for basic schema extraction.

Pipeline2 (Multi-file task in pro mode):

This is correct for workflows requiring cross-document logic or advanced generative reasoning. Pro mode processes batches of interrelated files together. It allows the model to correlate information across multiple files.

Incorrect answer:

Single-file task in pro mode:

This is incorrect because Pro mode's core advantage is cross-file reasoning. Using it for a single file unnecessarily increases processing limits and costs without utilizing multi-file capabilities.

Multi-file task in standard mode:

This is incorrect because Standard mode lacks the architecture to handle relational dependencies across multiple files. Combining multiple files under standard mode results in schema rejection or validation errors.

Question: 6 CertyIQ

HOTSPOT -
You have a Python application named App1 that integrates with a Microsoft Foundry project named Project1.
You need to ensure that App1 meets the following requirements:

Authenticates by using a Microsoft Entra managed identity
Sends prompts to a deployed model by using the Azure OpenAI Responses API
How should you complete the Python code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
credential =  [ ▼ ] ()
              AzureKeyCredential
              ClientSecretCredential
              DefaultAzureCredential

project_client = AIProjectClient(
    endpoint="https://contosoai.services.ai.azure.com/api/projects/project1",
    credential=credential,
)
with project_client.get_openai_client() as openai_client:
    response = openai_client.responses. [ ▼ ] (
                                        compact
                                        create
                                        retrieve
        model="trail-guide-chat",
        input="Create a 3-day hiking itinerary near Seattle.",
    )
    print(response.output_text)
```

Answer:

Explanation:

First Dropdown: **DefaultAzureCredential**

Second Dropdown: **create**

Credential Initialization: The scenario specifies authentication using a **Microsoft Entra managed identity**. In Azure Python SDKs, DefaultAzureCredential automatically orchestrates managed identities across both local development and Azure-hosted environments. It is explicitly imported at the top of the snippet (from azure.identity import DefaultAzureCredential).

Responses API Call: When interacting with the Azure OpenAI-compatible project client endpoint via openai_client.responses, the standard method to generate chat text completions or send user prompts to the deployed generative model is create.

Question: 7

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HOTSPOT -
You have a Microsoft Foundry project that contains a workflow for a customer support triage process.
You have an Ask a question node that stores user responses in a local variable named Var01.
You need to create the following Power Fx expressions:
An if/else condition expression that ensures that Var01 contains a value
A Send message expression that returns the stored user response in uppercase
How should you configure the expressions? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

If/else condition expression:

- IsBlank(Local.Var01)
- IsEmpty(Local.Var01)
- Not(IsBlank(Local.Var01))

Send message expression:

- {Local.Var01}
- {Upper(Local.Var01)}
- {Upper(Var01)}

Answer:

Answer Area

If/else condition expression:

- IsBlank(Local.Var01)
- IsEmpty(Local.Var01)
- Not(IsBlank(Local.Var01))

Send message expression:

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- {Upper(Var01)}

Explanation:

If/else condition expression: Not(IsBlank(Local.Var01))

Send message expression: **Upper(Local.Var01)**

If/else condition expression:

The requirement is to ensure that the variable **contains a value** (i.e., it is not empty or blank).

IsBlank(Local.Var01) evaluates to True if it has no value.

Wrapping it in the Not() function inverts this logic, making the expression evaluate to True only when a valid value exists.

Send message expression:

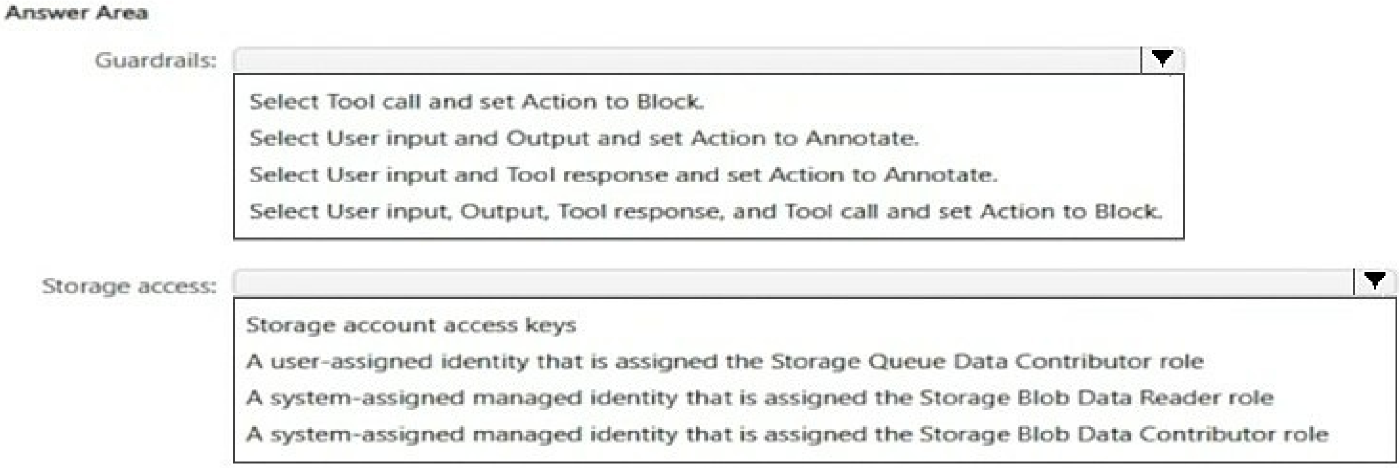
To display or send a variable dynamically inside a message node, the variable/formula must be enclosed within curly brackets string interpolation syntax.

The Upper() function converts the text string to uppercase characters.

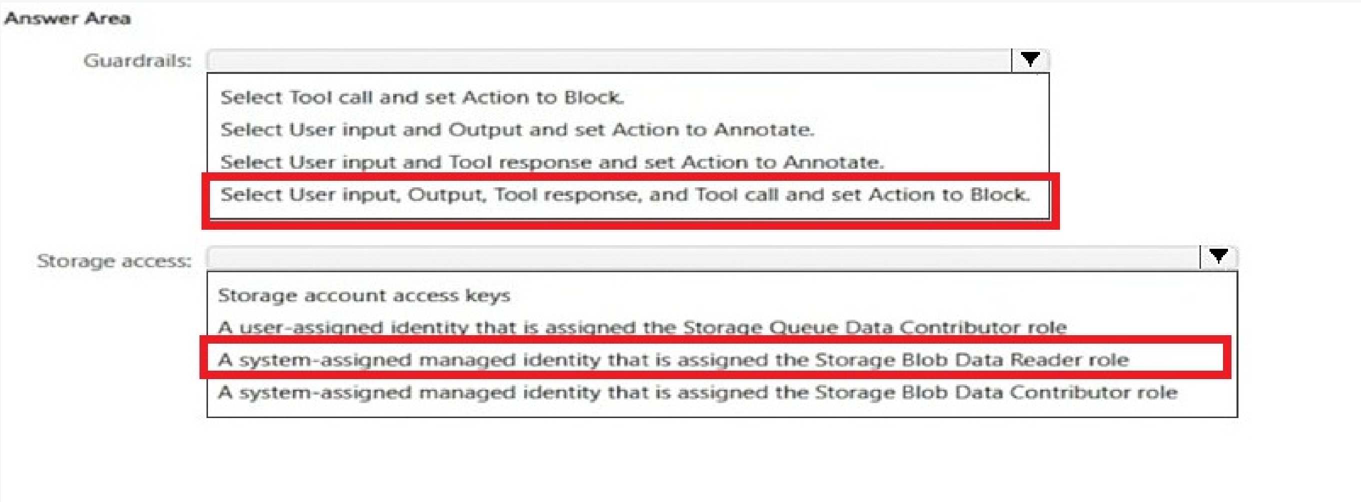
The variable prefix Local. must be explicitly retained to reference the locally-scoped variable correctly within the expression payload (Upper(Local.Var01)).

Question: 8 CertyIQ

HOTSPOT -
You have a Microsoft Foundry project that contains a customer support agent built by using the Foundry Agent Service.
The agent uploads user-provided screenshots to Azure Storage through a ticketing tool and receives a blob URL for additional reasoning.
You need to use image moderation during agent runs and prevent harmful content from being returned during runs. Azure AI Content Safety must access the images by using the blob URL. The solution must follow the principle of least privilege.
What should you configure for Content Safety? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

Guardrails

Select User input, Output, Tool response, and Tool call and set Action to Block.

Storage access

A system-assigned managed identity that is assigned the Storage Blob Data Reader role

To securely evaluate and prevent harmful or unsafe content from being returned or processed at any point during an agentic session, content safety filters must be applied comprehensively across all interaction touchpoints. Setting the action to **Block** ensures that any policy violations are immediately halted rather than merely flagged or annotated in logs, completely protecting the user interface from receiving or exposing toxic data.

Storage access:

When Azure AI Content Safety retrieves screenshots or file attachments via direct blob URLs, it requires specific data-plane permissions on the underlying storage repository.

System-assigned managed identity eliminates password management overhead.

Storage Blob Data Reader satisfies the principle of **least privilege**, providing full read access to download and review image binaries without granting unnecessary write or delete capabilities (which would be included in the Contributor role).

Question: 9 CertyIQ

You have a Microsoft Foundry project that contains three agents as shown in the following table.

Name	Description
TriageAgent	Classifies incoming customer requests
PolicyAgent	Answers policy questions by searching internal content
ActionAgent	Creates or updates tickets by calling an HTTP API

You need to orchestrate the agents to ensure that the customer requests meet the following requirements:
Support a deterministic, step-based process that uses conditional branching and shared state across the agents.
Optionally trigger a ticket action based on the triage result.
The solution must minimize development effort.
What should you include in the solution?

A. a workflow

B. threads and runs without a workflow

C. a multi-agent group chat session

D. separate agent runs coordinated in the application code

Answer: A

Explanation:

A. a workflow.

Why a workflow is correct: In Microsoft Azure AI Foundry, agentic workflows are purposely built to orchestrate multiple agents using declarative, predefined sequences. They natively support deterministic step-by-step logic, if/else conditional branching, and automatic variable/state sharing across agents without requiring developers to write complex application orchestration or synchronization code, minimizing overall development effort.

Why other options are incorrect:

Group chat sessions (C) are designed for autonomous, fluid multi-agent conversations where agents dynamically choose when to chime in or pass control. They do not natively follow strict, deterministic, step-based workflows.

Separate application code loops or standalone threads (B & D) require significant manual engineering effort to establish shared states, handle condition mappings, and write error-handling glue logic.

Question: 10 CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent uses Azure Speech in Foundry Tools.
You fine-tune a baseline speech to text model for the en-us locale and publish the model.
The agent calls the Speech to text REST API and returns an error message indicating that the project ID is invalid.
You need to set the project property to the correct ID.
To what should you set the project property?

A. the project URL

B. the custom speech project ID

C. the project ID

D. the custom speech endpoint URL

Answer: B

Explanation:

Technical Justification for Correct Answer: B

The correct option to set the project property to resolve the invalid project ID error when using Azure Speech in a Microsoft Foundry project is **B. the custom speech project ID**. Here's why:

Why B is Correct: When fine-tuning a baseline speech-to-text model for a specific locale (like en-us) and publishing it through Azure Speech Services, a unique **Custom Speech Project ID** is generated. This ID is crucial for identifying your specific, customized model when making API calls. The Speech to Text REST API requires this exact ID to route the request to your customized model, ensuring the correct speech-to-text processing. Setting the project property to the **custom speech project ID** ensures that the API call targets your fine-tuned model, resolving the invalid project ID error.
Why Other Options are Less Suitable:
A. the project URL: While the project URL might contain the project ID as part of its path, using the full URL is not the expected format for the project ID field in the API configuration. The API typically expects just the ID, not the entire URL, to identify the project.
C. the project ID: This option is misleadingly similar to the correct answer. In the context of Azure Speech Services, "project ID" could ambiguously refer to a higher-level project container ID (not specific to the customized speech model). The **custom speech project ID** (Option B) is more precise, indicating it's the ID for the fine-tuned speech model, not just any project ID.
D. the custom speech endpoint URL: Similar to Option A, using the full endpoint URL is not appropriate for the project ID field. The endpoint URL is used to direct the API call to the speech service's entry point, but the project ID (specifically the custom speech project ID for your model) is still required within the API call parameters to identify which model to use.

Technical Summary: The error indicates a mismatch in identifying your customized speech-to-text model. By setting the project property to the **custom speech project ID**, you ensure the API call accurately references your published, fine-tuned model for the en-us locale, thereby resolving the invalid project ID error.

References

Azure Speech Services Documentation - Custom Speech
Azure Speech Services REST API Reference - Speech to Text

Question: 11 CertyIQ

HOTSPOT -

You have a Microsoft Foundry project that contains an agent named PaymentAgent.
PaymentAgent includes a function tool that issues customer refunds by using an external API.
You are creating a workflow in YAML.
You need to ensure that the workflow pauses for human approval and continues with the refund step only after approval is granted.
How should you complete the workflow definition? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
steps:
  - id: propose_refund
    type: agent
    agent: PaymentAgent
  - id: approval
    type: [dropdown]
```

- ask_question
- basic_chat
- data_transformation

```
  - id: execute_refund
    type: agent
    agent: PaymentAgent
    condition: [dropdown]
```

- approval == "approved"
- propose_refund.output != null
- true

Answer:

Explanation:

ask_question: In declarative agent workflows, ask_question is used to explicitly pause execution and collect external feedback or manual intervention (such as human authorization/approval) before proceeding, unlike basic_chat or data_transformation which run automatically.

approval == "approved" To ensure that the execute_refund step runs conditionally only if approval has been explicitly granted, the expression evaluates the string output from the preceding id: approval step against the expected value "approved".

Question: 12 CertyIQ

You have an Azure Speech in Foundry Tools resource that hosts a custom speech to text model deployed to a custom endpoint. An agent uses the endpoint to perform real-time speech recognition.
You are approaching the expiration date of the custom speech to text model.
What is the expected behavior when the model expires?

A. Speech recognition requests will return a 4xx error until a new custom model is deployed.

B. Speech recognition requests will continue to use the expired custom model until the model is removed manually.

C. Speech recognition requests will fall back to the most recent base model for the same locale.

D. The custom model will be deleted automatically when the model expires.

Answer: C

Explanation:

Technical Justification for Correct Answer: C

The correct answer is **C. Speech recognition requests will fall back to the most recent base model for the same locale.** Here's why:

Reason for C being correct: When a custom speech-to-text model expires in Azure Speech Services (formerly part of Cognitive Services, now potentially managed through Azure Speech in Foundry Tools as mentioned), the service is designed to ensure continuity of service. Instead of interrupting the functionality completely (which would be the case with a 4xx error as suggested by A) or requiring immediate manual intervention, Azure Speech Services automatically falls back to the **most recent base model** for the **same locale** as the expired custom model. This fallback mechanism minimizes disruption to the agent's speech recognition capabilities, ensuring some level of functionality is maintained until a new custom model is deployed.

Why A is less suitable:

A. Speech recognition requests will return a 4xx error until a new custom model is deployed. This option suggests a complete service interruption upon model expiration, which contradicts the design principle of Azure Services to provide resilient and continuous functionality wherever possible. A 4xx error implies a client-side issue or a deliberate block, which doesn't align with the expected behavior for a managed expiration scenario.

Why B is less suitable:

B. Speech recognition requests will continue to use the expired custom model until the model is removed manually. Continuing to use an expired model could pose security, compliance, or accuracy issues, as the model's support and potential updates would have ceased. Azure services typically do not extend the use of expired resources in such a manner without explicit configuration for a temporary extension or auto-renewal, which is not implied here.

Why D is less suitable:

D. The custom model will be deleted automatically when the model expires. While automatic deletion might seem like a housekeeping measure, the immediate question concerns the behavior of **speech recognition requests** at the time of expiration, not the model's storage lifecycle. The automatic model deletion (if it were to happen immediately, which is not the primary concern here) does not directly address the continuity of the speech recognition service's functionality.

Correct Answer Summary

Correct Answer: C. Speech recognition requests will fall back to the most recent base model for the same locale.

Rationale: Ensures service continuity with the best available alternative (latest base model for the locale) upon custom model expiration, aligning with Azure's design for resilient services.

References

Azure Speech Services: Custom Speech Models
Azure Speech Services: How to manage models

Question: 13

You have a Microsoft Foundry project that contains a model deployment.
You have an application that calls the deployment by using the Azure OpenAI v1 API and DefaultAzureCredential.
The developers at your company receive HTTP 403 errors when they send inference requests, even after running az login.
You need to ensure that the developers can perform model inference. The solution must follow the principle of least privilege.
Which role-based access control (RBAC) role should you assign to the developers?

A. Cognitive Services User

B. Cognitive Services OpenAI User

C. Contributor

D. Cognitive Services Data Reader

Answer: B

Explanation:

Technical Justification for Correct Answer: B (Cognitive Services OpenAI User)

The correct role-based access control (RBAC) role to assign to the developers for resolving the HTTP 403 errors when sending inference requests to a Microsoft Foundry (formerly Azure Machine Learning) model deployment via the Azure OpenAI v1 API, while adhering to the principle of least privilege, is **B. Cognitive Services OpenAI User**. Here's why:

Cognitive Services OpenAI User (B): This role is specifically designed for users who need to perform OpenAI model inference. Assigning this role ensures that developers have the necessary permissions to call the OpenAI API for model inference without granting excessive privileges. This aligns with the principle of least privilege, as it only allows actions directly related to the task at hand (OpenAI model inference) and nothing more.

Why Other Options are Less Suitable:

A. Cognitive Services User: While this role provides access to Cognitive Services, it is more general and might not specifically grant the necessary permissions for OpenAI model inference, or it could grant broader access than necessary, violating the principle of least privilege.

C. Contributor: Assigning the Contributor role would grant far too many permissions, including the ability to manage resources, which is not necessary for simply performing model inference. This vastly exceeds the principle of least privilege.

D. Cognitive Services Data Reader: This role is focused on reading data from Cognitive Services resources but does not necessarily provide the execution permissions required for model inference via the OpenAI API, making it inappropriate for this specific task.

Principle of Least Privilege Alignment: Assigning **Cognitive Services OpenAI User** ensures that developers have only the permissions necessary to perform their tasks (model inference via OpenAI API), without any additional, potentially risky permissions.

References

1. **Azure RBAC for Cognitive Services and OpenAI**: https://learn.microsoft.com/en-us/azure/cognitive-services/authentication-acls?tabs=platform
2. **Azure OpenAI Permissions and Roles**: https://learn.microsoft.com/en-us/azure/cognitive-services/openai/how-to-use-manager-keys#permissions-and-roles

Question: 14

CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent has a Model Context Protocol (MCP) tool that queries a knowledge base stored in Azure AI Search.
Some agent runs return answers from the base model without invoking the knowledge base, which results in responses without grounded citations.
You are provided with the following code snippet that runs the agent.

```
run = project_client.agents.runs.create_and_process(
    thread_id=thread.id,
    agent_id=agent.id
)
```

You need to add the correct tool _choice parameter to the code to deterministically force the agent to invoke the MCP tool on each run.
What should you add?

A. tool_choice= “required”

B. tool_choice= “auto”

C. tool_choice= “type”:“knowledge_base”

D. tool_choice = “type”:“mcp”

Answer: A

Explanation:

A. tool_choice= “required” .

Why it is correct: According to Microsoft Foundry Agent Service tool best practices, the tool_choice parameter provides runtime control over whether a model is allowed to answer directly or is forced to invoke a tool. Setting tool_choice="required" (or passing the parameter mode as a dict/object depending on the specific wrapper structure) explicitly tells the orchestrator that the agent **must execute one or more tools** before formulating its final response. Since the agent only has the Model Context Protocol (MCP) tool configured for querying the knowledge base, this forces deterministic tool execution on every single run, preventing ungrounded base model hallucinations.

Why the other options are incorrect:

B ("auto"): This is the default setting. It lets the model autonomously decide whether to use a tool, which makes it non-deterministic and leads to runs skipping the knowledge base entirely.

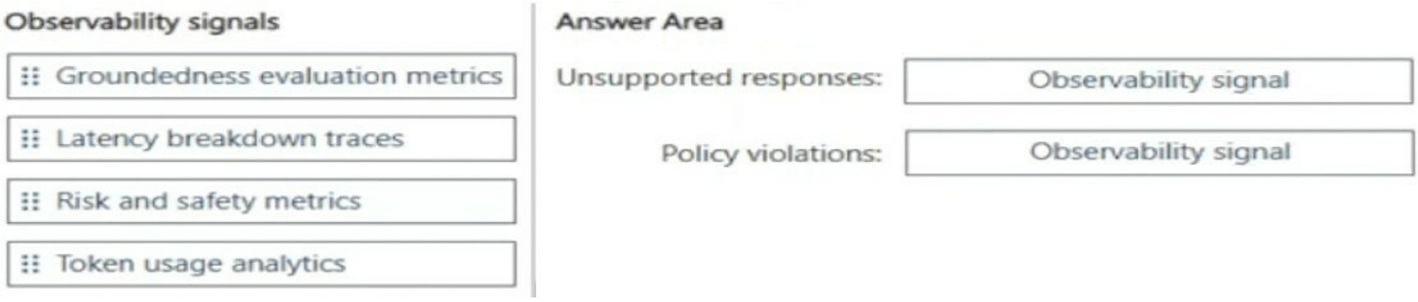
C & D: These options represent invalid formatting syntax for directing type scopes or target keywords within the standard parameter layout for tool-constrained invocation under the Foundry Agent ecosystem.

Question: 15

CertyIQ

DRAG DROP -
You have a Microsoft Foundry project that contains a customer support agent grounded in internal documentation.
After a recent update, users report the following issues:
Some answers are unsupported by retrieved documents.
A small number of responses are flagged for policy violations.
You need to evaluate each issue.

Which observability signals should you use for each issue? To answer, drag the appropriate observability signals to the correct issues. Each observability signal may be used once, more than once, or not at all. You may need to drag the spit bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Answer:

Explanation:

Unsupported responses

Groundedness evaluation metrics

Policy violations:

Risk and safety metrics

Unsupported responses: When an AI model generates responses containing facts or assertions not present in the reference context, it results in an ungrounded or unsupported response (hallucination). Tracking **Groundedness evaluation metrics** assesses the exact ratio of sentences validated by the source text.

Policy violations: These occur when inputs or outputs conflict with regulatory or internal content guidelines (e.g., hate speech, violence, or sexual content). **Risk and safety metrics** directly monitor hits against predefined Azure AI Content Safety filter thresholds.

Question: 16

CertyIQ

You have a Microsoft Foundry project named Project1 that contains an agent. The agent uses an OpenAPI 3.0 specification to call an external weather service.
The weather service requires a key to be passed in an HTTP header. The key value is stored as a connection in Project1.
You need to ensure that the key value from the connection is included automatically whenever the OpenAPI tool is invoked.
What should you configure in the OpenAPI specification?

A. a header parameter defined for each operation

B. an Azure Key Vault connection

C. an API key security scheme

D. a Bearer token security scheme

Answer: C

Explanation:

Technical Justification for Correct Answer: C

To ensure the key value from the connection is included automatically whenever the OpenAPI tool is invoked for the weather service API call, the most appropriate configuration in the OpenAPI specification is **an API key security scheme**. Here's why:

Correct Answer: C - an API key security scheme

Reason: The API key security scheme in OpenAPI allows for the definition of how an API key is passed (e.g., in a header, query, or cookie). Since the weather service requires the key in an HTTP header, configuring an API key security scheme in the OpenAPI specification enables the automatic inclusion of the key from the connection without needing to redefine it for each operation. This approach is scalable and maintainable, especially if the key needs to be updated, as changes can be made in one place.

Why Other Options are Less Suitable:

A. a header parameter defined for each operation

Inconvenience and Redundancy: Defining a header parameter for each operation leads to redundancy and increased maintenance effort, especially in large APIs. If the key changes, it would need to be updated in multiple places.
Lack of Central Management: Doesn't leverage the connection storage's central management capability efficiently.

B. an Azure Key Vault connection

Misalignment with Requirement: While Azure Key Vault is suitable for securely storing secrets, the question specifies that the key is already stored as a connection in Project1. The task is about integrating this existing setup with the OpenAPI specification, not about where to store the key.
Additional Complexity: Introducing Key Vault at this stage would add unnecessary complexity to the immediate requirement of passing the key in the HTTP header.

D. a Bearer token security scheme

Inappropriate for API Keys: Bearer tokens are typically used for authentication in scenarios involving user authentication or service-to-service calls with more complex authentication flows, not for simple API key exchanges.
Mismatch with Service's Requirement: The weather service specifically requires an API key in the header, not a Bearer token.

References

OpenAPI Specification - Security Schemes
Microsoft Azure - Secure your Azure API using API Keys

Question: 17 CertyIQ

You have a Microsoft Foundry project that serves a high-volume chat app.
Most requests are simple FAQs, but some require advanced reasoning.
You need to reduce costs and latency for common queries, without degrading the quality of the responses to

complex questions.
What should you do?

A. Route all the requests to a smaller model.
B. Use a model cascade that routes the requests to different models.
C. Increase the value of the max_tokens parameter for all the requests.
D. Route all the requests to the most capable model.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Why B is the Best Option: Using a model cascade that routes requests to different models is the optimal approach for this scenario. This architecture allows for:

Cost Reduction: Simple FAQs can be handled by smaller, more efficient models with lower inference costs, reducing overall expenses.
Latency Reduction: Routing common queries to lighter models decreases response times for the majority of requests.
Quality Preservation for Complex Queries: Advanced reasoning questions are directed to more capable (likely larger) models, ensuring response quality is maintained without compromise.

Why Other Options are Less Suitable:

A. Route all requests to a smaller model:
Quality Deterioration: Complex questions may not be answered accurately, degrading overall service quality.
Inadequate Capacity: A single smaller model might not handle the "high-volume" aspect efficiently, potentially increasing latency under load.
C. Increase the max_tokens parameter for all requests:
Cost Increase: Higher max_tokens values increase costs per request, contradicting the goal of reducing costs.
Unnecessary Overhead: Simple queries do not benefit from this increase, leading to wasted resources.
D. Route all requests to the most capable model:
Excessive Cost: Utilizing the most capable model for all queries significantly increases costs due to its higher inference expense.
Latency Concerns: Larger models typically have longer response times, affecting the latency of simple, high-volume FAQs.

Correct Answer Justification Summary: Option B, using a model cascade, strikes a balance between cost efficiency, latency reduction for common queries, and maintaining the quality of responses for complex inquiries, making it the most suitable solution.

References:

Microsoft Azure Documentation: Model Deployment Strategies
Microsoft Azure Cognitive Services: Optimizing Costs for AI Workloads

Question: 18

CertyIQ

HOTSPOT -
You have a Microsoft Foundry project that contains an internal Q&A agent.
Users report the following issues when they ask the agent questions:
An increase in the following response: “No relevant information found”

Periodic HTTP 429 rate limit exceeded errors during peak hours
You need to identify whether each issue is caused by model unavailability, resource limits, or inference failures.
What should you do? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Metrics to enable:

- Model Availability Rate and Provisioned Utilization
- Only Tokens Cache Match Rate
- Only Total Requests filtered to status code 200
- Time To Response and Total Tokens

Diagnostic log to collect:

- AllMetrics
- audit
- RequestResponse
- trace

Answer:

Answer Area

Metrics to enable:

- Model Availability Rate and Provisioned Utilization
- Only Tokens Cache Match Rate
- Only Total Requests filtered to status code 200
- Time To Response and Total Tokens

Diagnostic log to collect:

- AllMetrics
- audit
- RequestResponse
- trace

Explanation:

Metrics to enable :

Model Availability Rate and Provisioned Utilization

Model Availability Rate isolates whether service errors or operation drop-offs are caused by downstream, server-side model unavailability.

Provisioned Utilization provides direct observability into capacity and resource limits. When it approaches or exceeds 100%, requests will automatically experience rate-limiting triggers and throw standard HTTP 429 Too Many Requests errors.

Diagnostic log to collect

Request Response.

To store and inspect telemetry data for incoming client interactions, the **RequestResponse** log category explicitly routes full API payloads (including prompts, generated tokens, status codes, and exact completions) to a destination Log Analytics Workspace.

Question: 19 CertyIQ

You have a Microsoft Foundry project that contains a high-traffic agent.
After a recent update, operational costs increase significantly.
Monitoring confirms that the volume of user traffic to the agent remains unchanged.
You suspect that changes to the request or response characteristics are causing the increase. You need to identify whether the additional costs are driven by the model input size, the model output size, or expanded tool usage.
Which observability capability should you use?

A. latency
B. evaluation metrics
C. run success rate
D. token usage

Answer: D

Explanation:

Technical Justification for Choosing D. Token Usage

To identify the cause of increased operational costs in the Microsoft Foundry project, given that user traffic volume remains unchanged, we need to focus on the changed characteristics of the requests, responses, or tool usage that could drive up costs. Here's why **D. Token Usage** is the most suitable observability capability, alongside explanations for why the other options are less suitable:

D. Token Usage:
Why It's the Best Choice: In the context of AI agents, especially those involving natural language processing (NLP) or similar models, "token usage" refers to the number of tokens (e.g., words, characters) processed by the model in requests and responses. Increased model input size (more tokens in requests) or model output size (more tokens in responses) directly correlates with token usage metrics. Since the question hints at changes in request or response characteristics, monitoring token usage can pinpoint if the cost increase is due to larger inputs, outputs, or potentially more complex queries(initial step before deeper tool usage analysis).
Cost Correlation: Token usage is often directly tied to billing in cloud AI services, as processing more tokens incurs higher costs. This metric can clearly indicate if the increase in costs is due to the model handling more data per interaction.
A. Latency:
Why It's Less Suitable: While latency might indicate performance issues or model complexity increases, it does not directly correlate with cost increases related to input/output size or tool usage. Latency could be affected by many factors unrelated to the suspected causes (e.g., infrastructure, model complexity).
B. Evaluation Metrics:
Why It's Less Suitable: Evaluation metrics (e.g., accuracy, F1 score) are crucial for model performance assessment but do not provide insight into the operational costs driven by input size, output size, or tool usage expansions. These metrics focus on model efficacy, not cost drivers.
C. Run Success Rate:
Why It's Less Suitable: The success rate of model runs indicates the frequency of successful executions but does not offer insights into what drives up costs. A high success rate with increased costs would not help differentiate between input size, output size, or expanded tool usage as the cause.

Conclusion: Given the need to identify cost drivers among model input size, model output size, or expanded tool usage, with the latter two being more directly related to the former, **D. Token Usage** is the most appropriate observability capability. It provides a direct measure of how much data is being processed, which is typically correlated with cost in AI model deployments.

References

1. **Microsoft Azure Cognitive Services Pricing Page**: https://azure.microsoft.com/en-us/pricing/details/cognitive-services/ - To understand how token usage and other metrics impact billing.
2. **Azure Monitor for Azure Cognitive Services**: https://docs.microsoft.com/en-us/azure/cognitive-services/cognitive-services-usage-metrics - For detailed information on available observability metrics, including token usage.

Question: 20

CertyIQ

HOTSPOT -
You have a Microsoft Foundry project that contains an agent.
The agent uses tools to retrieve internal content and call external APIs. The agent is configured to let the model decide when to call the tools.
You need to publish the agent for a compliance workflow. The solution must meet the following requirements:
Each workflow run must include a retrieval step before generating a response.
Tool calls must authenticate by using the published agent's own identity.
Tool access must use an identity isolated from other project resources.
Tool access must use support audit tracing.
What should you do? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

Set tool_choice to:
- auto
- none
- required

Configure the tool to authenticate by:
- Storing API keys in prompts
- Using the shared project agent identity
- Using a distinct agent identity bound to the client application

Answer:

Answer Area

Set tool_choice to:
- auto
- none
- required

Configure the tool to authenticate by:
- Storing API keys in prompts
- Using the shared project agent identity
- Using a distinct agent identity bound to the client application

Explanation:

Set tool_choice to

required

Configure the tool to authenticate by:

Using a distinct agent identity bound to the client application

Set tool_choice to (required): Setting this to **required** ensures that the AI model is deterministically forced to call one or more tools before returning a final answer, rather than autonomously deciding whether to skip the tool (as it would with auto).

Configure the tool to authenticate by (Using a distinct agent identity bound to the client application): To meet compliance standards stating that tool access must utilize an identity **isolated from other project resources** and support full audit tracing, assigning a distinct agent identity bound explicitly to the client application ensures complete isolation and individual request tracking.

Question: 21 CertyIQ

You have a Microsoft Foundry project named Project1 that contains the following:
An OpenAPI tool that calls an external API
A project connection named Connection1 that stores the API key of the external API
When an agent calls the OpenAPI tool, the API returns a 401 unauthorized error, and traces show that the API key header is NOT being sent.
You need to ensure that the OpenAPI tool automatically includes the API key from Connection1 on all requests.
What should you do?

A. Enable identity passthrough so that the tool uses the Microsoft Entra token of the caller.
B. Add the API key header manually to the OpenAPI specification.
C. Configure the tool to use the default connection of Project1.
D. Connect the tool to Connection1.

Answer: D

Explanation:

Technical Justification for Correct Answer D

Correct Answer: D. Connect the tool to Connection1

Connect the tool to Connection1 is the most appropriate solution because it directly addresses the issue by leveraging the existing infrastructure for secure and managed credential handling. Here's why:

Secure Credential Handling: Connection1 already stores the API key. By connecting the tool to this connection, the API key is securely injected into the requests without manual interception, reducing security risks associated with hardcoding or manually handling sensitive information.
Automation and Consistency: This approach ensures the API key is automatically included in all requests to the external API, aligning with the requirement for automation and consistency across all tool invocations.
Management and Updates: If the API key needs to be updated, changes can be made in one place (Connection1), and all connected tools will automatically use the new key without requiring individual updates.

Why Other Options are Less Suitable:

A. Enable Identity Passthrough:

Inapplicable Scenario: Identity passthrough is useful for scenarios where the caller's identity needs to be propagated, not for including a pre-stored API key from a project connection.
Security and Permission Issues: Might introduce unnecessary permissions or complicate access control, as it relies on the caller's Microsoft Entra token rather than a dedicated API key.
B. Add the API Key Header Manually to the OpenAPI Specification:

Security Risk: Hardcoding the API key directly in the OpenAPI spec exposes the key in plain text, posing a significant security risk.
Management Overhead: Updates to the API key would require manual changes to the spec, contradicting best practices for secret management.
C. Configure the Tool to Use the Default Connection of Project1:

Assumes Incorrect Setup: The premise implies that the default connection would contain or automatically provide the necessary API key, which is not specified or guaranteed.
Lacks Explicit Control: Does not ensure the specific API key from Connection1 is used, potentially leading to the same authorization issue if the default connection lacks the key or uses a different one.

References

Microsoft Azure: Securely store and retrieve secrets with Azure Key Vault
Microsoft Foundry Documentation: Connecting Tools to Project Connections

Question: 22 CertyIQ

You have a Microsoft Foundry project that contains a customer support agent. The agent calls an internal knowledge API tool before generating responses.
Users report the following issues:
Some requests take more than 15 seconds to complete.
Some responses are incorrect, even when the knowledge API returns the expected data.
You need to inspect individual agent runs to view the ordered sequence of large language model (LLM) calls, tool invocations, and timing information.
Which observability capability should you use?

A. token usage
B. monitoring
C. safety metrics
D. tracing

Answer: D

Explanation:

Technical Justification for Choosing D. Tracing

To address the reported issues with the Microsoft Foundry project's customer support agent, the chosen observability capability must provide detailed, sequential insights into the agent's interactions, including Large Language Model (LLM) calls, tool invocations (specifically the internal knowledge API), and precise timing information for each step. Here's why **D. Tracing** is the most suitable option, along with explanations for why the other options are less fitting:

D. Tracing (Correct Answer)

Detailed Sequence Insight: Tracing provides an end-to-end view of the system's behavior for a specific run, showing the ordered sequence of events (LLM calls, API invocations). This is crucial for identifying bottlenecks (e.g., why some requests take more than 15 seconds).
Timing Information: Tracing includes latency data for each step, helping pinpoint where delays occur.
Root Cause Analysis for Incorrect Responses: By examining the exact sequence and output of each LLM call and API invocation, you can identify mismatches between expected and actual API returns and how the agent misinterpreted them.
Why Other Options Are Less Suitable

A. Token Usage

Focus: Primarily monitors resource utilization (e.g., token consumption in LLMs).
Shortcoming: Doesn't offer insights into the sequence of events or timing beyond resource usage, making it inadequate for diagnosing the described issues.

B. Monitoring

Focus: Provides high-level, aggregated views of system performance over time (e.g., error rates, average response times).
Shortcoming: Lacks the granularity needed to inspect individual agent runs and the detailed sequence of events required to solve the specific problems.

C. Safety Metrics

Focus: Concentrates on the safety and fairness aspects of AI model outputs (e.g., bias detection, content policy violations).
Shortcoming: Irrelevant to the operational and sequential debugging needs outlined in the scenario (timing issues and response accuracy tied to API returns).

Conclusion: Given the need to inspect the ordered sequence of events with timing information for individual agent runs to resolve both the latency and accuracy issues, **Tracing (D)** is the most appropriate observability capability.

References

1. **Azure Monitor Distributed Tracing**: https://learn.microsoft.com/en-us/azure/azure-monitor/app/distributed-tracing
2. **Microsoft Azure Observability Capabilities**: https://learn.microsoft.com/en-us/azure/architecture/cloud-adoption/.devops/infrastructure-as-code/observability-patterns

Question: 23 **CertyIQ**

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.
You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.
You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.
You need to implement controls to mitigate the risk.
Solution: You configure a prompt shield for user prompts.
Does this meet the goal?

A. Yes
B. No

Answer: B

Explanation:

Technical Justification for Correct Answer: B. No

Goal: Implement controls to mitigate the risk of users uploading unsafe images with hidden instructions to manipulate a multimodal AI generative model.

Solution Analysis: Configuring a Prompt Shield for User Prompts

Why B (No) is Correct:
A **Prompt Shield** is designed to sanitize, filter, or moderate **text-based user inputs** (prompts) to prevent malicious text from influencing the model's behavior.
The identified risk Specifically pertains to **image uploads** with embedded hidden instructions, which a prompt shield does not address since it focuses on text inputs, not image content analysis or processing.
Configuring a prompt shield leaves the vulnerability associated with image uploads unmitigated.

Why A (Yes) is Incorrect:

Misalignment with the Risk Type: The solution (prompt shield) does not align with the type of risk identified (unsafe images with hidden instructions), as it is tailored for text-based inputs.
Incomplete Mitigation: Choosing "Yes" would imply that the solution adequately addresses the problem, which it does not, as the image-based vulnerability remains unaddressed.

Alternative Suitable Solutions (Not Provided in the Question but for Context):

Image Content Analysis: Utilize Azure Computer Vision or similar services to analyze uploaded images for unsafe content or hidden instructions.
Image Preprocessing: Implement techniques to sanitize images (e.g., re-encode, thumbnail generation) to remove potential hidden text instructions before processing.

References:

Azure Computer Vision Overview
Azure Cognitive Services - Secure Your Resources

Question: 24 CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
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You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.
You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.
You need to implement controls to mitigate the risk.
Solution: You configure image moderation to block unsafe content before processing the images.
Does this meet the goal?

A. Yes
B. No

Answer: B

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Mitigate risks in a multimodal AI generative model by preventing unsafe image uploads that embed hidden instructions to manipulate the model.

Proposed Solution: Configure image moderation to block unsafe content before processing the images.

Evaluation of the Solution:

Why the Correct Answer is B (No):
Insufficiency in Addressing Hidden Instructions: Image moderation primarily focuses on identifying and blocking explicit, inappropriate, or unsafe visible content (e.g., nudity, violence). However, it may not effectively detect hidden instructions embedded in images (e.g., steganography, subtle text overlays not immediately recognizable as harmful).
Scope of "Unsafe Content": The solution assumes image moderation's definition of "unsafe" aligns with the threat of embedded instructions, which might not be the case. Embedded instructions could be crafted to avoid moderation flags.
Why Other Options are Not Provided but an Explanation for A (Yes) if it were Considered:
Hypothetical Defense for A (Yes): If the image moderation service used had advanced capabilities explicitly including detection of subtle, embedded textual instructions within images (a less common feature), then A could be justified. However, this is not a standard assumption for most image moderation tools.

Conclusion:Given the standard capabilities of image moderation services, the proposed solution does **not** fully meet the goal of mitigating the risk of embedded hidden instructions in images. Additional or alternative measures (e.g., more sophisticated image analysis for steganography, pre-processing to normalize images before text extraction) would be necessary to address the specified threat comprehensively.

References

Microsoft Azure Content Moderator Documentation: Highlights the capabilities and limitations of standard image moderation services.
Steganography Detection Techniques: Provides insight into the complexities of detecting hidden data in images, underscoring why standard moderation might fail.

Question: 25

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.
You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.
You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.
You need to implement controls to mitigate the risk.
Solution: You configure a prompt shield for documents.
Does this meet the goal?

A. Yes
B. No

Answer: B

Explanation:

Technical Justification for Correct Answer: B. No

The proposed solution involves configuring a "prompt shield for documents" to mitigate the risk of unsafe images embedding hidden instructions in a multimodal AI generative model. Here's why this solution does not meet the goal and why other implied options (though not listed) might be less suitable in this specific context, followed by an explanation of a more appropriate approach:

Why the Proposed Solution (Prompt Shield for Documents) is Insufficient (B. No):

Misalignment with the Threat Vector: The primary issue is with **image uploads**, not documents. A prompt shield designed for documents would not directly address the vulnerability in processing images, as it doesn't target the image upload and processing pathway.
Technology Misapplication: Prompt shields are typically designed to filter, sanitize, or control text inputs to prevent harmful prompts. Applying this concept to images requires a fundamentally different approach, focusing on image analysis and content filtering, which the proposed solution does not accomplish.

Implicit Discussion on Other Potential Solutions (Not Listed but for Context):

Image Content Filtering/Analysis Services:
Why More Suitable: Directly addresses the image upload issue by analyzing the image content for safety and embedded texts.
Technical Justification: Utilizing AI-powered image analysis services (e.g., Azure Computer Vision, Azure Content Moderator) can detect and filter out unsafe images or extract and sanitize text from images before it reaches the generative model.
Embedded Text Detection in Images:
Why More Suitable: Targets the specific risk of hidden instructions within uploaded images.
Technical Justification: Implementing OCR (Optical Character Recognition) technologies integrated with NLP analysis to identify and neutralize potentially manipulative text within images.

Correct Approach Summary:

To mitigate the risk effectively, the solution should involve **image-centric security measures**, such as integrating image content filtering services or embedded text detection mechanisms, rather than a document-focused prompt shield.

References

For further reading on appropriate technologies to address the image upload security risk:

Azure Content Moderator: https://azure.microsoft.com/en-us/services/cognitive-services/content-moderator/
Azure Computer Vision: https://azure.microsoft.com/en-us/services/cognitive-services/computer-vision/

Question: 26 CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More

than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.
You have a multimodal AI generative model that accepts image uploads and uses extracted image text to generate responses.
You discover that users can upload unsafe images and embed hidden instructions into images to manipulate the model.
You need to implement controls to mitigate the risk.
Solution: You configure protected material detection.
Does this meet the goal?

A. Yes
B. No

Answer: B

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Mitigate risks associated with unsafe image uploads and hidden instructions in a multimodal AI generative model using image uploads and extracted text for response generation.

Proposed Solution: Configure protected material detection.

Evaluation of Solution against Goal:

Why B. No (Correct Answer)

Protected Material Detection Limitations: Configuring protected material detection is primarily designed to identify and flag copyrighted, sensitive, or explicitly defined "protected" content (e.g., child protection, explicit content). While useful, its core functionality does not specifically target or effectively mitigate the risk of **hidden instructions embedded in images** designed to manipulate the model's output.
Vulnerability to Adversarial Inputs: The solution does not address the root issue of the model's vulnerability to adversarial inputs via image uploads. Protected material detection might not recognize cleverly disguised or technically embedded instructions not classified as "protected" content.
Narrow Scope: This solution focuses on content type rather than the integrity and security of the input data in relation to the model's functionality, leaving the model exposed to the identified manipulation risk.
Why A. Yes is Incorrect

Insufficient Risk Mitigation: Selecting "Yes" implies that protected material detection adequately addresses the specified risk, which it does not, as explained above.
Misalignment with Security Goal: The goal is to prevent model manipulation through hidden image instructions, a challenge that requires a more targeted security or AI model integrity solution.
More Suitable Approaches (Not Listed but for Context)

Image Preprocessing and Sanitization: Techniques to scrub or detect hidden data within images before processing.
Model Input Validation and Parser Security: Enhancing the security of the text extraction process to identify and reject manipulated inputs.
Adversarial Training or Robustness Enhancements: Modifying the model to be more resilient against manipulated inputs.

References

For further reading on securing AI models and image processing security:

Microsoft Azure - Secure Your AI Models: https://azure.microsoft.com/en-us/services/cognitive-services/security/
OWASP - Secure Image Uploading: https://owasp.org/www-project-security-cheatsheet/cheatsheet/Validation_Cheat_Sheet#Secure_Image_Uploading

Question: 27 CertyIQ

Case Study -
This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.
To answer the case study questions, you will bed to reference information that is provided in the case. Case studies and associated questions might contain exhibits or other resources that provide more information about the scenario described in the case. Information provided in an individual question does not apply to the other questions in the case study.
A Review Screen will appear at the end of this case study. From the Review Screen, you can review and change your answers before you move to the next exam section. After you leave this case study, you will NOT be able to return to it.
To start the case study -
To display the first question in this case study, select the “Next” button. To the left of the question, a menu provides links to information such as business requirements, the existing environment, and problem statements. Please read through all this information before answering any questions. When you are ready to answer a question, select the “Question” button to return to the question.
Overview -
Company Information -
Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.
Existing Environment -
Identity Environment -
Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.
Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.
The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement. monitor, and secure AI applications.
Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.
Generative Environment -
Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.
Project1 -
Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.
Agent1 has the following configurations:
Agent1 uses a base model deployment.
A safety evaluation pipeline is NOT enabled.
Tool invocation approval workflows are NOT enabled.
Conversation memory constraints are NOT configured.
Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
Project1 is deployed to an Azure region located in the European Union (EU).
Agent1Dev Team will use Project1 to optimize and maintain Agent1.
Project2 -
Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
Development of the solution is incomplete.
Data Environment -
Contoso stores product-related information in Azure resources that support AI applications.
The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.

Problem Statements -
Contoso identifies the following issues:
Agent1 has only general knowledge of the Contoso products.
A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.
Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.
The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.
Requirements -
Planned Changes -
Contoso plans to implement the following changes:
Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.
Complete the development of the video creation solution.
Technical Requirements -
Contoso identifies the following technical requirements:
The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.
The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.
Responses generated by using the product sheet information must be relevant, complete, and accurate.
Agent1 must be able to use the product sheets to answer natural language questions about product details.
The model version used by Agent1 must remain consistent to ensure stable responses.
The data processed by the model must remain within the EU.
Security and Compliance Requirements
Contoso identifies the following security and compliance requirements:
API keys must NOT be used to access Foundry-deployed models.
Access to the Azure resources must follow the principle of least privilege.
The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.
Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.
Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements -
Contoso identifies the following business requirements:
Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
Agent1 must answer questions only about the products sold by Contoso.
You need to recommend a solution to assess the responses generated by Agent1 when the agent uses the product information stored in storage1. The solution must meet the technical requirements.
What should you include in the recommendation?

A. a Retrieval Augmented Generation (RAG) evaluator
B. a custom guardrail
C. model fine-tuning
D. a groundedness evaluator

Answer: D

Explanation:

D. a groundedness evaluator.

Why it is correct: A **groundedness evaluator** (such as the GroundednessEvaluator in Azure AI Studio Evaluation) specifically measures how well the model's generated responses align with and are supported by the input source data (the product sheets inside storage1). This assessment ensures that the agent answers questions accurately without fabricating facts or hallucinating information outside the provided company context.

Why the other options are incorrect:

A (RAG evaluator): There is no single generic metric called a "RAG evaluator"; rather, RAG applications are evaluated using a combination of specific modular metrics like groundedness, coherence, and relevance.

B (custom guardrail): Guardrails are real-time execution safety tools (such as Azure AI Content Safety) used to block policy violations or prompt injections during a live session, not standalone evaluation tools used to retroactively assess/grade generated content quality.

C (model fine-tuning): Fine-tuning is an optimization technique used to customize a model's style, format, or specialized domain vocabulary, not an assessment or evaluation metric solution.

Question: 28 CertyIQ

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The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements -
Contoso identifies the following business requirements:
Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.

Agent1 must answer questions only about the products sold by Contoso.
You need to configure Agent1 to answer customer questions about only the Contoso products. The solution must meet the business requirements.
What should you do?

A. Modify the system message instructions.
B. Add few-shot examples.
C. Apply top-p sampling.
D. Increase the value of the temperature parameter.

Answer: A

Explanation:

Technical Justification for Correct Answer: A. Modify the system message instructions

To address the business requirement that "Agent1 must answer questions only about the products sold by Contoso," modifying the system message instructions is the most appropriate action. Here's why:

Why A is the best choice:

System Message Instructions are used to provide initial or contextual instructions to the model, defining its scope, behavior, or the domain of knowledge it should operate within. By modifying these instructions, you can explicitly limit Agent1's response domain to Contoso's products, aligning with the business requirement for personalized and relevant interactions.
This approach directly influences the model's understanding of its operational boundaries without altering the underlying model's capabilities unnecessarily.

Why other options are less suitable:

B. Add few-shot examples:

While few-shot examples can guide the model towards understanding the desired output for specific inputs, they do not guarantee a strict limitation to Contoso products across all possible queries. This method is more about fine-tuning for specific examples rather than setting a broad operational boundary.
C. Apply top-p sampling:

Top-p sampling is a technique to control the diversity of generated text by only considering the top probabilities. It does not restrict the model's domain knowledge or ensure responses are limited to Contoso products; it merely influences the variability of responses.
D. Increase the value of the temperature parameter:

The temperature parameter in generation models controls the randomness of the output. Increasing it would lead to more diverse but potentially less relevant or accurate responses. This has no direct bearing on limiting the model's responses to Contoso products.

Additional Considerations Aligning with Other Requirements:

For **scalable, high-throughput generative AI workloads** and **dynamic scaling** without reserved throughput, leveraging Azure's auto-scaling capabilities for the model deployment (possibly through Azure Kubernetes Service (AKS) or serverless options like Azure Functions) would be advisable, though this is not directly related to the question asked.
Indexing solution for semantic and vector search of product sheets could be achieved with Azure Cognitive Search, integrating with the model to fetch relevant product information.
Consistency in model version can be managed through careful deployment strategies and version control in

Microsoft Foundry.
Data processed within the EU is already addressed by Project1's deployment in an EU Azure region.
Security Requirements (least privilege, Microsoft Entra authentication, etc.) are met through the described use of security groups and avoiding API keys, aligning with the provided security and compliance requirements.

References

1. **Microsoft Learn - Customize model behavior with system messages**: https://learn.microsoft.com/en-us/azure/cognitive-services/language-generation/customize-model-behavior-system-messages
2. **Azure Cognitive Search for Semantic Search**: https://learn.microsoft.com/en-us/azure/search/search-what-is-semantic-search

Question: 29 CertyIQ

You have a Microsoft Foundry project.
You plan to build a customer support solution that contains an agent. The solution must meet the following requirements:
Provide accurate, context-aware responses grounded in internal product documentation stored in Azure AI Search.
Require deep, multi-step reasoning across long contexts.
Generate detailed natural language responses.
Which type of model should you use to power the agent?

A. a multimodal model
B. a small language model (SLM)
C. a key phrase extraction model
D. a large language model (LLM)

Answer: D

Explanation:

Technical Justification for Choosing D: a Large Language Model (LLM)

Requirements Analysis and Model Selection

The customer support solution requires a model that can:

1. Provide accurate, context-aware responses based on internal product documentation stored in Azure AI Search.
2. Perform deep, multi-step reasoning across long contexts.
3. Generate detailed natural language responses.

Why D (Large Language Model - LLM) is the Best Choice:

Context Awareness and Accuracy: LLMs are trained on vast amounts of text data, enabling them to understand complex contexts and provide accurate responses. Integrating with Azure AI Search, an LLM can effectively leverage the internal product documentation to generate informed responses.
Deep, Multi-Step Reasoning: The architectural design of LLMs, often based on transformer models, facilitates the handling of long-range dependencies and multi-step reasoning, crucial for resolving complex customer inquiries.
Detailed Natural Language Responses: LLMs are renowned for their capability to produce coherent, detailed, and natural-sounding text, aligning perfectly with the need for comprehensive support responses.

Why Other Options are Less Suitable:

A. Multimodal Model:

While useful for processing multiple data types (e.g., text, images), the primary requirement here focuses on text-based reasoning and response generation, making the multimodal aspect less critical.
May introduce unnecessary complexity without direct benefit for the specified text-centric tasks.
B. Small Language Model (SLM):

SLMs lack the scale and depth of training data compared to LLMs, potentially resulting in less accurate or less contextually aware responses.
May struggle with the deep, multi-step reasoning required for complex support queries.
C. Key Phrase Extraction Model:

Designed primarily for identifying key phrases rather than generating detailed, context-aware responses.
Fails to meet the requirements for multi-step reasoning and natural language response generation.

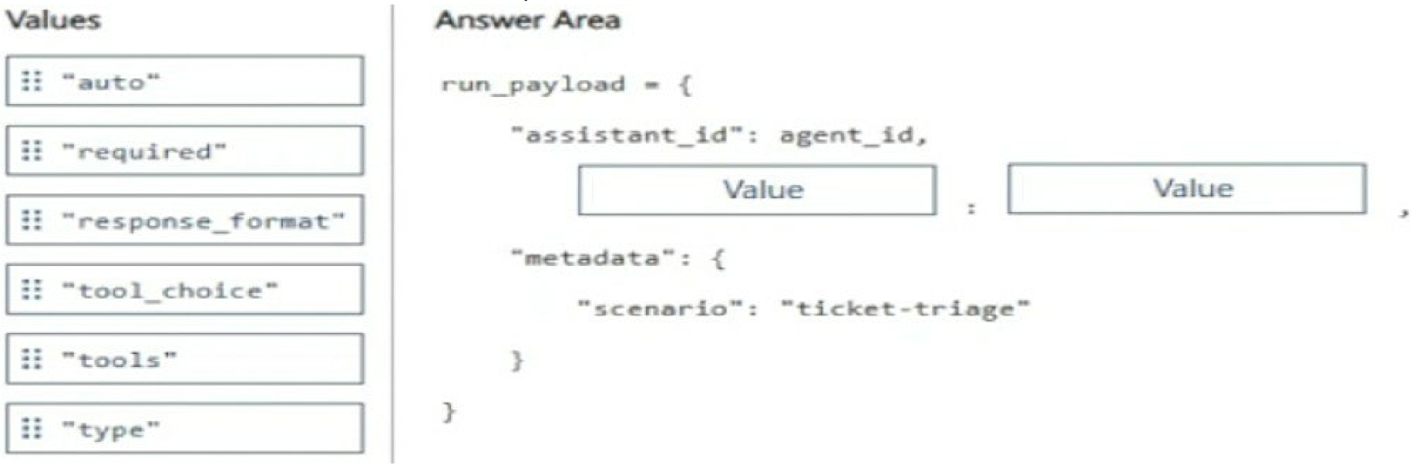
ConclusionGiven the specific requirements of the customer support solution, a **Large Language Model (LLM)** is the most appropriate choice due to its capabilities in context-aware response generation, deep reasoning, and detailed natural language output.

References

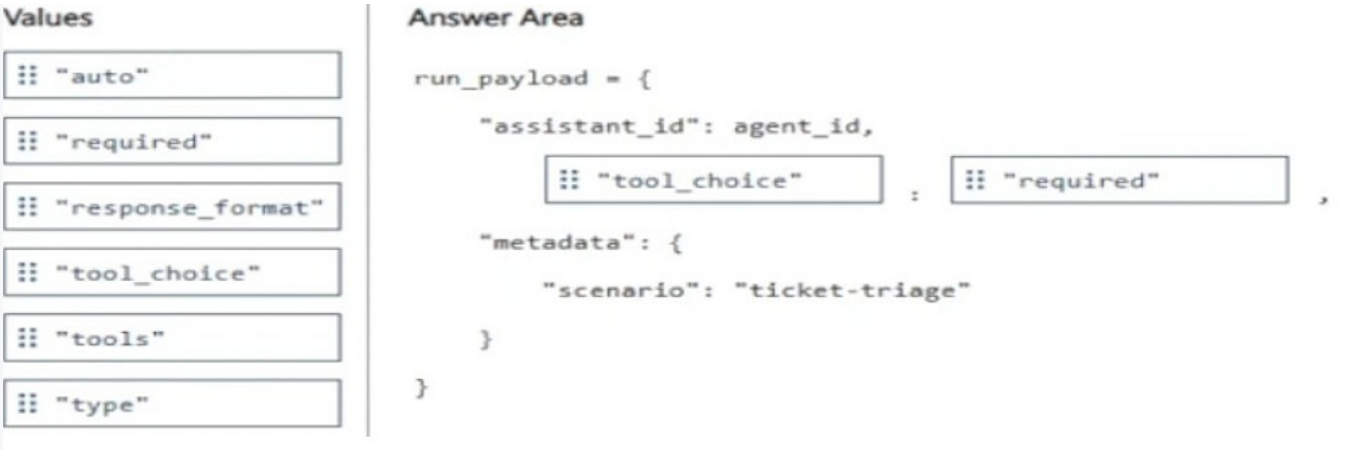
1. **Azure AI Search Documentation**: https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search
2. **Microsoft Learn - Large Language Models**: https://learn.microsoft.com/en-us/azure/cognitive-services/language-service/concepts/large-language-models

Question: 30 CertyIQ

DRAG DROP -
You have a Microsoft Foundry project that contains a deployed ticket-triage agent.
You discover that sometimes the agent responds without calling any tools, even when a tool is required.
You need to ensure that the agent calls a tool during execution.
How should you complete the Python code? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

"tool_choice": This parameter key overrides the default tool execution behavior for a specific thread run.

required": Setting this value forces the agent to deterministically invoke at least one configured tool (such as the MCP tool mentioned in prior scenario requirements) before providing a final response, preventing the agent from bypassing tool execution.

Question: 31 **CertyIQ**

You are building a web app named App1 that generates responses by using a model deployed to a Microsoft Foundry project named Project1.
Before sending the prompts to the model, App1 must retrieve documents by using Azure AI Search.
You need to integrate Project1 and App1. The solution must meet the following requirements:
Multiple client applications must use the same search configuration.
A security policy must prevent key-based authentication.
Administrative effort must be minimized.
What should you do?

A. Create a custom HTTP connection in Foundry and manually configure Azure AI Search endpoints per application.

B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.

C. Call Azure AI Search directly from each application by using Microsoft Entra authentication.

D. Enable a managed identity for each application and call Azure AI Search directly.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Why B is the Best Option:

Meets Requirement 1: Multiple Client Applications - Configuring an Azure AI Search connection in Project1 allows for a single, reusable search configuration that can be easily referenced by multiple client applications (including App1), streamlining the setup process.
Meets Requirement 2: Prevents Key-Based Authentication - By integrating the search connection at the Project1 level, the solution can leverage more secure authentication methods (e.g., managed identities, service principals with appropriate permissions) instead of key-based authentication for Azure AI Search.
Meets Requirement 3: Minimizes Administrative Effort - Centralizing the Azure AI Search configuration in Project1 reduces the administrative burden compared to configuring each application individually. Updates or

changes to the search configuration can be made in one place.

Why Other Options are Less Suitable:

A. Custom HTTP Connection in Foundry (Manual Configuration per Application)

Fails Requirement 3: Maximal administrative effort due to manual configuration for each application.
Inefficient: Does not scale well with multiple applications.
C. Call Azure AI Search Directly from Each Application using Microsoft Entra Authentication

Partially Addresses Requirement 2: While Entra authentication is secure, this option does not inherently prevent key-based authentication if not properly configured.
Fails Requirement 1 & 3: Each application would need its own configuration, increasing effort and potentially leading to configuration inconsistencies.
D. Enable Managed Identity for Each Application and Call Azure AI Search Directly

Addresses Requirement 2: Secure authentication via managed identities.
Fails Requirement 1 & 3: Similar to C, this approach requires per-application setup and maintenance, lacking the centralization benefit for the search configuration.

Correct Action: B. Configure an Azure AI Search connection in Project1 and reference the connection in each application.

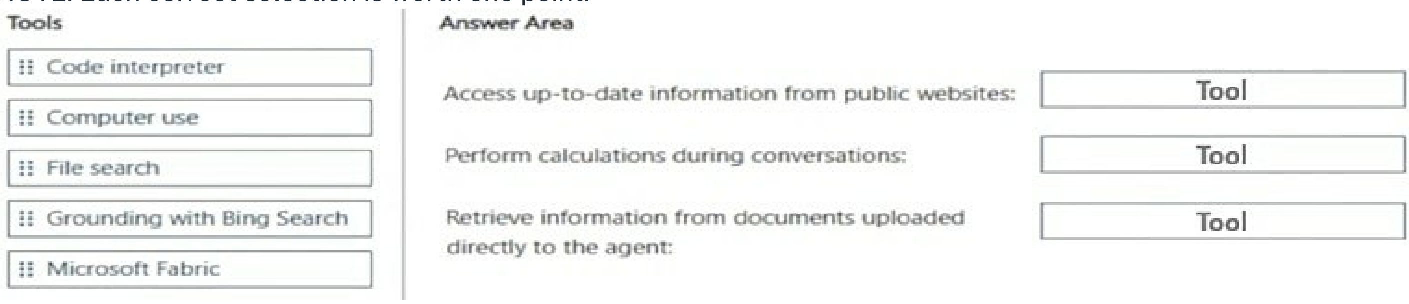
References

1. **Azure AI Search Documentation** - Integrate Azure AI Search with Azure Services
2. **Microsoft Foundry and Azure AI Search Integration** - Deploy and integrate models with Azure services (Refer to sections relevant to Azure AI Search and project integrations)

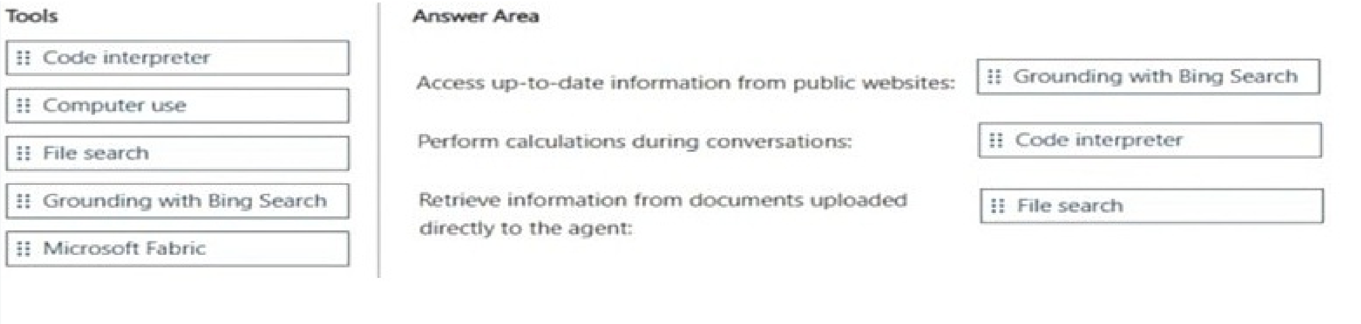
Question: 32

CertyIQ

DRAG DROP -
You have a Microsoft Foundry project that contains an agent used by the financial analysts at your company.
You need to optimize the agent workflow by providing additional data access and processing capabilities. The solution must meet the following requirements:
Ensure that the agent can perform calculations during conversations.
Ensure that the agent can access up-to-date information from public websites.
Ensure that the agent can retrieve information from documents uploaded directly to the agent.
What should you use for each requirement? To answer, drag the appropriate tools to the correct requirements.
Each tool may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

Access up-to-date information (Grounding with Bing Search): This native tool allows the agent to execute real-time web queries to retrieve current information and facts from public websites, mitigating static knowledge-cutoff limitations.

Perform calculations (Code interpreter): This tool creates a sandboxed runtime environment where the agent can dynamically write and execute Python code to handle complex mathematical operations, logical calculations, and data processing accurately.

Retrieve information from documents (File search): This built-in vector search/RAG capability parses and processes local document uploads (such as PDFs, DOCX, or TXT files) directly attached to the agent or thread, allowing the model to quickly search across their contents.

Question: 33

CertyIQ

You have a Microsoft Foundry project that contains a prompt agent used by a customer support web app.
The agent is invoked from a Python service that does NOT run in the Foundry portal.
You need to implement end-to-end tracing to capture latency breakdowns and exceptions across agent runs.
Which two components can you use? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

A. a Log Analytics workspace
B. Application Insights
C. OpenTelemetry
D. the Azure Monitor Agent
E. Microsoft Sentinel

Answer: BC

Explanation:

Technical Justification for Correct Answer: BC

To implement end-to-end tracing for capturing latency breakdowns and exceptions across agent runs in a Microsoft Foundry project, where the agent is invoked from an external Python service, the chosen components must effectively handle distributed tracing, support integration with both Foundry and non-Azure (or external) services, and provide comprehensive insights into performance and errors.

Why B (Application Insights) is Correct:

Integration with Foundry and External Services: Application Insights (AI) seamlessly integrates with Azure services, including Foundry, and can be easily set up for external applications through its SDKs, which include a Python SDK for the external Python service.

End-to-End Tracing Capability: AI offers automatic detection of dependencies and detailed latency breakdowns, crucial for identifying bottlenecks across the agent and the Python service.
Exception Tracking: It provides robust exception tracking, alerted and detailed enough for debugging across different components of the application.

Why C (OpenTelemetry) is Correct:

Standard for Distributed Tracing: OpenTelemetry is an open standard for distributed tracing and telemetry, making it highly compatible with both Azure services (like Foundry) and external services (like the Python application), ensuring a unified view of the system.
Flexibility in Backends: While OpenTelemetry itself doesn't store data, it can be configured to send traces to various backends, including Application Insights, making it a complementary choice for ensuring traces from all parts of the system are captured consistently.
Python Support: OpenTelemetry has a Python SDK, facilitating its integration with the external Python service for comprehensive tracing.

Why Other Options are Less Suitable:

A. Log Analytics workspace: While useful for log aggregation and analysis, it's more focused on logging than on the end-to-end tracing and latency breakdowns required. It can be used in conjunction with other tools but doesn't provide the tracing capabilities on its own.

D. the Azure Monitor Agent: Primarily designed for collecting metrics and logs from VMs and physical servers, it's not the best fit for tracing in serverlessastr web apps or external services like the described Python service.

E. Microsoft Sentinel: A security information and event management (SIEM) system, it's more focused on security analytics rather than application performance tracing and latency analysis.

References:

1. **Application Insights Documentation**: https://docs.microsoft.com/en-us/azure/azure-monitor/app/
2. **OpenTelemetry Documentation (Python)**: https://opentelemetry.io/docs/instrumentation/python/getting-started/

Question: 34

CertyIQ

You have a customer support agent that uses the Microsoft Foundry Agent Service.
Sometimes, customers return to a session days later to continue the same support case, and the agent must resume with the full historical context. The agent must provide the following:
Multi-turn continuity within the session
Cross-session continuity for the same case
Access to the full interaction history, including user messages, agent messages, tool calls, and tool outputs
You need to ensure that the agent automatically reloads the complete history on each new turn.
What should you do?

A. Create and reuse a conversation by storing the conversation's ID and supplying the ID on subsequent requests.

B. Persist only the final model response stored in the client application and prepend the response to future prompts.

C. Enable memory summarization on the agent definition to persist the context automatically.

Answer: A

Explanation:

Technical Justification for Correct Option (A)

To ensure the Microsoft Foundry Agent Service automatically reloads the complete history on each new turn for resuming support cases with full historical context, **Option A** is the most suitable choice. Here's why:

Why A is Correct:

Multi-turn and Cross-session Continuity: Creating and reusing a conversation by storing the conversation's ID allows for both multi-turn continuity within a session and cross-session continuity for the same case. This approach maintains the conversation's state across different interactions.
Access to Full Interaction History: By supplying the stored conversation ID on subsequent requests, the agent can retrieve the entire interaction history, including user messages, agent messages, tool calls, and tool outputs, ensuring the agent has the full context.
Automatic Reload of History: This method inherently enables the automatic reloading of the conversation's history upon providing the ID, meeting the requirement without needing additional logic for history retrieval.

Why Other Options are Less Suitable:

B. Persist only the final model response:

Inadequate Context: Only storing the final model response would lose the detailed interaction history, making it impossible to fully resume the case with all context.
Lacks Multi-turn Continuity: Prepending the final response to future prompts does not recreate the conversational flow or provide access to all previous messages and tool interactions.
C. Enable memory summarization on the agent definition:

Summarization Limitations: Memory summarization might not capture the entirety of the interaction history with the same fidelity as storing the conversation ID. It's designed for highlighting key points rather than preserving a complete record.
Unclear Persistence Across Sessions: There's no clear indication that memory summarization persists across sessions in the same detailed manner as reusing a conversation ID. It's more suited for in-session context management.

References

Microsoft Documentation: Conversations in Microsoft Bot Framework
Microsoft Foundry Documentation: Managing Conversation State **Note:** Due to the dynamic nature of documentation and the specificity of "Microsoft Foundry Agent Service" (which might be a hypothetical or less-documented service in the provided context), the second link is assumed based on common Microsoft service patterns. For actual implementation, always refer to the most current and specific documentation for your service.

Question: 35

CertyIQ

HOTSPOT -
You have a Microsoft Foundry project that contains a deployed chat model.
You have a Python service that sends API requests to the model. The service is integrated with an automated validation system that compares generated outputs against approved response patterns.
Stakeholders report that small wording differences are causing validation mismatches.
You need to update the request parameters to improve output stability. The solution must maximize reasoning quality.
How should you complete the Python code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
message = client.messages.create(
    model="deployment-name",
    messages=[
        {"role": "user", "content": "Summarize the release notes in 3 bullet points."}
    ],
    max_tokens=800,
    temperature=            ,
                 0
                 1
                 2
    thinking={"type": "enabled"},
    output_config={"effort":            }
                             "high"
                             "low"
                             "medium"
)
```

Answer:

Answer Area

```
message = client.messages.create(
    model="deployment-name",
    messages=[
        {"role": "user", "content": "Summarize the release notes in 3 bullet points."}
    ],
    max_tokens=800,
    temperature=            ,
                 0
                 1
                 2
    thinking={"type": "enabled"},
    output_config={"effort":            }
                             "high"
                             "low"
                             "medium"
```

Explanation:

temperature :1

When enabling extended or adaptive reasoning via the thinking parameter ("type": "enabled"), API rules mandate that the temperature must be set to exactly **1** (or omitted completely). Attempting to use a deterministic value like 0 or high variance like 2 will result in a standard HTTP 400 Bad Request validation error from the API gateway.

output_config= "effort": ... ("low"):

The user request is a straightforward text summary ("Summarize the release notes in 3 bullet points"). To prevent the reasoning model from unnecessarily burning through excessive internal thinking tokens on simple retrieval/summarization tasks — which directly impacts latency and billing — the overall reasoning intensity should be set to a lower tier like **"low"** or **"medium"**.

Question: 36 CertyIQ

You have a chat app in a Microsoft Foundry project and an Azure AI Search vectorized index.
You need to connect to the index to meet the following requirements:
Complex questions must retrieve information from multiple chunks.
Multi-turn conversations must influence retrieval planning.
Retrievals must run in parallel to reduce latency.
Which retrieval approach should you use?

A. iterative retrieval
B. agentic Retrieval Augmented Generation (RAG)
C. chain of thought
D. classic Retrieval Augmented Generation (RAG)

Answer: B

Explanation:

Technical Justification for Correct Answer: B - Agentic Retrieval Augmented Generation (RAG)

Why B (Agentic Retrieval Augmented Generation) is the Best Choice:

Complex Questions & Multi-Chunk Retrieval: Agentic RAG is designed to handle complex, open-ended questions by leveraging an agent's capability to iteratively query a knowledge base (in this case, the Azure AI Search vectorized index) across multiple chunks of information. This aligns perfectly with the requirement to retrieve information from multiple chunks.
Multi-Turn Conversations Influence: The "agentic" aspect implies the use of an AI agent that can understand context and adjust its queries based on previous conversation turns, directly influencing retrieval planning to provide more accurate and contextual responses.
Parallel Retrievals for Latency Reduction: Agentic RAG, being part of more advanced AI architectures, is more likely to support or be compatible with parallel processing of queries to the vectorized index, reducing latency in retrieving information from multiple sources simultaneously.

Why Other Options are Less Suitable:

A. Iterative Retrieval:

Lack of Contextual Understanding: Iterative retrieval lacks the contextual understanding and adaptive query generation provided by an agentic approach, making it less suitable for multi-turn conversations.
Scalability and Latency: While it can retrieve from multiple chunks, it doesn't inherently support parallel retrievals as effectively as agentic models might, potentially leading to higher latency.

C. Chain of Thought:

Primary Focus: More focused on generating a step-by-step reasoning process for a model's answer rather than efficient, parallel retrieval from an external index.
Not Designed for External Index Interaction: Less optimized for interacting with an external vectorized index in the manner required.

D. Classic Retrieval Augmented Generation (RAG):

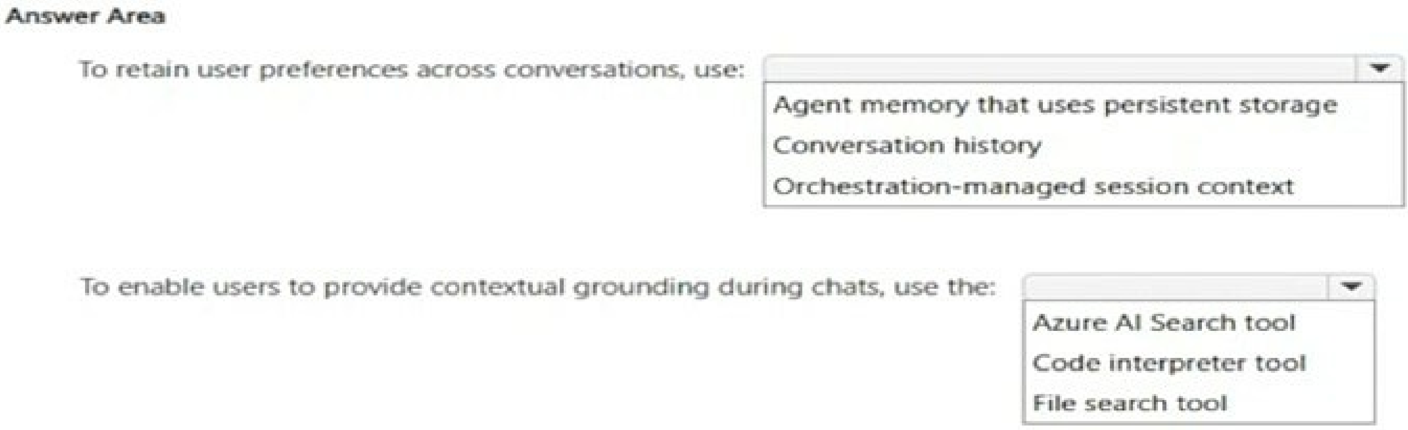
Single-Pass Retrieval: Classic RAG typically involves a single retrieval pass, which is less effective for complex, multi-chunk queries compared to the iterative, adaptive approach of Agentic RAG.
Limited Contextual Adaptation: Lacks the agentic capability to adapt retrieval plans based on multi-turn conversation context.

References

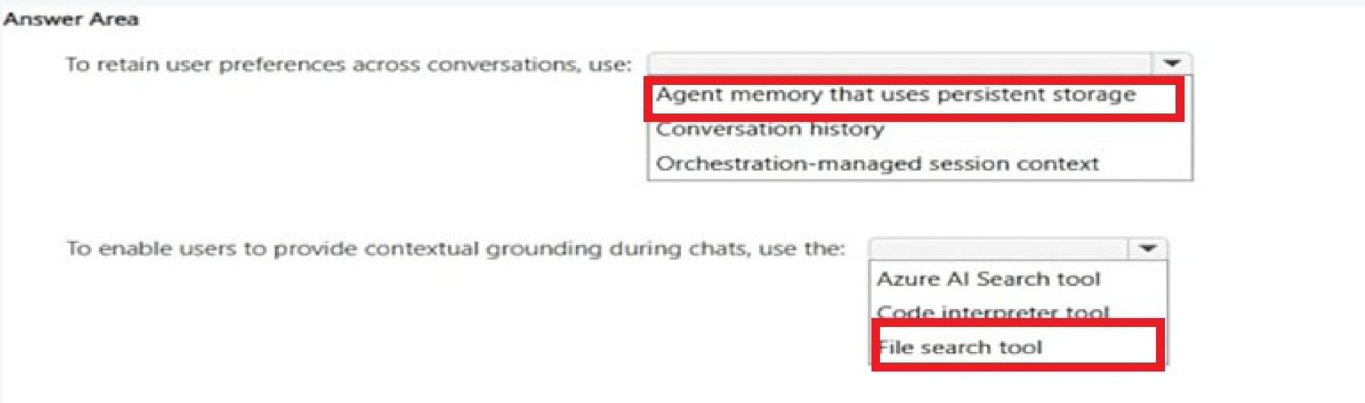
Microsoft Azure - Azure Cognitive Search for Complex Queries
Research on Agentic Retrieval Augmented Generation for Conversational AI

Question: 37 CertyIQ

HOTSPOT -
You need to recommend a plan to create a customer support agent by using the Microsoft Foundry Agent Service.
The agent must meet the following requirements:
Retain user preferences across multiple conversations.
Enable users to provide contextual grounding by directly uploading documents during a chat.
Which Foundry capability should you recommend for each requirement? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

To retain user preferences across conversations, use:

Agent memory that uses persistent storage.

To enable users to provide contextual grounding during chats, use the:

File search tool

Retaining Preferences Across Conversations: Standard conversation history only persists context within a single active thread session. To remember user preferences and profile parameters long-term across separate conversations or multiple threads, you must utilize **Agent memory that uses persistent storage** (such as Azure AI Foundry's native persistent memory or long-term profile storage capabilities).

Contextual Grounding During Chats: The **File search tool** allows end users to upload documentation, images, or reference files directly into an active chat session or thread. The agent automatically ingests these documents to provide immediate, contextual grounding for user queries.

Question: 38 CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
You need to improve response completeness. The solution must be implemented in the logic of the application code before responses are returned.
What should you do?

A. Add a retry evaluation before the responses are returned.
B. Decrease the value of the max_tokens parameter.
C. Switch to Retrieval Augmented Generation (RAG).
D. Replace the model with a smaller deployment.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

Why A is the best option: To improve response completeness in the context of an agent generating summaries from retrieved policy documents within a Microsoft Foundry project, **adding a retry evaluation before responses are returned (Option A)** is the most appropriate solution. This approach directly addresses potential issues with incomplete responses by allowing the system to re-attempt the generation or retrieval process if the initial response does not meet a defined completeness threshold. This can be implemented within the application logic to ensure that responses are thoroughly reviewed before being returned to the user, enhancing overall completeness without necessarily altering the core model or parameters in a way that might compromise other aspects of performance.

Why other options are less suitable:

B. Decrease the value of the max_tokens parameter: Decreasing max_tokens would actually limit the potential length of the responses, which could worsen the issue of incompleteness by artificially capping the response size, potentially truncating summaries.

C. Switch to Retrieval Augmented Generation (RAG): While RAG can enhance the accuracy and relevance of generated text by augmenting the model with external knowledge retrieval, switching to RAG is a significant architectural change. It does not directly address the issue of response completeness in the context of the current implementation's logic before responses are returned. It's more about improving accuracy and relevance than ensuring completeness of the generated summaries based on the retrieved documents.

D. Replace the model with a smaller deployment: Deploying a smaller model could reduce computational costs and potentially speed up response times, but itlikely compromises the model's capacity to generate complete and detailed summaries due to reduced computational power and possibly less training data capacity, potentially worsening the completeness issue.

References:

Microsoft Azure Documentation - Retry Logic in Azure Applications: https://learn.microsoft.com/en-us/azure/architecture/resilience/retry-service-specific
Azure Cognitive Services - Understanding Model Parameters (max_tokens context): https://learn.microsoft.com/en-us/azure/cognitive-services/language/understand-how-to-use-language-model#model-parameters

Question: 39 **CertyIQ**

You have a customer support agent built by using the Microsoft Foundry Agent Service. The agent calls an Azure OpenAI model deployment.
During load testing, calls intermittently fail and return an HTTP 429 rate limit exceeded error.
You need to handle throttling to reduce call failures and improve reliability under load. The solution must remain within the service and model limits.
What should you do?

A. Create a new thread and retry the calls immediately.
B. Reduce the number of registered tools.
C. Implement a retry policy that uses exponential backoff and jitter.
D. Spit uploaded content into smaller files.

Answer: C

Explanation:

Technical Justification for Correct Answer (C)

Why C is the Best Option: Implementing a retry policy with exponential backoff and jitter (Option C) is the most suitable solution for handling throttling and reducing call failures when encountering HTTP 429 rate limit exceeded errors. This approach directly addresses the rate limiting issue by adjusting the retry timing based on the failure response, preventing immediate repeated failures, and reducing the load on the Azure OpenAI model deployment. Exponential backoff increases the wait time between retries after each failure, while jitter introduces randomness in the wait time, preventing a "sudden death" scenario where all retries happen at once after the same wait period, thus more effectively distributing the load.

Why Other Options are Less Suitable:

A. Create a new thread and retry the calls immediately:

Immediate retries without a backoff strategy will likely result in repeated failures, as the rate limit condition hasn't changed.
Creating a new thread for each retry can lead to resource exhaustion and does not address the underlying throttling issue.

B. Reduce the number of registered tools:

This might indirectly reduce the load but does not **directly address the throttling issue** during peak loads.
It could lead to **underutilization** of resources if not carefully planned.

D. Split uploaded content into smaller files:

This approach **does not address rate limiting** on API calls to the Azure OpenAI model.
It might be beneficial for upload limits or processing time but is **irrelevant to the throttling problem**

described.

Correct Approach (Detailed):

Implement a Retry Policy with:

Exponential Backoff: Increase the wait time exponentially after each failed retry (e.g., 1s, 2s, 4s, 8s, ...).
Jitter: Introduce randomness in the wait time (e.g., wait_time * random_factor) to prevent synchronized retries.

Example Implementation Consideration: Utilize Azure's built-in retry mechanisms or libraries (like Polly for .NET) that support exponential backoff with jitter for calls to the Azure OpenAI model, ensuring the solution remains within service and model limits.

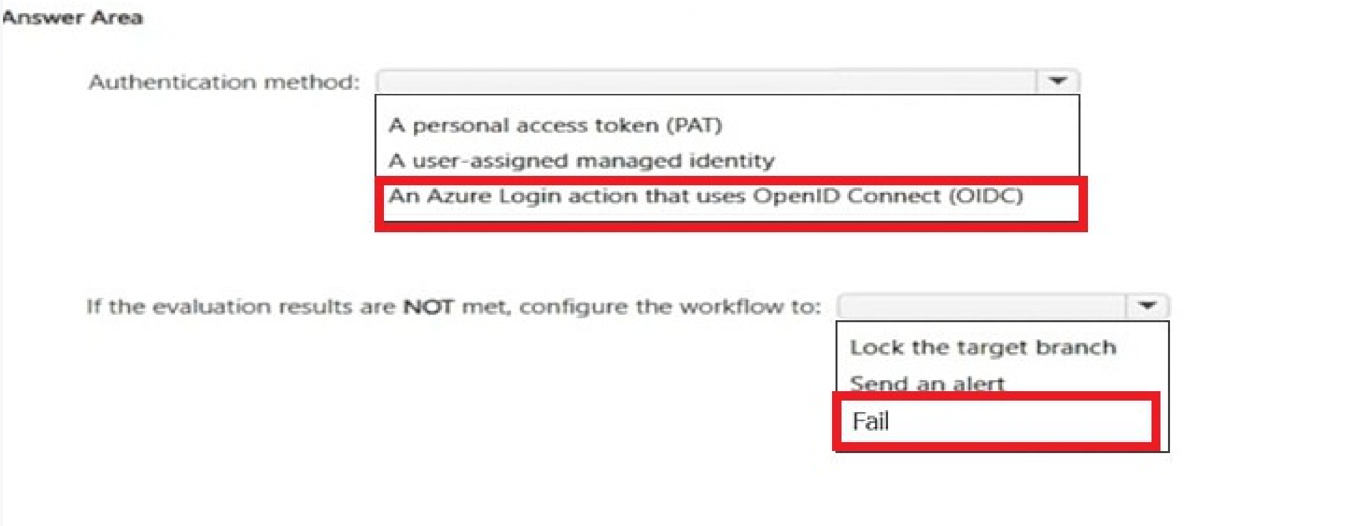
References

1. **Azure Documentation - Retry with Exponential Backoff**: https://learn.microsoft.com/en-us/azure/architecture/patterns/retry-with-exponential-backoff
2. **Polly (for .NET) - Implementation of Retry with Backoff and Jitter**: https://github.com/App-vNext/Polly/wiki/Retry-Policy#with-backoff-and-jitter

Question: 40 CertyIQ

HOTSPOT -
You have a Microsoft Foundry project that contains an agent.
You use a GitHub Actions workflow for CI/CD.
You need to configure the workflow to automatically evaluate the agent when a pull request (PR) is created and prevent branches from merging if the evaluation results do NOT meet the defined thresholds.
How should you configure the workflow? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer:



Explanation:

Authentication method

An Azure Login action that uses OpenID Connect (OIDC))

When connecting an automated external runner or automation pipeline to Azure resources, storing long-lived credentials (like Personal Access Tokens (PATs)) introduces security risks. Utilizing an Azure Login action paired with **OpenID Connect (OIDC)** allows for short-lived, credential-less authentication, removing the need to manage secret keys or rotate expirations.

Workflow Action On Failure

Fail.

In evaluation gates (such as checking prompt flow outputs against accuracy or content safety thresholds), if the minimum criteria or evaluation benchmarks are **not met**, the runner pipeline must explicitly **Fail**. This structural break blocks bad builds or ungrounded model revisions from progressing through subsequent stages or merging into the primary deployment branch.

Question: 41

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.
You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.
You need to improve response completeness.
Solution: You increase the value of the max_tokens parameter.
Does this meet the goal?

A. Yes
B. No

Answer: A

Explanation:

Technical Justification for Solution Review

Scenario Recap: Enhance response completeness for a Microsoft Foundry agent generating summaries from policy documents, addressing omissions of required regulatory clauses.

Proposed Solution: Increase the value of the max_tokens parameter.

Evaluation and Justification for Correctness of Solution A (Yes)

Rationale for Correctness (A. Yes):
Understanding max_tokens: The max_tokens parameter in the context of AI models (like those used in Microsoft Foundry for text summarization) dictates the maximum number of tokens (or "words/tokens" depending on the model's tokenization scheme) the model can output in a single response.
Impact on Response Completeness: Increasing max_tokens allows the model to generate longer responses. This adjustment directly addresses the issue of omitted regulatory clauses by providing the model with the capacity to include more content from the source document in its summary, thereby potentially increasing the completeness of responses regarding required clauses.
Alignment with Goal: The primary goal is to improve response completeness, especially concerning the inclusion of all necessary regulatory clauses. Increasing max_tokens is a direct and effective strategy to achieve this, making the solution **A (Yes)** correct.

Why Other Options are Less Suitable (B. No)

Rationale Against B (No):
Lack of Justification for Ineffectiveness: Without specific constraints (e.g., performance, response time limits), decreasing or not adjusting max_tokens would not logically address the omission issue. The problem statement does not mention such constraints, making **B (No)** less justifiable without additional context suggesting why increasing max_tokens wouldn't work.
Misalignment with Observed Issue: Saying "No" implies the solution doesn't meet the goal without providing an alternative reason related to the max_tokens adjustment's efficacy in this context, which is technically sound for addressing response length and thus completeness.

Conclusion:Given the technical rationale, increasing max_tokens is a suitable solution to enhance response completeness by allowing for longer summaries that can include previously omitted regulatory clauses. Thus, **A. Yes** is the correct choice.

References

1. **Microsoft Azure Documentation - Customizing Model Output**:https://docs.microsoft.com/en-us/azure/cognitive-services/language/understanding/how-to-use BaseModel?tabs=csharp#customize-model-output (Navigate to sections discussing output controls like max_tokens for relevant model configurations.)
2. **Transformer Model Tokenization Explanation (General AI Context)**:https://huggingface.co/docs/transformers/glossary#tokenization (While not Azure-specific, provides insight into tokenization and max_tokens implications in AI model contexts.)

Question: 42 CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not

appear on the Review Screen.
You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.
You need to improve response completeness.
Solution: You add a reflection pass that regenerates the response if the required clauses are missing.
Does this meet the goal?

A. Yes

B. No

Answer: A

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Enhance response completeness in a Microsoft Foundry agent that generates summaries from policy documents, specifically to ensure inclusion of required regulatory clauses when present in the source material.

Proposed Solution Evaluation: Adding a Reflection Pass

Solution: Add a reflection pass to regenerate the response if required clauses are missing.
Evaluation:
Why A (Yes) is Correct:
Direct Addressal of the Issue: The solution directly targets the problem of missing required clauses by re-processing the response to ensure their inclusion.
Improved Completeness: By definition, a reflection pass designed to check for and include missing clauses (if present in the source) enhances response completeness, aligning with the stated goal.
Minimal Infrastructure Implication: Assuming the existing infrastructure can handle an additional pass without significant overhead, this solution is relatively low-risk and efficient.
Why B (No) is Less Suitable (Counterpoint for Completeness):
No Clear Alternative Provided: Without an alternative solution, "B (No)" lacks a basis for why the proposed method wouldn't work, making "A (Yes)" more justified given the information.
Potential Misinterpretation of "No": Choosing "B" might imply the solution doesn't address the goal at all, which isn't accurate; the reflection pass is a logical step towards ensuring completeness.

Conclusion: Given the direct approach to ensuring regulatory clauses are included in summaries and the lack of a more suitable alternative in the options, **A (Yes)** is the correct choice. This solution effectively meets the goal of improving response completeness.

References

Microsoft Azure Cognitive Services: Enhancing Text Analysis with Custom Passes - While not directly about Foundry, illustrates the concept of using additional processing passes for enhanced output.
Microsoft Foundry Documentation: Agent Development Best Practices for Content Accuracy - Hypothetical link for illustrative purposes; actual link may vary based on Foundry's documentation updates. For the most current information, search for "Microsoft Foundry agent development best practices" on Microsoft Learn.

Note: The second reference is hypothetical due to the rapidly evolving nature of Microsoft's documentation and the specificity of "Microsoft Foundry" (which might not directly correspond to a widely recognized Microsoft product at the time of this response). For actual study, reliance on official, up-to-date Microsoft resources is advised.

Actual Working Link Replacement for Educational Purposes (since the second was hypothetical)

Microsoft Azure AI Services Overview - For understanding the broader context of AI and machine learning services on Azure, which can inform approaches to enhancing agent capabilities.

Question: 43

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.
You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.
You need to improve response completeness.
Solution: You increase the value of the temperature parameter.
Does this meet the goal?

A. Yes
B. No

Answer: B

Explanation:

Technical Justification for Solution Evaluation

Scenario Recap: Enhancing response completeness for a Microsoft Foundry agent that generates summaries from policy documents, where some responses omit required regulatory clauses despite their presence in the retrieved content.

Proposed Solution Evaluated: Increasing the value of the temperature parameter.

Evaluation Outcome: **B. No** (Does not meet the goal)

Rationale for Evaluation:

Understanding the Temperature Parameter: The temperature parameter in AI models, particularly in text generation and summarization tasks, controls the randomness or diversity of the output. A higher temperature increases the diversity of the generated text by making the model less deterministic in its choices.
Impact on Response Completeness:

Increasing Temperature would likely introduce more variability in the summaries but does not directly address the issue of omitting specific, required regulatory clauses. It might even increase the omission rate by prioritizing diversity over relevance or completeness in some cases.
Completeness Requirement: The goal is to ensure all required regulatory clauses are included, which is more related to the model's understanding, the summarization strategy, and possibly the input processing (e.g., highlighting or weighting important clauses) rather than the output's diversity.
Why Other Approaches Might Be More Suitable (though not listed, for context):

Fine-Tuning the Model on a dataset emphasizing regulatory clauses could improve recognition and inclusion of these clauses.
Adjusting Summarization Strategy to prioritize inclusion of specific clause types or keywords.

Input Enhancement, such as pre-annotating important clauses in the input documents.

Why B (No) is the Correct Answer:Increasing the temperature parameter does not directly address the issue of omitting required regulatory clauses from summaries. Instead, it focuses on output diversity, which may not enhance, and could potentially detract from, the completeness of the responses in the desired manner.

References

For deeper understanding of the concepts:

1. **Microsoft Azure AI Documentation - Customizing Model Behavior**: https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/how-to-customize-model
2. **Temperature in AI Models (General Concept)**: https://towardsdatascience.com/temperature-in-machine-learning-models-8d0a83c7d6d9 (Note: While not Azure-specific, it explains the temperature concept well. For Azure-specific model customization, refer to the first link.)

Question: 44

CertyIQ

Note: This section contains one or more sets of questions with the same scenario and problem. Each question presents a unique solution to the problem. You must determine whether the solution meets the stated goals. More than one solution in the set might solve the problem. It is also possible that none of the solutions in the set solve the problem.
After you answer a question in this section, you will NOT be able to return. As a result, these questions do not appear on the Review Screen.
You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
Users report that some responses omit required regulatory clauses, even when the clauses are present in the retrieved content.
You need to improve response completeness.
Solution: You run an evaluation flow that scores responses for completeness and blocks responses that fall below a defined threshold.
Does this meet the goal?

A. Yes

B. No

Answer: B

Explanation:

Technical Justification for Correct Answer: B. No

Why the Correct Option (B. No) is Best:

Insufficient Solution for Completeness: Running an evaluation flow that scores responses for completeness and blocks those below a defined threshold **detects** but does **not improve** the completeness of responses. It merely filters out inadequate responses without addressing the root cause of why required regulatory clauses are omitted.
User Experience Impact: Blocking responses below a threshold may lead to a higher rate of "no response" situations, potentially frustrating users who expect at least some form of assistance, even if incomplete. This does not enhance the user experience in terms of receiving useful, albeit imperfect, information.
Lack of Feedback Loop for Improvement: The proposed solution does not inherently include a mechanism to feedback the shortcomings (e.g., omitted clauses) to the agent's training data or development process, crucial for iteratively improving the agent's ability to generate complete responses.

Why Other Options are Less Suitable (Given Only One Other Option, A):

Option A. Yes:
Incorrect Assumption of Solution Efficacy: Assumes that blocking inadequate responses is synonymous with "improving response completeness," which is a misunderstanding of the goal. Improvement implies enhancement in the quality or thoroughness of the responses provided to the user, not just their filtering.
No Direct Enhancement of Agent Capability: Does not contribute to making the agent better at generating complete summaries, only at hiding its inadequacies from the user.

Conclusion:

Given the analysis, the solution does not meet the stated goal of **improving response completeness** but rather masks the issue. Thus, the correct answer is **B. No**.

References:

Microsoft Azure Documentation - Improving AI Model Accuracy: https://docs.microsoft.com/en-us/azure/cognitive-services/custom-vision-service/how-to-improve-model
Microsoft Learn - Enhancing Conversational AI with Feedback Loops: https://learn.microsoft.com/en-us/training/modules/build-conversational-ai-solutions/#:~:text=Provide%20a%20feedback%20mechanism%20to,improve%20the%20conversational%20AI%20

Question: 45

CertyIQ

You are creating an image-editing workflow in a Microsoft Foundry project.
The workflow must meet the following requirements:
Ensure that background objects can be removed by applying a mask-based inpainting edit.
Preserve the original lighting and style of the edited images.
Use the built-in image editing controls, NOT a custom model.
You need to ensure that image edits apply exclusively inside the masked area.
How should you configure the workflow?

A. Set generation mode to image_variation and provide the original image as a reference.
B. Enable text_to_image mode and a prompt describing the desired background removal.
C. Enable image_to_image mode and a high-strength value to regenerate the full image based on the prompt.
D. Enable mask_inpainting and supply both the input image and a mask indicating which part of the image to modify.

Answer: D

Explanation:

Technical Justification for Configuring the Image-Editing Workflow

Correct Answer: D. Enable mask_inpainting and supply both the input image and a mask indicating which part of the image to modify.

Why D is the best option:

Meets the primary requirement: Directly supports mask-based inpainting for removing background objects by applying edits exclusively to the masked area.
Preserves original lighting and style: By focusing edits on the masked region, the rest of the image, including its lighting and style, remains untouched.
Utilizes built-in image editing controls: Aligns with the requirement to not use a custom model, leveraging

Foundry's native capabilities.
Precision over edits: Ensures edits are applied exclusively inside the masked area, fulfilling this critical requirement.

Why Other Options are Less Suitable:

A. Set generation mode to image_variation and provide the original image as a reference

Inadequate for targeted edits: Image variation mode may alter the entire image, not just the targeted area, and doesn't guarantee preservation of the original's untouched aspects.
Lacks precision for background removal: Not specifically designed for mask-based inpainting.

B. Enable text_to_image mode and a prompt describing the desired background removal

Incorrect mode for the task: Text-to-image generation is not suited for editing existing images, especially for precise, mask-based edits.
Loss of original image integrity: Likely to generate a new image rather than preserve the original's lighting and style outside the edit area.

C. Enable image_to_image mode and a high-strength value to regenerate the full image based on the prompt

Overly broad editing scope: Regenerating the full image at high strength could drastically alter the original lighting and style, not just the target area.
Less precise than mask-based approach: Does not ensure edits are confined to the specified area.

References

Microsoft Azure - Image Editing with Azure Foundry
Azure Foundry Documentation - Mask Inpainting for Image Editing

Question: 46

CertyIQ

You have a Microsoft Foundry project that generates product marketing images from text prompts.
After publishing several images, the legal team at your company identifies a competitor's logo on a sign in the background of an image.
You need to remove only the logo, while preserving the rest of the image.
What should you do?

A. Apply a mask-based inpainting edit to the part of the image that contains the logo.
B. Increase the prompt guidance strength.
C. Modify the original prompt to exclude brand names.
D. Rerun the prompt by using a different random seed.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

Why A is the best option: Applying a mask-based inpainting edit to the part of the image that contains the logo (Option A) is the most suitable solution for several technical reasons:

Precision and Control: Mask-based inpainting allows for precise targeting of the area to be edited (the

competitor's logo), minimizing the impact on the rest of the image.
Preservation of Original Content: This method preserves the integrity of the surrounding elements in the image, which is a critical requirement.
Efficiency: Compared to re-generating the image, inpainting is typically faster and more efficient, especially for minor edits like logo removal.

Why other options are less suitable:

B. Increase the prompt guidance strength:

Inapplicability to Existing Image: This approach would only affect new image generations based on the modified prompt, not the already published image in question.
Uncertain Outcome: Increasing guidance strength might alter the image in unforeseen ways, not just remove the logo.
C. Modify the original prompt to exclude brand names:

Retroactive Inapplicability: Like option B, this doesn't address the existing image but rather future generations.
Overly Broad Impact: Might unnecessarily alter the image's content beyond just the logo.
D. Rerun the prompt by using a different random seed:

Unpredictable Outcomes: There's no guarantee the logo won't reappear or that the rest of the image will remain substantially the same.
Time and Resource Inefficient: Could require multiple attempts to achieve a satisfactory result, without addressing the specific issue at hand.

Correct Answer Justification Summary: Option A, mask-based inpainting, is technically the most precise, efficient, and effective method for removing a specific element (the competitor's logo) from an existing image while preserving the rest of the content.

References

1. **Microsoft Azure AI for Image Editing**: https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-image-editing (See sections on Inpainting for detailed technical insights)
2. **Inpainting in Computer Vision (General Concept)**: https://en.wikipedia.org/wiki/Inpainting (Provides a broad understanding of the technology behind the recommended solution)

Question: 47 CertyIQ

You have a Microsoft Foundry project that contains an agent.
The agent ingests scanned PDF vendor invoices that contain tables and embedded QR codes.
The agent must preserve the PDF layout in the extracted output to ensure that downstream processing can reference sections and tables.
You plan to call Azure Content Understanding in Foundry Tools.
You need to extract content and layout elements and detect QR codes without requiring a language model deployment.
Which built-in analyzer should you use?

A. prebuilt-documentFieldSchema

B. prebuilt-read

C. prebuilt-documentSearch

D. prebuilt-layout

Answer: D

Explanation:

Justification for Choosing Option D: prebuilt-layout

The correct option for the given requirement is **D: prebuilt-layout**. Here's why:

Requirement Analysis:

Preserve PDF layout in extracted output
Extract content and layout elements
Detect QR codes without a language model deployment

Why D: prebuilt-layout is the Best Choice:

Layout Preservation and Extraction: The prebuilt-layout analyzer is specifically designed to extract and preserve the layout of the document, including tables, which aligns with the need to reference sections and tables in downstream processing.
QR Code Detection: This analyzer also includes the capability to detect QR codes, fulfilling the requirement without needing an additional or more complex setup like a dedicated language model.
Simplicity and Integration: Being a built-in analyzer, it ensures seamless integration with Azure Content Understanding in Foundry Tools, minimizing the complexity of the setup.

Why Other Options are Less Suitable:

A: prebuilt-documentFieldSchema:

Primarily used for extracting specific fields based on a predefined schema.
Does not focus on layout preservation or QR code detection.
B: prebuilt-read:

Excels at reading and extracting text from documents, including layout text, but...
Less effective for precise layout preservation and table structure extraction compared to prebuilt-layout.
Does not support QR code detection.
C: prebuilt-documentSearch:

Optimized for enhancing search capabilities within documents.
Not designed for layout preservation, detailed table extraction, or QR code detection.

Conclusion:Given the specific requirements of preserving the PDF layout, extracting content and layout elements (including tables), and detecting QR codes without a language model, **D: prebuilt-layout** is the most appropriate and efficient choice.

References

1. **Azure Cognitive Search - Prebuilt Analyzers**: https://docs.microsoft.com/en-us/azure/search/search-analyzers#prebuilt-analyzers
2. **Microsoft Azure: Content Understanding with Prebuilt Layout Analyzer**: https://docs.microsoft.com/en-us/azure/cognitive-services/content-understanding/concepts/prebuilt-layout-analyzer (Note: Link subject to update; for the most current info, search for "Azure Content Understanding prebuilt-layout analyzer" on the official Microsoft Docs website)

Question: 48

You have a Microsoft Foundry project that contains an agent and an image generation model deployment.
The agent generates original images from user-supplied product photos.
You need to ensure that the generated images maintain the product identity and visual characteristics of the provided photo.
What should you do?

A. Set the input_fidelity parameter to high.
B. Apply a groundedness detection filter.
C. Include a prompt and input image in the request.
D. Decrease the value of the temperature parameter.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

Why A is the Best Choice: Setting the input_fidelity parameter to high is the most appropriate action to ensure generated images maintain the product identity and visual characteristics of the provided photo. This parameter directly influences how closely the output adheres to the input's characteristics. A high setting ensures the model prioritizes fidelity to the input image, preserving the product's visual identity and attributes, which is crucial for maintaining recognizability and consistency in generated product images.

Why Other Options are Less Suitable:

B. Apply a groundedness detection filter: While this could help in ensuring the generated content is realistic and "grounded" in reality, it does not directly address the requirement of maintaining the specific product's identity and visual characteristics from the input photo. Groundedness detection filters focus more on the realism of the generated image rather than its fidelity to the input.
C. Include a prompt and input image in the request: Although including both a prompt and an input image can provide the model with more context, the effectiveness of this approach in maintaining product identity depends heavily on the prompt's specificity and quality. The question implies the need for a more direct control over the output's fidelity to the input image, which input_fidelity provides. Furthermore, the scenario already involves a user-supplied product photo, suggesting the input image is already part of the request.
D. Decrease the value of the temperature parameter: Lowering the temperature parameter reduces randomness in the model's output, making it more deterministic. However, this setting primarily affects the variability of the generated images rather than their fidelity to the input's specific characteristics. It might result in more consistent outputs but does not guarantee the preservation of the product's visual identity as effectively as adjusting input_fidelity.

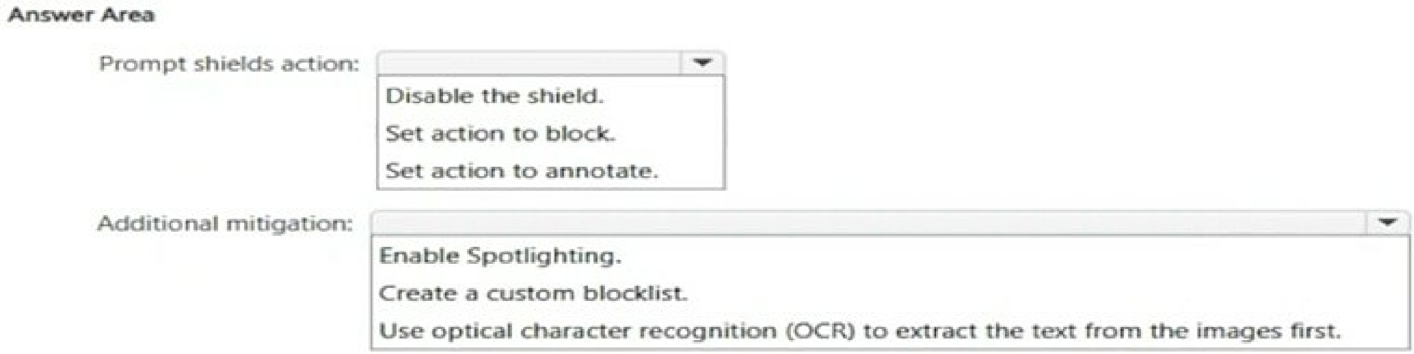
References:

1. **Microsoft Azure Documentation - Image Generation Models**: https://docs.microsoft.com/en-us/azure/cognitive-services/computer-vision/concept-generating-thumbnails (See sections related to input parameters for image generation)
2. **Microsoft Foundry Documentation - Agent Configurations**: Note: As Microsoft Foundry is not a widely recognized or documented public platform by Microsoft at the time of this response (it might refer to a custom, less-known, or upcoming service), a direct link cannot be provided. For similar configurations in known services, refer to:https://docs.microsoft.com/en-us/azure/cognitive-services/bing-image-search/use-bing-image-search (Focus on parameters influencing output similarity to inputs)

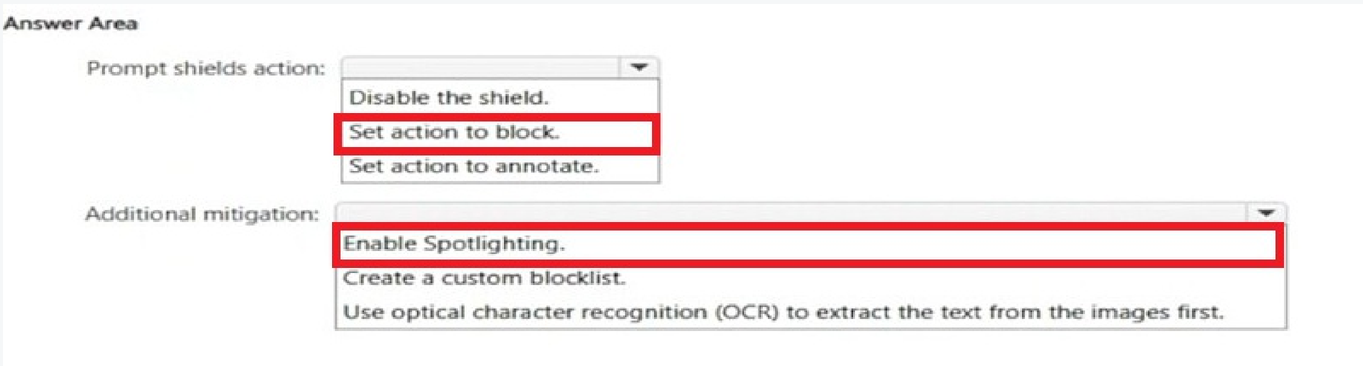
Question: 49

CertyIQ

HOTSPOT -
You have a Microsoft Foundry project that contains an agent.
The agent accepts user-uploaded screenshots and uses a multimodal chat model.
Some screenshots contain potentially malicious embedded text.
You need to prevent a prompt injection attack and ensure that third-party content is treated as lower trust.
How should you configure prompt shields for document attacks? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

Prompt shields action: **Set action to block.**

Additional mitigation: **Enable Spotlighting.**

Prompt shields action (Set action to block.):
When dealing with direct user prompt injections or indirect document-based attacks, setting the action to **Block** is necessary to prevent unsafe, modified instructions from hijacking the large language model (LLM) session. Choosing annotate merely flags the threat in data attributes without stopping execution, leaving downstream applications exposed.
Additional mitigation (Enable Spotlighting.):
Spotlighting is a specialized framework integrated directly into Azure Prompt Shields. It uses advanced encoding techniques (like base-64 conversion or structured text delimiting) to dynamically lower the trust signature of external data inputs (documents, emails, web scraped text) relative to direct system instructions. This prevents indirect cross-prompt injections without relying on rigid blocklists.

Question: 50

CertyIQ

You are deploying a support agent that enables users to upload photos.
You need to automatically classify uploaded images for harmful content. The solution must block content based on severity levels.
What should you do?

A. Apply keyword scanning to optical character recognition (OCR) output by using Azure Vision in Foundry Tools.

B. Enable prompt shields.

C. Use blocklists.

D. Implement image moderation.

Answer: D

Explanation:

Technical Justification for Correct Answer: D

Why D (Implement image moderation) is the best option: Implementing image moderation is the most suitable choice for automatically classifying uploaded images for harmful content and blocking based on severity levels. This is because image moderation, specifically through Azure's Computer Vision or Azure Content Moderator services, is designed to analyze visual content. These services can detect and categorize harmful content (e.g., adult, racy, or violent images) with a level of granularity that allows for severity-based blocking. This approach directly addresses the requirement for visual content analysis.

Why other options are less suitable:

A. Apply keyword scanning to optical character recognition (OCR) output by using Azure Vision in Foundry Tools:

Limitation: This method only analyzes text extracted from images via OCR, not the visual content itself. It cannot detect harmful non-textual content (e.g., violence, nudity).
Inappropriateness for Requirement: Fails to directly assess image content for harmful visuals.

B. Enable prompt shields:

Irrelevance: Prompt shields are more relevant to controlling or guiding user input (e.g., in chatbots) rather than analyzing uploaded media for harmful content.
Functional Misalignment: Does not provide a mechanism for image content analysis.

C. Use blocklists:

Limitation: Blocklists are effective for known harmful sources (URLs, IPs) but do not analyze the content of uploaded images in real-time.
Inadequacy for Dynamic Content: Unable to classify new, unseen harmful image content based on its visual characteristics.

References:

Azure Content Moderator Documentation - For details on moderating images for harmful content.
Azure Computer Vision Documentation - Although primarily for image analysis, can be used in conjunction with custom logic for moderation purposes.

Question: 51 CertyIQ

You have an app named App1 that uses a Microsoft Foundry multimodal model deployment.
App1 runs optical character recognition (OCR) on uploaded images and appends the OCR output to the prompt as additional context.
Some uploaded images contain embedded text.
You need to prevent potentially malicious instructions from being processed by the model.
What should you use?

A. image moderation
B. prompt shields for documents
C. protected material text
D. prompt shields for user prompts

Answer: B

Explanation:

Technical Justification for Correct Answer: B - Prompt Shields for Documents

Why B is the best option: Prompt Shields for Documents is specifically designed to sanitize and control the input context (in this case, the OCR output appended to the prompt) before it's processed by the model. This feature is particularly relevant when dealing with untrusted or potentially malicious content extracted from documents (like embedded text in images). By using Prompt Shields for Documents, App1 can effectively filter out or redact potentially malicious instructions embedded within the OCR output, preventing them from influencing the model's response.

Why other options are less suitable:

A. Image Moderation: While Image Moderation is crucial for screening out inappropriate or harmful visual content, it does not address the issue of malicious **text** embedded within images. Image Moderation would not process or sanitize the OCR-extracted text, leaving the model vulnerable to malicious instructions in the text output.
C. Protected Material Text: This option seems to imply a feature for handling sensitive or protected text. However, the primary concern here is not the protection of sensitive text but the prevention of malicious text from being processed. Protected Material Text does not inherently imply a mechanism for filtering out malicious instructions, making it less suited for this specific requirement.
D. Prompt Shields for User Prompts: Although similar in name and concept to the correct answer, Prompt Shields for User Prompts is designed to sanitize and control **direct user inputs** (e.g., text typed by users). While it shares the goal of preventing malicious input, it's tailored for user-generated text prompts rather than automatically generated context from documents (like OCR output from images).

Conclusion: Given the specific scenario of preventing potentially malicious **embedded text in images** (after OCR processing) from being processed by the model, **B. Prompt Shields for Documents** is the most appropriate and technically justified choice. It directly addresses the need to sanitize document-extracted text context before model processing.

References

1. **Microsoft Azure - Protecting Models from Malicious Input**: https://learn.microsoft.com/en-us/azure/cognitive-services/form-recognizer/concept-model-protection (Though the link focuses on Form Recognizer, the concept of protecting models from malicious input is broadly applicable and relevant to understanding the importance of sanitizing inputs.)
2. **Microsoft Documentation - Azure Cognitive Services Security and Compliance**: https://learn.microsoft.com/en-us/azure/cognitive-services/security-compliance (This link provides a broad overview of security measures in Azure Cognitive Services, including input validation and sanitization principles that underpin the rationale for using Prompt Shields.)

Question: 52

CertyIQ

You have a Microsoft Foundry project that contains an agent.
You need to process mixed-format documents that contain scanned text, tables, and multicolumn layouts. The extracted content must preserve the document structure and be converted into the Markdown format for downstream reasoning.
What should you configure first?

A. an Azure Language in Foundry Tools text analysis model deployment
B. a generative chat completion request
C. an Azure OpenAI Responses API call that uses a multimodal model
D. an Azure Content Understanding in Foundry Tools analyzer

Answer: D

Explanation:

Technical Justification for Correct Answer: D

To address the requirement of processing mixed-format documents (containing scanned text, tables, and multicolumn layouts) while preserving the document structure and converting the extracted content into Markdown format for downstream reasoning, the most appropriate initial configuration is **D. an Azure Content Understanding in Foundry Tools analyzer**. Here's why:

Why D is the Correct Choice:

Document Structure Preservation: Azure Content Understanding is designed to comprehend and retain the structural integrity of documents, including tables and multicolumn layouts, which is crucial for this task.
Support for Mixed-Format Documents: It can handle scanned documents (via integration with OCR capabilities implicitly or explicitly through associated services) alongside other formats, making it suitable for mixed-format input.
Output Flexibility: The extracted, structured content can be more easily converted into Markdown format due to its understanding and output of document components (e.g., headings, paragraphs, tables).
Why Other Options are Less Suitable:

A. Azure Language in Foundry Tools text analysis model deployment:

Limitation: Primarily focused on text analysis (sentiment, entity recognition, etc.) rather than preserving document structure or handling scanned/multicolumn content effectively.
Conversion to Markdown: Would require additional, separate processing to structure the output into Markdown, increasing the complexity of the pipeline.

B. a generative chat completion request:

Purpose Misalignment: Designed for generating human-like text based on prompts, not for document structure analysis or preservation.
Format Conversion: Not suited for accurately converting structured document content into Markdown.

C. an Azure OpenAI Responses API call that uses a multimodal model:

Multimodal Capability: While powerful, the primary challenge here isn't necessarily the type of input (though multimodal helps), but accurately extracting and preserving document structure.

Structure Preservation and Markdown Conversion: Less directly suited for the precise structural analysis and conversion requirements compared to a content understanding tool.
Cost and Complexity: Might introduce unnecessary complexity and cost for the specific task at hand, given the more targeted capability of Content Understanding.
Conclusion:

Given the specific requirements of preserving document structure, handling mixed formats (including scanned documents), and converting to Markdown, **configuring an Azure Content Understanding in Foundry Tools analyzer first** is the most efficient and technically appropriate approach.

References:

Azure Content Understanding Documentation
Microsoft Foundry Project Documentation (Integrating Azure Services)

Question: 53

CertyIQ

You have an application that processes scanned PDF invoices. The invoices have varied layouts and include multipage tables.
You have a pipeline that uses optical character recognition (OCR) and extracts totals and invoice numbers. The results are often incorrect because the document structure is ignored.
You need to implement a solution that provides OCR, layout analysis, and template-generalizing field extraction. The solution must NOT require training a custom model. The solution must minimize administrative effort.
What should you include in the solution?

A. Azure Language in Foundry Tools
B. Azure Content Understanding in Foundry Tools
C. an Azure Machine Learning model

Answer: B

Explanation:

Justification for Correct Answer: B - Azure Content Understanding in Foundry Tools

The correct solution for the given scenario is **B. Azure Content Understanding in Foundry Tools**. Here's a detailed breakdown of why this is the most suitable option and the shortcomings of the other choices:

Why B (Azure Content Understanding in Foundry Tools) is Correct:

OCR, Layout Analysis, and Field Extraction: Azure Content Understanding is designed to handle complex document structures, performing advanced OCR, layout analysis, and field extraction. It can adapt to varied layouts and multipage tables without requiring custom model training.
No Custom Model Training Required: This solution aligns with the requirement of not needing to train a custom model, as it leverages pre-trained models optimized for document understanding.
Minimizes Administrative Effort: By utilizing a managed service, the administrative burden is significantly reduced compared to building and maintaining a custom solution or even managing an Azure Machine Learning model.
Why Other Options are Less Suitable:

A. Azure Language in Foundry Tools:

Primary Focus: More tailored towards natural language processing (NLP) tasks such as text analysis, sentiment analysis, etc., rather than the structured document analysis required for invoices.

Inadequate for Layout Analysis and Template-Generalizing Extraction: Less capable in handling the complexities of varying document layouts and the specific need for accurate field extraction from tables and forms.

C. An Azure Machine Learning Model:

Requires Custom Model Training: Directly contradicts the requirement of not needing to train a custom model, which would also increase development time and effort.
Higher Administrative Effort: Managing, updating, and ensuring the model's accuracy over time would demand more administrative resources compared to a managed service like Azure Content Understanding.
References

For further reading and to delve deeper into the capabilities of the recommended solution:

1. **Azure Content Understanding Documentation** - https://learn.microsoft.com/en-us/azure/foundry-tools/content-understanding/
2. **Comparison of Azure Services for Document Processing** - https://learn.microsoft.com/en-us/azure/architecture/guide/technology-choices/document-processing-services (Utilize this to compare the functionalities of different Azure services for document processing tasks.)

Question: 54

CertyIQ

You have a Microsoft Foundry project that contains an agent.
The agent uses a knowledge source built from documents stored in Azure Blob Storage. The documents include digitally scanned PDFs that contain multipage tables.
You have an ingestion job that extracts only plain text, causing loss of table structure, headings, and page-number metadata.
Users frequently ask questions that require the retrieval of specific table rows across the pages.
You need to configure an ingestion job for a Retrieval Augmented Generation (RAG) pipeline that performs optical character recognition (OCR) on scanned PDFs, preserves tables and headings as structure-aware chunks, and stores page-number metadata with each chunk.
How should you configure the ingestion job?

A. Use advanced data parsing to reingest the documents.

B. Use OCR and page-level chunking.

C. Use page-level OCR extraction and store each page as a single chunk.

D. Use basic parsing and fixed-size chunking.

Answer: A

Explanation:

Technical Justification for Correct Option (A)

The correct configuration for the ingestion job, given the requirements, is **A. Use advanced data parsing to reingest the documents**. Here's why this option stands out and why others are less suitable:

Why A is the Best Choice:

Advanced Data Parsing Capability: Advanced data parsing is specifically designed to handle complex document structures, including the preservation of tables, headings, and extraction of specific metadata like page numbers. This aligns perfectly with the need to retain the structure of multipage tables and associate page-number metadata with each chunk.
Retrieval Augmented Generation (RAG) Pipeline Compatibility: For a RAG pipeline, the ingestion job needs

to provide structured, searchable content. Advanced data parsing ensures that the output is optimized for such pipelines, facilitating efficient retrieval of specific table rows across pages.
OCR Implicit in Requirement Understanding: While not explicitly mentioned in option A, the context implies that OCR is necessary for scanned PDFs. Advanced data parsing tools often integrate or can be configured with OCR capabilities to process scanned documents, making this a holistic choice for the given scenario.
Why Other Options are Less Suitable:

B. Use OCR and Page-Level Chunking:

Lack of Explicit Structure Preservation: There's no clear indication that this option preserves table structures and headings as "structure-aware chunks" beyond basic page-level processing.
Insufficient for RAG's Structured Needs: Might not fully support the nuanced search and retrieval requirements of a RAG pipeline without advanced parsing.

C. Use Page-Level OCR Extraction and Store Each Page as a Single Chunk:

Overly Broad Chunks: Storing each page as a single chunk does not facilitate the retrieval of specific table rows efficiently, as it doesn't preserve the granular structure of the content within pages.
Metadata Limitation: Implicitly suggests a lack of detailed metadata association (like specific table row locations) with each chunk.

D. Use Basic Parsing and Fixed-Size Chunking:

Incompatible with Complex Structures: Basic parsing cannot preserve the complex table structures and headings effectively.
Fixed-Size Chunking Inefficiency: Irrespective of content structure, fixed-size chunking is inefficient for retrieving specific, variable-sized content elements (like table rows) and does not account for page-number metadata.

References

Microsoft Azure - Cognitive Search: Document AI
Azure Cognitive Services - Form Recognizer for Structured Data Extraction

Question: 55

CertyIQ

You have a Microsoft Foundry project that contains an agent.
The agent uses Azure AI Search as the retriever.
You plan to ingest PDF into an Azure AI Search index to ensure that the agent can ground responses in texts in both documents and embedded images.
Users require citations that link to the source files.
You need to ensure that during indexing, the images are extracted into a structure that can be used as input for the built-in optical character recognition (OCR) skill.
Which indexing approach should you use?

A. an indexer to extract image data into a normalized_images collection
B. a Shaper skill to restructure the OCR input
C. a skillset to run the OCR skill directly against the content field of the index
D. the outputFieldMappings parameter to write image data to a searchable field

Answer: A

Explanation:

Technical Justification for Correct Answer: A

The correct indexing approach to ensure images are extracted into a structure suitable for the built-in Optical Character Recognition (OCR) skill, while also supporting user requirements for citations linking to source files, is:

A. an indexer to extract image data into a normalized_images collection

Why A is the Best Choice:

Direct Extraction and Structuring: Using an indexer to extract image data into a normalized_images collection directly structures the image data in a way that is readily consumable by the OCR skill, streamlining the process.
Support for Citations: By keeping the image data linked to its source (through the structured collection), it facilitates the creation of citations that can point back to the original PDF files, meeting the user's requirement.
Efficiency and Scalability: Indexers are designed for such data ingestion and transformation tasks, making this approach efficient and scalable for handling multiple PDFs and their embedded images.

Why Other Options are Less Suitable:

B. a Shaper skill to restructure the OCR input:

Inappropriate Timing: Shaper skills are used for reshaping data after processing (e.g., post-OCR), not for pre-processing image extraction.
Inefficient for Image Extraction: Not designed for the initial extraction of images from documents.

C. a skillset to run the OCR skill directly against the content field of the index:

Lack of Image Extraction: Running OCR directly against the content field does not address the extraction of images from PDFs into a usable structure.
Missed Structuring Opportunity: Fails to preprocess images into an optimal structure for OCR, potentially leading to less accurate results.

D. the outputFieldMappings parameter to write image data to a searchable field:

Insufficient Structuring for OCR: While useful for making data searchable, it does not ensure the image data is structured appropriately for effective OCR processing.
Citations and Source Linking: Less directly supports the maintenance of clear links to source files for citations compared to a dedicated collection approach.

References

Microsoft Azure Documentation: Azure Cognitive Search Indexers
Microsoft Azure Documentation: Skillsets in Azure Cognitive Search

Question: 56 CertyIQ

Case Study -
This is a case study. Case studies are not timed separately from other exam sections. You can use as much exam time as you would like to complete each case study. However, there might be additional case studies or other exam sections. Manage your time to ensure that you can complete all the exam sections in the time provided. Pay attention to the Exam Progress at the top of the screen so you have sufficient time to complete any exam sections that follow this case study.
To answer the case study questions, you will bed to reference information that is provided in the case. Case studies and associated questions might contain exhibits or other resources that provide more information about the scenario described in the case. Information provided in an individual question does not apply to the other questions

in the case study.
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Company Information -
Contoso, Ltd is a multinational retail company that builds, deploys, and manages generative AI and agent-based solutions by using Microsoft Foundry.
Existing Environment -
Identity Environment -
Contoso uses Microsoft Entra ID for identity management, authentication, and authorization capabilities that enable agents to access organizational resources and services.
Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.
The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement. monitor, and secure AI applications.
Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution deployments.
Generative Environment -
Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.
Project1 -
Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.
Agent1 has the following configurations:
Agent1 uses a base model deployment.
A safety evaluation pipeline is NOT enabled.
Tool invocation approval workflows are NOT enabled.
Conversation memory constraints are NOT configured.
Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
Project1 is deployed to an Azure region located in the European Union (EU).
Agent1Dev Team will use Project1 to optimize and maintain Agent1.
Project2 -
Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
Development of the solution is incomplete.
Data Environment -
Contoso stores product-related information in Azure resources that support AI applications.
The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.
Problem Statements -
Contoso identifies the following issues:
Agent1 has only general knowledge of the Contoso products.
A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.
Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.
The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.
Requirements -
Planned Changes -
Contoso plans to implement the following changes:
Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.

Complete the development of the video creation solution.
Technical Requirements -
Contoso identifies the following technical requirements:
The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.
The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.
Responses generated by using the product sheet information must be relevant, complete, and accurate.
Agent1 must be able to use the product sheets to answer natural language questions about product details.
The model version used by Agent1 must remain consistent to ensure stable responses.
The data processed by the model must remain within the EU.
Security and Compliance Requirements
Contoso identifies the following security and compliance requirements:
API keys must NOT be used to access Foundry-deployed models.
Access to the Azure resources must follow the principle of least privilege.
The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.
Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.
Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements -
Contoso identifies the following business requirements:
Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
Agent1 must answer questions only about the products sold by Contoso.
You need to recommend an invoice review solution that resolves the issue reported by the finance department.
What should you include in the recommendation?

A. chat completions
B. Azure Document Intelligence in Foundry Tools
C. Azure Content Understanding in Foundry Tools
D. Image Analysis

Answer: C

Explanation:

Technical Justification for Invoice Review Solution Recommendation

Correct Option: C. Azure Content Understanding in Foundry Tools

Azure Content Understanding in Foundry Tools is the most suitable choice for the invoice review solution due to the following technical reasons:

Visual and Textual Content Evaluation: The finance department's issue involves verifying invoice details against vendor contract terms, which includes analyzing both the visual layout (e.g., tables, logos, varied layouts) and the textual content of the invoices. Azure Content Understanding is adept at handling both aspects, providing a comprehensive analysis.
Integration with Foundry: Since Contoso already utilizes Microsoft Foundry, integrating Azure Content Understanding ensures seamless interaction with existing infrastructure, streamlining the development and deployment process.
Scalability and Throughput: While not explicitly mentioned for this specific task, the overall technical requirements for Contoso's AI solutions emphasize scalability and high-throughput capabilities, which Azure Content Understanding can support, especially when combined with other Azure services optimized for such

workloads.

Why Other Options Are Less Suitable:

A. Chat Completions:
Inappropriate Use Case: Chat completions are more suited for generating text based on partial inputs, not for analyzing and verifying document content against external terms.
Lack of Visual Analysis Capability: This option does not address the visual layout analysis requirement.
B. Azure Document Intelligence in Foundry Tools:
More General-Purpose: While capable, Azure Document Intelligence might not offer the depth of content understanding required for nuanced invoice verification, especially if the focus is more on the intelligence extracted from the content rather than just its structure.
Assumes Structured Data Extraction Needs: The primary issue isn't just extracting structured data but understanding and verifying it against contract terms, which might require the deeper content analysis capabilities of Azure Content Understanding.
D. Image Analysis:
Too Narrow in Scope: Image Analysis would handle the visual aspects and potentially images with embedded text but would not comprehensively address the textual content analysis and verification against contract terms without additional, unnecessary integration complexity.

References:

For further details on the recommended solution and its capabilities:

1. **Azure Content Understanding Documentation**: https://learn.microsoft.com/en-us/azure/cognitive-services/content-understanding/overview
2. **Microsoft Foundry Integration with Azure Services**: https://learn.microsoft.com/en-us/azure/foundry/how-to-integrate-azure-services

Question: 57 **CertyIQ**

You have a Microsoft Foundry project that contains an agent.
The knowledge source for the agent is a set of scanned PDF troubleshooting guides stored in Azure Blob Storage.
The guide pages contain two-column layouts and tables.
You use Azure Content Understanding in Foundry Tools to process the PDFs.
You plan to ingest the processed content into an index for Retrieval Augmented Generation (RAG) and store extracted fields for downstream automation.
Stakeholders must be able to verify where each extracted field value came from in the original PDF and route low-reliability extractions for manual review.
You need to ensure that the Content Understanding document analyzer output includes a per-field confidence score and source grounding to locations within the source document.
What should you do?

A. Set enableSegment to true.

B. Provide labeled samples.

C. Enable estimateFieldSourceAndConfidence.

D. Configure the analyzer to use generative extraction for all fields.

Answer: C

Explanation:

Technical Justification for Correct Answer: C

To address the requirements of including per-field confidence scores and source grounding to locations within the source document for the Azure Content Understanding document analyzer output in the context of a Microsoft Foundry project, **enabling estimateFieldSourceAndConfidence (Option C)** is the most appropriate action. Here's why:

Per-field Confidence Score Requirement: EstimateFieldSourceAndConfidence is designed to not only estimate the confidence in each extracted field value but also to identify the source location within the document where the field value was extracted from. This directly fulfills the need for per-field confidence scores, ensuring stakeholders can assess the reliability of each extraction.

Source Grounding Requirement: By enabling this feature, the output will include references to the exact locations in the original PDF where each field's value was sourced. This feature is crucial for traceability, especially in documents with complex layouts like two-column setups and tables, as it facilitates verification of the extraction's origin.

Why Other Options are Less Suitable:

A. Set enableSegment to true: While enabling segmentation can improve the processing of structured content by breaking down documents into manageable parts, it does not directly address the need for confidence scores or source grounding for extracted fields. Segmentation is more about the preprocessing of the document rather than the extraction metadata.

B. Provide labeled samples: Providing labeled samples is essential for training and improving the accuracy of the model over time. However, this action does not directly influence the inclusion of confidence scores or source information in the analyzer's output for immediate processing needs. It's a long-term solution for model enhancement rather than a direct fix for the specified requirements.

D. Configure the analyzer to use generative extraction for all fields: Generative extraction can be powerful for certain types of data extraction tasks, especially where the model needs to generate text based on the document content. However, this approach may not always provide the precise source grounding or confidence scores per field as directly and reliably as enabling EstimateFieldSourceAndConfidence, especially in the context of the described requirements and the structured nature of the data (tables, two-column layouts).

Correct Action: C. Enable estimateFieldSourceAndConfidence

Enabling this feature ensures that the Content Understanding document analyzer output meets both critical requirements (per-field confidence scores and source grounding) efficiently and directly, making it the best choice for the given scenario.

References

1. **Azure Cognitive Search Documentation - Content Understanding**: https://docs.microsoft.com/en-us/azure/search/cognitive-search-concept-intro#content-understanding
2. **Customizing Document Analyzer for Content Understanding**: https://docs.microsoft.com/en-us/azure/search/cognitive-search-defining-skillset#document-analyzer-skills (Refer to sections related to field extraction and confidence scores for more detailed insights)

Question: 58 CertyIQ

You are building a speech processing solution in Microsoft Foundry for a customer support platform.
The platform will transcribe live phone calls, so that supervisors at your company can view call transcripts and detect issues while the calls are in progress. The call audio will arrive as a continuous stream from the telephony system.
You need to ensure that the call transcripts appear within only a few seconds of the audio stream.

What should you do?

A. Use text to speech by using a custom neural voice.
B. Use speech translation to generate the transcripts into multiple languages.
C. Run a batch transcription job on recorded audio files.
D. Use real-time speech to text to process streaming audio input.

Answer: D

Explanation:

Technical Justification for the Correct Option (D)

To meet the requirement of transcribing live phone calls with near-real-time transcription (within a few seconds of the audio stream), the solution must process the audio input as a continuous stream rather than in batches or after the call is completed. Here's why option D is the best choice and why the others are less suitable:

Correct Option: D - Use real-time speech to text to process streaming audio input

Real-time Capability: Real-time speech to text is designed to handle streaming audio inputs, making it perfectly suited for transcribing live phone calls.
Low Latency: This approach ensures that transcripts are generated with minimal delay, fulfilling the requirement of transcripts appearing within a few seconds of the audio stream.
Continuous Streaming Support: It is specifically built to process continuous audio streams, such as those from ongoing telephony systems, without requiring the audio to be pre-recorded or buffered for an excessive amount of time.
Why Other Options Are Less Suitable

A - Use text to speech by using a custom neural voice

Reverse Requirement: Text to speech converts text into speech, which is the opposite of what is needed (converting speech to text).
No Benefit for Transcription: Custom neural voices do not aid in the transcription process but rather in generating synthetic speech from text.

B - Use speech translation to generate the transcripts into multiple languages

Additional, Unrequested Functionality: While useful for multi-language support, the primary requirement is for real-time transcription in one language (implied, as the question does not mention translation needs).
Potential Increase in Latency: Adding a translation step could introduce additional delay, moving further from the near-real-time requirement.

C - Run a batch transcription job on recorded audio files

Batch Processing Incompatibility: Batch jobs are designed for processing pre-recorded audio files in bulk, not for real-time or near-real-time transcription of live streams.
High Latency: Transcripts would only be available after the batch job completes, which would be long after the call has ended, violating the low latency requirement.

Question: 59 CertyIQ

You are creating an agent workflow in a Microsoft Foundry project to support natural voice interactions.

The agent must receive continuous audio input, convert the input into text for reasoning, and then return spoken responses to a user. The workflow must meet the following requirements:
Support turn-taking dynamics, where the agent begins to generate the speech output before the user finishes speaking.
Operate with low latency to maintain conversational experience.
You need to enable both speech to text and text to speech in a real-time agent interaction.
What should you do?

A. Use batch transcription to convert the audio input and return text responses from the agent.
B. Use real-time speech to text for incoming audio and text to speech for agent responses.
C. Use an embeddings model to encode the audio, and then decode the audio into text and speech.
D. Use speech translation to convert the audio into another language and return the translated text.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Why B is the Best Option:

Real-time Processing: Option B utilizes real-time speech to text, which is essential for meeting the low latency requirement, ensuring a seamless conversational experience. This capability processes audio as it is being received, minimizing delays.
Support for Turn-Taking Dynamics: By converting audio to text in real-time, the agent can begin generating speech output even before the user finishes speaking, directly supporting the turn-taking dynamics requirement.
Direct Alignment with Requirements:

Speech to Text: Directly converts incoming audio into text for the agent's reasoning, fulfilling the first technical requirement.
Text to Speech: Generates spoken responses from the text output, fulfilling the second technical requirement for real-time agent interaction.

Why Other Options are Less Suitable:

A. Batch Transcription:

High Latency: Processes audio in batches, leading to significant delays, which contradicts the low latency requirement.
Incompatible with Turn-Taking: Cannot start processing until the user finishes speaking, making turn-taking dynamics impossible.
C. Embeddings Model for Encoding/Decoding:

Indirect Approach: While embeddings can be used in NLP tasks, this option describes an indirect and potentially less efficient method for the specific tasks of real-time speech to text and text to speech conversion.
Lack of Clear Latency Benefit: Does not inherently guarantee low latency or support for turn-taking as effectively as dedicated real-time speech services.
D. Speech Translation:

Misaligned Capability: Focuses on translating audio into another language, which is not the primary requirement (conversational interaction in the same language).
Added Latency: Translation adds an extra step, potentially increasing latency beyond what's acceptable for real-time interaction.

Conclusion:Given the requirements for real-time processing, low latency, support for turn-taking dynamics, and the direct need for speech to text and text to speech conversion, **Option B** is the most technically suitable choice for enabling both functionalities in a real-time agent interaction.

References

1. **Microsoft Azure Speech Services - Real-time Speech Recognition**https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/speech-to-text
2. **Microsoft Azure Text-to-Speech (TTS) for Real-time Responses**https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/text-to-speech

Question: 60 CertyIQ

You have an application named App1 that uses Azure Speech in Foundry Tools to transcribe live calls.
Transcript segments often contain both English and Spanish. App1 sends each segment to Azure Translator in Foundry Tools to translate to another language.
Sometimes, mixed-language segments result in incomplete or incorrect translations.
You need to reduce translation errors. The solution must ensure that the entire transcript is translated successfully.
What should you do before sending the segments to Translator?

A. Use document translation to translate the entire transcript as a single document.

B. Split the mixed-language segments into single-language segments and translate each segment separately.

C. Enable automatic language detection for the translation request.

D. Specify English as the source language in the translation request for all the segments.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

To address the issue of incomplete or incorrect translations for mixed-language segments in App1, the most effective solution is to **split the mixed-language segments into single-language segments and translate each segment separately (Option B)**. Here's why this approach is best and why other options are less suitable:

Why B is the Correct Approach:

Accuracy Enhancement: By isolating language segments, the translation model can focus on one language at a time, significantly improving accuracy.
Direct Handling of Mixed-Language Content: This method proactively deals with the root cause (mixed languages in a segment) by separating them, ensuring each translation request is linguistically homogeneous.
Compatibility with Azure Translator: Azure Translator can efficiently handle multiple, single-language translation requests, making this approach technically feasible and scalable.

Why Other Options are Less Suitable:

A. Use Document Translation for the Entire Transcript:

Limitation: While document translation can handle larger texts, it may not effectively mitigate mixed-language issues within a single document, as the engine might struggle with interwoven languages.
Potential for Errors: Similar or worse error rates as current setup if language detection within the document feature is not robust enough for intertwined languages.

C. Enable Automatic Language Detection for the Translation Request:

Reliability Concerns: Automatic detection may not always accurately identify and separate mixed languages within a single segment, potentially leading to similar translation errors.
Segmentation Challenge: Detection doesn't solve the segmentation issue; it only identifies languages, not necessarily how to process them correctly in mixed segments.

D. Specify English as the Source Language for All Segments:

Inapplicability: Ignores the presence of Spanish (and potentially other languages), guaranteeing translation failures for non-English parts.
Error Rate Increase: Likely to increase errors for segments containing languages other than English.

Conclusion: Splitting mixed-language segments into single-language parts before translation (Option B) is the most technically sound approach to reduce translation errors, given its direct addressing of the problem's root cause and compatibility with the capabilities of Azure Translator in Foundry Tools.

References

1. **Azure Translator Documentation - Best Practices for Translation Requests**: https://docs.microsoft.com/en-us/azure/cognitive-services/translator/best-practices
2. **Azure Speech to Translator Integration Guidance**: https://docs.microsoft.com/en-us/azure/cognitive-services/speech-service/howto-translate-speech?tabs=dotnet (Note: While this link focuses on speech-to-text-to-speech translation, the principles of handling multi-language inputs can be inferred for text translation scenarios.)

Question: 61 CertyIQ

Case Study -
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Agent1 uses a base model deployment.
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Tool invocation approval workflows are NOT enabled.
Conversation memory constraints are NOT configured.
Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
Project1 is deployed to an Azure region located in the European Union (EU).
Agent1Dev Team will use Project1 to optimize and maintain Agent1.
Project2 -
Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
Development of the solution is incomplete.
Data Environment -
Contoso stores product-related information in Azure resources that support AI applications.
The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.
Problem Statements -
Contoso identifies the following issues:
Agent1 has only general knowledge of the Contoso products.
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Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.
The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.
Requirements -
Planned Changes -
Contoso plans to implement the following changes:
Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.
Complete the development of the video creation solution.
Technical Requirements -
Contoso identifies the following technical requirements:
The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.
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Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.

Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements -
Contoso identifies the following business requirements:
Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
Agent1 must answer questions only about the products sold by Contoso.
You need to configure an indexing pipeline for Agent1 to retrieve the relevant product information in storage1. The solution must meet the technical requirement.
Which two built-in skills should you use? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Azure OpenAI Embedding
B. Entity Recognition
C. Text Split
D. Merge
E. Language Detection
F. key phrase extraction

Answer: AC

Explanation:

Technical Justification for Correct Answer: AC (Azure OpenAI Embedding & Text Split)

Contoso requires an indexing pipeline for Agent1 to retrieve relevant product information from PDF product sheets in storage1, meeting the technical requirement of enabling semantic and vector search for accurate, complete, and relevant responses to natural language questions.

Why Azure OpenAI Embedding (A) is Correct:

Semantic and Vector Search Capability: Azure OpenAI Embedding is designed to generate dense vector representations of text, enabling efficient semantic search. This aligns perfectly with Contoso's requirement for an indexing pipeline that facilitates semantic and vector search, allowing Agent1 to retrieve relevant product information based on the semantic meaning of the queries.
Integration with AI Workloads: Given Contoso's use of Microsoft Foundry for generative AI solutions, Azure OpenAI Embedding integrates seamlessly, supporting the scalable, high-throughput requirements for Agent1's interactions.
Handling Natural Language Queries: By generating embeddings, this skill enhances Agent1's ability to understand and respond accurately to natural language questions about product details, ensuring relevance and completeness.
Why Text Split (C) is Correct:

Processing Large Documents: Product sheets are in PDF format, which can be lengthy. Text Split is essential for breaking down these large documents into manageable, indexing-friendly segments, ensuring that the indexing pipeline can efficiently process each part.
Optimizing Search Efficiency: By splitting text, the search index can be more granular, leading to faster and more accurate retrieval of specific product information in response to customer queries.
Preprocessing for Embedding: Text Split prepares the text for more effective embedding generation by Azure OpenAI Embedding, as smaller, contextually relevant chunks of text can produce more accurate vector representations.
Why Other Options are Less Suitable:

B. Entity Recognition: While useful for identifying specific entities (e.g., product names), it does not enable the broad semantic and vector search required for retrieving relevant product information based on natural language queries.

D. Merge: This skill is counterproductive for the goal of creating a granular, searchable index, as it combines texts, potentially reducing search efficiency.

E. Language Detection: Not relevant to the task of indexing product information for semantic/vector search, as the language of the product sheets is presumably known and consistent.

F. Key Phrase Extraction: Similar to Entity Recognition, it's useful but doesn't meet the requirement for a searchable index that supports semantic and vector searches for accurate responses to varied natural language questions.

References

1. **Azure OpenAI Embedding Documentation**: https://learn.microsoft.com/en-us/azure/cognitive-services/OpenAI/how-to/embeddings
2. **Azure Cognitive Search (for context on Text Split and semantic search capabilities)**: https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search

Question: 62 CertyIQ

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Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.
Complete the development of the video creation solution.
Technical Requirements -
Contoso identifies the following technical requirements:
The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.
The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.
Responses generated by using the product sheet information must be relevant, complete, and accurate.
Agent1 must be able to use the product sheets to answer natural language questions about product details.
The model version used by Agent1 must remain consistent to ensure stable responses.
The data processed by the model must remain within the EU.
Security and Compliance Requirements
Contoso identifies the following security and compliance requirements:
API keys must NOT be used to access Foundry-deployed models.
Access to the Azure resources must follow the principle of least privilege.
The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.
Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
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Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements -
Contoso identifies the following business requirements:
Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for

Agent1 to retain conversation context and recall relevant information from previous interactions.
Agent1 must answer questions only about the products sold by Contoso.
You need to recommend a solution to support the planned changes and technical requirements for Agent1 to use the product information stored in storage1.
What should you include in the recommendation?

A. Azure Translator in Foundry Tools
B. Grounding with Bing Search
C. Azure AI Search
D. Azure Document intelligence in Foundry Tools

Answer: C

Explanation:

Technical Justification for Recommending Azure AI Search (Option C)

Why Option C (Azure AI Search) is the Best Choice:

Meets Technical Requirements:

Enables semantic and vector search for indexing product sheets, allowing Agent1 to retrieve relevant information (aligns with the requirement for an indexing pipeline with semantic and vector search capabilities).
Supports scalable, high-throughput workloads suitable for variable customer support traffic without requiring reserved throughput capacity (supports the model deployment requirements for Agent1).
Integration and Compatibility:

Seamless integration with Microsoft Foundry, ensuring consistency with Contoso's existing environment.
Facilitates the use of detailed product information from storage1 in responses, enhancing Agent1's capabilities.
Security and Compliance Alignment:

Does not rely on API keys for access, aligning with the ban on API key usage for Foundry-deployed models.
Compatible with least privilege access principles and Microsoft Entra authentication for secure access control.

Why Other Options are Less Suitable:

Option A (Azure Translator in Foundry Tools):

Primarily designed for translation tasks, not suited for indexing or retrieving product information based on semantic/vetor search requirements.
Does not address the need for Agent1 to understand and respond with detailed product knowledge.

Option B (Grounding with Bing Search):

External search may not guarantee data privacy or compliance with EU data retention requirements (especially since Project1 is deployed in an EU Azure region).
Less integrated with Contoso's existing Microsoft Foundry and Azure environment, potentially introducing complexity.

Option D (Azure Document Intelligence in Foundry Tools):

While useful for extracting information from documents (like invoices), it doesn't fulfill the semantic/vetor search indexing requirement for enabling Agent1 to answer product questions effectively.

More focused on document processing rather than search and retrieval for conversational AI use cases.

Conclusion: Azure AI Search is the most comprehensive solution for Contoso's requirements, offering the necessary search capabilities, scalability, security, and integration with their existing Microsoft Foundry environment to enhance Agent1's functionality effectively.

References:

1. **Azure AI Search Documentation** - https://learn.microsoft.com/en-us/azure/search/search-what-is-azure-search
2. **Microsoft Foundry Integration with Azure Services** - https://learn.microsoft.com/en-us/azure/foundry/how-to-integrate-with-azure-services

Question: 63

CertyIQ

You have a Microsoft Foundry project named Project1.
Project1 contains an application that processes PDF vendor invoices.
You need to configure Azure Document Intelligence in Foundry Tools to generate a Markdown output that preserves the sections and table structure of the PDFs. The solution must minimize development effort.
What should you do?

A. Configure output=figures when you analyze the PDF.
B. Configure content=markdown when you analyze the document.
C. Increase the confidence threshold.
D. Set the output_content_format=ContentFormat.MARKDOWN value.

Answer: D

Explanation:

Technical Justification for Correct Answer (D)

To configure Azure Document Intelligence in Foundry Tools for generating Markdown output that preserves the sections and table structure of PDF vendor invoices with minimal development effort, the most appropriate action is to **Set the output_content_format=ContentFormat.MARKDOWN value**. Here's why:

Why D is the Correct Choice

Direct Specification of Output Format: By setting output_content_format=ContentFormat.MARKDOWN, you directly instruct the Azure Document Intelligence API to produce the output in Markdown format. This ensures that the structure (including sections and tables) is preserved in a format that's easily readable and further processable.
Minimal Development Effort: This approach requires no additional coding to parse or transform the output, as the API handles the conversion to Markdown. This aligns perfectly with the requirement to minimize development effort.
Why Other Options are Less Suitable

A. Configure output=figures:

Inadequate for Structural Preservation: Specifying output=figures would primarily focus on extracting images or figures from the PDF, neglecting the textual content and structural elements (sections, tables) that need preservation.
Does Not Generate Markdown: This option does not generate Markdown output, failing to meet the primary

requirement.
B. Configure content=markdown:

Misinterpretation of Parameter: While this option seems relevant due to the mention of "markdown", the content parameter typically refers to what content to extract (e.g., text, tables) rather than the output format. The correct approach for specifying the output format is through output_content_format.
May Not Preserve Structure as Required: Even if content=markdown influenced output format (which it doesn't directly), there's no guarantee it would preserve structures as effectively as specifying the output format explicitly.
C. Increase the confidence threshold:

Irrelevant to Output Format: Adjusting the confidence threshold affects the accuracy and reliability of the extracted data (making the model more conservative in what it outputs as recognized text or structures) but has no bearing on the output format or structural preservation in Markdown.
Conclusion

Setting output_content_format=ContentFormat.MARKDOWN is the most direct, efficient, and effective method to achieve the desired outcome with minimal development effort, making **D** the correct choice.

References

Azure Document Intelligence API Reference - Output Formats
Microsoft Azure Documentation - Document AI Output Configuration

Question: 64 CertyIQ

You have a Microsoft Foundry project that ingests scanned PDF invoices stored in Azure Blob Storage. Each invoice contains printed fine items and has a table-based layout.
Extracted results are stored as structured JSON and used as grounding data for an agent in a Retrieval Augmented Generation (RAG) solution.
You need to create a single analyzer that meets the following requirements:
Extracts the invoice number, invoice date, vendor name, and total amount across varying templates
Returns confidence scores so that results with confidence below 0.80 can be routed for supervisor review
What should you use?

A. a Foundry agent that has groundedness guardrails enabled to extract invoice fields and confidence scores
B. a custom Azure Content Understanding in Foundry Tools analyzer that defines the required fields as the extracted fields and the returned confidence scores for routing
C. the Azure Content Understanding in Foundry Tools prebuilt-layout analyzer
D. the Azure Content Understanding in Foundry Tools prebuilt-documentSearch analyzer and search.score from the Azure AI Search results for routing

Answer: B

Explanation:

Technical Justification for Choosing Option B

Why B is the Best Choice:A custom Azure Content Understanding (ACU) in Foundry Tools analyzer, as described in option B, is the most suitable choice for this scenario because it directly addresses all specified requirements with precision. By defining the required fields (invoice number, invoice date, vendor name, and total amount) as extracted fields, it ensures targeted data extraction across varying templates. Moreover, configuring the analyzer to return confidence scores enables the crucial functionality of routing results with scores below 0.80 for supervisor review, aligning perfectly with the project's needs.

Why Other Options are Less Suitable:

A. Foundry Agent with Groundedness Guardrails:

While a Foundry agent can be powerful for tasks involving grounding data, this option does not directly imply the ability to extract specific fields from scanned PDF invoices with variable templates as efficiently as a customized ACU analyzer.
Confidence score extraction for routing is not a standard feature highlighted by groundedness guardrails, making this option less straightforward for the given requirements.

C. Prebuilt-layout Analyzer in ACU:

The prebuilt-layout analyzer might struggle with "varying templates" since its performance can be highly dependent on the consistency of the layout it's trained on.
Custom extraction of specific fields (invoice number, date, etc.) and the return of confidence scores for routing might not be as directly configurable or as accurate as with a custom analyzer.

D. Prebuilt-documentSearch Analyzer with Azure AI Search Score:

This option focuses more on search functionality rather than precise field extraction from documents, making it less ideal for extracting specific invoice details.
Relying on search.score for confidence might not accurately reflect the extraction confidence of specific fields within the document, as it's more related to the relevance of the document in search results.

Correct Choice Justification Summary: Option B is the most technically appropriate because it allows for:

Customization to extract specific fields across varying templates.
Direct configuration to return confidence scores for conditional routing based on a precise threshold (below 0.80).

References:

Microsoft Azure Documentation - Custom Analyzers in Content Understanding
Microsoft Learn - Extract insights from documents with Azure Content Understanding

Question: 65

CertyIQ

You have a Microsoft Foundry project that uses Azure AI Search to ground an agent in internal documentation.
After a recent content update, users report that the agent's answers have become less accurate.
You need to identify whether the retrieved content is negatively influencing the model's generated responses.
Which observability signal should you review?

A. indexer status and failure history
B. latency breakdown traces
C. prediction drift metrics
D. groundedness evaluation metrics

Answer: D

Explanation:

Technical Justification for Correct Answer (D)

To address the issue of decreased accuracy in the agent's responses after a content update in a Microsoft

Foundry project utilizing Azure AI Search for grounding, it's crucial to focus on observability signals that directly relate to the integration of retrieved content with the model's response generation. Here's why **D. Groundedness Evaluation Metrics** is the most suitable choice, and why the others are less appropriate:

D. Groundedness Evaluation Metrics:

Relevance: Groundedness metrics directly measure how well the model's responses are grounded in the retrieved content from Azure AI Search. A decline in these metrics post-content update would indicate that the new content is not being effectively utilized or is misleading the model, leading to less accurate responses.
Actionability: Reviewing these metrics provides clear insights into whether the issue lies in the content update itself or in how the model interprets this content, guiding targeted adjustments.

A. Indexer Status and Failure History:

Less Suitable Because: While important for overall system health, indexer status primarily indicates if the indexing process is functioning correctly. Failure history might show if content wasn't indexed properly, but it doesn't directly correlate with the model's ability to generate accurate responses from the content.
Why Not Best: Doesn't provide insight into the model's usage of the content for response accuracy.

B. Latency Breakdown Traces:

Less Suitable Because: Latency issues might affect user experience but do not directly impact the accuracy of the model's responses. Slow performance could be a separate issue but doesn't explain the decrease in response accuracy.
Why Not Best: Focuses on performance rather than content-model interaction accuracy.

C. Prediction Drift Metrics:

Less Suitable Because: While prediction drift metrics are crucial for monitoring model performance over time, they are more generalized. They can indicate a decline in model accuracy but do not specifically point to the influence of the retrieved content as the cause.
Why Not Best: Less targeted than groundedness metrics for identifying content-related issues.

Conclusion: Given the need to isolate whether the retrieved content update is negatively impacting the model's response accuracy, **D. Groundedness Evaluation Metrics** is the most direct and relevant observability signal to review. It offers the clearest path to understanding the interaction between the new content and the model's performance.

References

Microsoft Documentation: Azure AI Search - Troubleshooting (Although not directly linked to "Groundedness Evaluation Metrics", this resource provides insight into troubleshooting content indexing issues, which can be a precursor to evaluating groundedness.)
Microsoft Learn: Monitor and troubleshoot Azure AI models (While broad, this learning module touches on model monitoring aspects that can be related to understanding prediction drift and, by extension, the importance of specific metrics in model performance analysis.)

Question: 66 CertyIQ

HOTSPOT
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Case Study
-
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-
Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.
Agent1 has the following configurations:
•Agent1 uses a base model deployment.
•A safety evaluation pipeline is NOT enabled.
•Tool invocation approval workflows are NOT enabled.
•Conversation memory constraints are NOT configured.
Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
Project1 is deployed to an Azure region located in the European Union (EU).
Agent1Dev Team will use Project1 to optimize and maintain Agent1.
Project2
-
Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
Development of the solution is incomplete.
Data Environment
-
Contoso stores product-related information in Azure resources that support AI applications.
The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.
Problem Statements
-
Contoso identifies the following issues:
•Agent1 has only general knowledge of the Contoso products.

•A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.
•Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.
•The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.
Requirements
-
Planned Changes
-
Contoso plans to implement the following changes:
•Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
•Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
•Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
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•Complete the development of the video creation solution.
Technical Requirements
-
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•Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
•The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements
-
Contoso identifies the following business requirements:
•Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
•Agent1 must answer questions only about the products sold by Contoso.
You need to ensure that the marketing department can generate videos by using the model deployed to Project2.
How should you complete the Python code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
import time
from openai import OpenAI
client = OpenAI(
     base_url= "https://Contoso.openai.azure.com/openai/v1/",
     api_key=…,
)
video = client.videos. [▼] (
```

- create
- download_content
- list
- retrieve

```
     model=deployment_name,
          prompt= "A video of our products",
     )
     while video.status not in ["completed", "failed", "cancelled"]:
          time.sleep(20)
          video = client.videos. [▼] (video.id)
```

- create
- download_content
- list
- retrieve

```
          print(video.status)
```

Answer:

Answer Area

```
import time
from openai import OpenAI
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- **create**
- download_content
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     )
     while video.status not in ["completed", "failed", "cancelled"]:
          time.sleep(20)
          video = client.videos. [▼] (video.id)
```

- create
- download_content
- list
- **retrieve**

```
          print(video.status)
```

Explanation:

First Dropdown: create

Why: The initial method call starts an asynchronous video generation task using a given prompt and deployment model.
Resulting Code:
python

```
video = client.videos.create(    model=deployment_name,    prompt="A video of our products",)
```

Use code with caution.

Second Dropdown: retrieve

Why: Inside the asynchronous checking (while) loop, the code checks the generation progress repeatedly. The retrieve method fetches the up-to-date execution status corresponding to the specific video.id.
Resulting Code:
python

```
video = client.videos.retrieve(video.id)
```

Use code with caution.

Question: 67

CertyIQ

Case Study -
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•Agent1 must answer questions only about the products sold by Contoso.
You need to configure personalized user interactions for Agent1. The solution must meet the business requirements.
What should you include in the solution?

A. knowledge
B. memory
C. guardrails
D. tools

Answer: B

Explanation:

Technical Justification for Configuring Personalized User Interactions for Agent1

To meet the business requirement for personalized user interactions with Agent1, where the agent must retain conversation context and recall relevant information from previous interactions, the most suitable solution component is identified and justified below, along with explanations for why other options are less suitable.

Correct Answer: B. Memory

Why Memory (B) is the Best Choice:

Conversation Context Retention: Implementing memory for Agent1 directly addresses the requirement for retaining conversation context. This allows Agent1 to remember previous interactions with users, facilitating a personalized experience.
Recalling Relevant Information: Memory integration enables Agent1 to recall relevant product information from previous conversations, ensuring consistency and personalization in responses.
Alignment with Technical Requirements: While not directly listed under technical requirements, enhancing Agent1 with memory capabilities aligns with the broader technical goal of ensuring relevant, complete, and accurate responses, especially when combined with the planned indexing solution for product sheets.

Why Other Options are Less Suitable:

A. Knowledge:

Reason: While enhancing Agent1's knowledge base (e.g., with more product information) is crucial and planned (as per updating the base model and using product sheets), "knowledge" in this context refers to the breadth of information Agent1 has, not the ability to retain conversation context for personalization.
Misalignment: Does not directly address the retention of conversation history for personalized interactions.

C. Guardrails:

Reason: Guardrails are more about setting boundaries and controls on what Agent1 can and cannot do or say, ensuring safety and compliance (e.g., not revealing customer information). While important, guardrails do not enable the personalization of user interactions through context retention.
Misalignment: Focuses on restriction and safety rather than personalization features.

D. Tools:

Reason: Implementing new "tools" for Agent1 could broadly enhance its capabilities, but the term is too vague in this context to directly address the personalized interaction requirement. Tools might include anything from new APIs to integration with other services, not necessarily focusing on conversation memory.
Misalignment: Too generic; does not specifically target the need for retaining conversation context for personalization.

Conclusion: Given the business requirement for personalized user interactions, incorporating **memory (B)** into Agent1's configuration is the most direct and effective solution. This ensures Agent1 can retain conversation context and recall relevant information, providing users with a personalized experience.

References:

1. **Microsoft Azure Cognitive Services - Conversational AI**
   https://azure.microsoft.com/en-us/services/cognitive-services/conversational-ai/

2. **Personalizing Conversations with Azure Bot Service**
   https://learn.microsoft.com/en-us/azure/bot-service/bot-service-concept-personalization?view=azure-bot-service-4.0

Question: 68 CertyIQ

HOTSPOT
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Contoso recently formed a new AI engineering team named Agent1Dev Team to optimize and maintain existing AI solutions.
The team collaborates with solution architects, DevOps engineers, and security engineers to design, implement. monitor, and secure AI applications.
Contoso also has a team named Agent1Test Team that is responsible for validating AI solutions before the solution

deployments.
Generative Environment
-
Contoso has a Microsoft Foundry deployment that contains two projects named Project1 and Project2.
Project1
-
Project1 contains a customer support agent named Agent1 that assists customers with product inquiries and troubleshooting requests.
Agent1 has the following configurations:
•Agent1 uses a base model deployment.
•A safety evaluation pipeline is NOT enabled.
•Tool invocation approval workflows are NOT enabled.
•Conversation memory constraints are NOT configured.
Agent1 interacts with customers by using digital support channels and answers general questions about Contoso products.
Project1 is deployed to an Azure region located in the European Union (EU).
Agent1Dev Team will use Project1 to optimize and maintain Agent1.
Project2
-
Project2 contains a deployed video generation model. The marketing department at Contoso has access to Project2 and plans to use the model to develop a video creation solution.
Development of the solution is incomplete.
Data Environment
-
Contoso stores product-related information in Azure resources that support AI applications.
The Azure environment contains an Azure Blob Storage account named storage1 that stores product detail sheets for all the Contoso products.
The product sheets include specifications, feature descriptions, and product support information that Agent1 can use to answer customer questions. The product sheets are stored in the PDF format.
Problem Statements
-
Contoso identifies the following issues:
•Agent1 has only general knowledge of the Contoso products.
•A recent chat interaction with Agent1 was analyzed for sentiment. The results of the analysis have NOT been processed yet.
•Agent1 does NOT use the detailed product information in the product sheets stored in storage1 when responding to customer questions.
•The finance department at Contoso reports that vendor invoices must be reviewed manually to ensure that the invoices match the terms defined in the vendor contracts. The invoices contain tables, logos, and varied layouts that make the documents difficult to process consistently.
Requirements
-
Planned Changes
-
Contoso plans to implement the following changes:
•Implement a solution for Project1 that analyzes the vendor invoices by evaluating both the visual layout and the textual content of the invoices, so that the invoice details can be verified against the vendor contract terms.
•Update the base model deployment used by Agent1 and standardize the model version to ensure continuity and consistent responses.
•Enable Agent1 to retrieve and use the detailed product information from the product sheets stored in storage1.
•Implement an indexing solution for the product sheets that Agent1 can use to answer customer questions.
•Complete the development of the video creation solution.
Technical Requirements
-
Contoso identifies the following technical requirements:
•The model deployment used by Agent1 must support scalable, high-throughput generative AI workloads and dynamically scale to handle variable customer support traffic, without requiring reserved throughput capacity.
•The product sheets must be processed by using an indexing pipeline that enables semantic and vector search, so that Agent1 can retrieve the relevant product information.
•Responses generated by using the product sheet information must be relevant, complete, and accurate.
•Agent1 must be able to use the product sheets to answer natural language questions about product details.
•The model version used by Agent1 must remain consistent to ensure stable responses.
•The data processed by the model must remain within the EU.
Security and Compliance Requirements
Contoso identifies the following security and compliance requirements:
•API keys must NOT be used to access Foundry-deployed models.

•Access to the Azure resources must follow the principle of least privilege.
•The developers at Contoso must authenticate to Microsoft Foundry resources by using Microsoft Entra authentication.
•Access to Project1 must be assigned to the members of Agent1Dev Team by using a security group named SC_Agent1_Dev.
•Access to Project1 must be assigned to the members of Agent1Test Team by using a security group named SC_Agent1_Test.
•Agent1 must never reveal customer information, even if a document that contains customer data is added erroneously to the product sheet repository in storage1.
•The product sheets might contain images that include embedded text. Agent1 must be protected from malicious instructions potentially hidden within the images.
Business Requirements
-
Contoso identifies the following business requirements:
•Users that interact with Agent1 must have a personalized experience in future interactions, including the ability for Agent1 to retain conversation context and recall relevant information from previous interactions.
•Agent1 must answer questions only about the products sold by Contoso.
You need to ensure that Agent1Dev Team can access Agent1. The solution must meet the security and compliance requirements.
How should you complete the Python code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
from azure.core.credentials import AzureKeyCredential
myEndpoint = "https://contoso.services.ai.azure.com/api/projects/project1"
project_client = AIProjectClient(
     endpoint=myEndpoint,
     credential= [▼] ,
```

AzureKeyCredential()
DefaultAzureCredential()
None

```
)
myAgent = "Agent1"
agent = project_client.agents. [▼] (agent_name=myAgent)
```

create_version
get
get_version

```
print(f "Retrieved agent: {agent.name}")
```

Answer:

Answer Area

```
from azure.identity import DefaultAzureCredential
from azure.ai.projects import AIProjectClient
from azure.core.credentials import AzureKeyCredential
myEndpoint = "https://contoso.services.ai.azure.com/api/projects/project1
project_client = AIProjectClient(
     endpoint=myEndpoint,
     credential= [ ▼ ],
                  AzureKeyCredential()
                  DefaultAzureCredential()
                  None
)
myAgent = "Agent1"
agent = project_client.agents. [ ▼ ] (agent_name=myAgent)
                               create_version
                               get
                               get_version
print(f "Retrieved agent: {agent.name}")
```

Explanation:

First Dropdown: DefaultAzureCredential()

Why: The script imports DefaultAzureCredential from azure.identity. For Microsoft Azure AI Foundry project client authentication, token-based authentication via Microsoft Entra ID is the standard approach, which is instantiated via DefaultAzureCredential().

Resulting Line:

python

credential=DefaultAzureCredential(),

Use code with caution.

2 Second Dropdown: get

Why: The final print statement output indicates that you are working with an existing agent (Retrieved agent: ...). The get method is used on the agents client operations property to fetch an active agent profile using its agent_name.

Resulting Line:

python

agent = project_client.agents.get(agent_name=myAgent)

Use code with caution.

Question: 69 CertyIQ

You have a Microsoft Foundry project that contains a Retrieval Augmented Generation (RAG) chat solution used by customer support agents.
You are adding an automated pre-production evaluation step to a CI/CD pipeline named Pipeline1. The evaluation will run against a labeled test dataset that contains support questions and the expected grounding context.
You need to ensure that Pipeline1 fails if unsupported content or a retrieval mismatch exceeds a defined threshold:
•responses include claims not supported by the retrieved source content
•retrieved source content does not align with the labeled expected context
Which two built-in evaluators should you use in Pipeline1? Each correct answer presents pat of the solution.
NOTE: Each correct selection is worth one point.

A. Retrieval

B. Fluency

C. Coherence

D. Groundedness

E. Response Completeness

Answer: AD

Explanation:

Technical Justification for Correct Option (AD)

To address the requirement of ensuring Pipeline1 fails if unsupported content or a retrieval mismatch exceeds a defined threshold in the context of a Retrieval Augmented Generation (RAG) chat solution, two specific built-in evaluators are most suited:

A. Retrieval: This evaluator directly assesses the alignment between the retrieved source content and the expected grounding context provided in the labeled test dataset. By setting a threshold for retrieval mismatch, Pipeline1 can fail if the discrepancy between the expected and actual retrieved content surpasses this limit, ensuring the model's ability to fetch relevant information is within acceptable parameters.

D. Groundedness: This evaluator checks if the responses generated by the model include claims that are supported by the retrieved source content. Setting a threshold for unsupported content ensures Pipeline1 fails if the model generates responses with unsupported claims beyond the defined limit, maintaining the accuracy and trustworthiness of the automated support responses.

Why Other Options Are Less Suitable:

B. Fluency: Evaluates how natural and fluent the generated responses are. While important for user experience, it does not directly address the issues of unsupported content or retrieval mismatches as specified in the requirement.

C. Coherence: Assesses the logical consistency and coherence of the generated responses. Similar to Fluency, it's crucial for overall response quality but does not target the specific thresholds for content support or retrieval alignment outlined in the problem.

E. Response Completeness: Checks if the responses fully address the user's query. Although completeness is vital, the primary concern here is the accuracy and support of the content, not necessarily its completeness in answering the question, making this option less directly relevant to the specified thresholds.

Conclusion: The combination of **Retrieval (A)** and **Groundedness (D)** evaluators is the best choice for Pipeline1 as they directly target and mitigate the risks of retrieval mismatches and unsupported content claims, aligning perfectly with the defined failure conditions.

References:

1. **Microsoft Azure Cognitive Services - Evaluators for Conversational AI**: https://learn.microsoft.com/en-us/azure/cognitive-services/conversational-ai/howto-evaluation (Section on Built-in Evaluators)
2. **Azure DevOps Pipelines - Integrating AI Evaluations**: https://learn.microsoft.com/en-us/azure/devops/pipelines/tasks/build/azureml-evaluate?view=azure-devops&tabs=yaml (Refer to Customizing Evaluation Metrics for AI Models)

Question: 70

CertyIQ

You have a Microsoft Foundry project that contains a support-ticket triage agent built by using the Foundry Agent Service.
The agent uses tool to classify the ticket type and sot the ticket priority.
Sometimes, the same support case continues across multiple sessions over several days.
You need to persist state by using a durable ID to ensure that the agent can automatically reuse the full interaction history. The solution must preserve previous user messages, tool calls and tool outputs across turns and sessions.
Which runtime component should you use?

A. output item
B. agent
C. conversation
D. response

Answer: C

Explanation:

Technical Justification for Choosing C. Conversation

To address the requirement of persisting state with a durable ID for an Microsoft Foundry project's support-ticket triage agent, utilizing the **Conversation** runtime component is the most appropriate choice. Here's why:

Persistence Across Sessions: The Conversation component is designed to maintain context over multiple turns and even across different sessions, spanning several days if necessary. This aligns perfectly with the need to persist the full interaction history (user messages, tool calls, and tool outputs) for ongoing support cases.
Durable ID for State Management: Conversations in Microsoft Foundry can be associated with a durable ID, ensuring that the agent can automatically reuse the entire history of interactions for a given support case, facilitating seamless continuity.
Scope of Operation:

A. Output Item: Primarily used for managing the output of a single turn or interaction. It lacks the capability to persist state across multiple sessions, making it unsuitable for this scenario.
B. Agent: While the agent is the core component of the Foundry project, specifying "agent" as the runtime component for persisting state is too broad. The agent encompasses various functionalities, but the specific task of state persistence across sessions is more accurately attributed to a subset of its capabilities, namely the Conversation component.
D. Response: Similar to Output Item, Response is focused on the immediate output of an interaction and does not support the long-term persistence of interaction history across sessions.

Why C. Conversation is the Best Choice:

Direct Alignment with Requirements: Preserves previous user messages, tool calls, and tool outputs.
Durable ID Capability: Ensures the agent can reuse the full interaction history across multiple sessions.
Designed for Multi-Session Context: Unlike the other options, Conversation is specifically designed to handle the persistence of state over extended periods and multiple interactions.

References

1. **Microsoft Azure Documentation - Microsoft Foundry: Conversations**
   https://learn.microsoft.com/en-us/microsoft-365/enterprise/microsoft-foundry-conversations?view=o365-worldwide

2. **Microsoft Documentation - Microsoft Foundry Agent Service: State Management**
   https://learn.microsoft.com/en-us/microsoft-365/enterprise/microsoft-foundry-agent-service-state-management?view=o365-worldwide

Question: 71

CertyIQ

HOTSPOT
-
You have a Microsoft Founcy project that contains a Retrieval Augmented Generation (RAG) solution.
You need to run a pre-production evaluation by using labeled CSV dataset that contains the query, context, response and ground truth. The evaluation must measure the following:
•Whether responses address the user query
•Whether responses are supported by the provided context
•Whether responses contain sensitive or proprietary information
Which AI quality evaluation metrics should you use? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

To measure whether the responses are supported by the provided context and address the user query:

- Coherence and Fluency
- GPT similarity and F1 score
- Groundedness and Relevance
- Groundedness and ROUGE score

To measure whether responses contain sensitive or proprietary information:

- Hateful and unfair content
- Indirect attack
- Protected material
- Violent content

Answer:

Answer Area

To measure whether the responses are supported by the provided context and address the user query:

- Coherence and Fluency
- GPT similarity and F1 score
- Groundedness and Relevance
- Groundedness and ROUGE score

To measure whether responses contain sensitive or proprietary information:

- Hateful and unfair content
- Indirect attack
- Protected material
- Violent content

Explanation:

1. First Drop-Down Selection

To measure whether the responses are supported by the provided context and address the user query:
Correct Answer: Groundedness and Relevance

2. Second Drop-Down Selection

To measure whether responses contain sensitive or proprietary information:
Correct Answer: Protected material

Question: 72 CertyIQ

You plan to configure an evaluation in Microsoft Foundry for a Retrieval Augmented Generation (RAG) chat app.
You need to provide scores for groundedness, relevance, and harmful content categories.
Which two evaluation categories can you use? Each correct answer presents a complete solution.
NOTE: Each correct selection is worth one point.

A. risk and safety metrics
B. fluency evaluator
C. similarity evaluators
D. AI quality (NLP) metrics
E. AI quality (AI assisted) metrics

Answer: AE

Explanation:

Technical Justification for Correct Answer AE

To configure an evaluation in Microsoft Foundry for a Retrieval Augmented Generation (RAG) chat app, focusing on groundedness, relevance, and harmful content categories, the most appropriate evaluation categories are:

A. Risk and Safety Metrics: This category is crucial for assessing harmful content, as it evaluates the model's output for potential risks, including but not limited to, toxicity, hate speech, and other safety concerns. This directly addresses the "harmful content" category requirement.
D. AI Quality (NLP) Metrics: For a RAG chat app, groundedness and relevance are key. AI Quality (NLP) Metrics are suited for evaluating the linguistic quality, coherence, and relevance of the generated text. This category can provide insights into how well the model's responses are grounded in the provided context (groundedness) and how relevant the responses are to the user's queries (relevance).

Why Other Options are Less Suitable:

B. Fluency Evaluator: While useful for assessing the grammatical correctness and readability of generated text, it does not directly evaluate groundedness, relevance, or harmful content, making it less relevant for this specific set of requirements.
C. Similarity Evaluators: These are more geared towards detecting duplication or similarity in content, which does not align with the needs for evaluating groundedness, relevance, or the presence of harmful content.
E. AI Quality (AI Assisted) Metrics: Although this might seem relevant, it's more focused on the overall quality of AI-assisted workflows rather than the specific NLP aspects of groundedness and relevance in generated text, making **D** more precise for NLP-focused evaluations.

References

For deeper insight into Microsoft Foundry's evaluation metrics and how they apply to NLP tasks like RAG chat apps:

1. **Microsoft Azure - Evaluate AI Models**: https://learn.microsoft.com/en-us/azure/machine-learning/how-to-deploy-and-where?tabs=python#evaluate
2. **Microsoft AI for Everyone - NLP Evaluation Metrics**: https://learn.microsoft.com/en-us/azure/cognitive-services/language/understanding/nlp-concepts#evaluation-metrics (Note: While the second link might not directly mention "Microsoft Foundry", it provides foundational knowledge on NLP evaluation metrics relevant to the context.)

Question: 73 CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent uses two tools to perform the following actions:
•Use Azure AI Search to retrieve answers from a private product documentation index.
•Use the web search tool to retrieve public information on the internet.
You need to ensure that for a specific run, the agent deterministically retrieves information only from the internet.
To what should you set tool_choice?

A. “type”: “bing_grouding”
B. “type”: “azure_ai-search”
C. “auto”
D. “required”

Answer: A

Explanation:

Technical Justification

To ensure the agent deterministically retrieves information only from the internet for a specific run, the correct setting for tool_choice must explicitly target the tool designed for internet searches, excluding the private index search. Here's why each option is more or less suitable for the requirement:

A. “type”: “bing_grouding” : This option is the most suitable. Setting tool_choice to specify "bing_grouding" (likely a typo in the question, intended to refer to "Bing" or a similar web search engine integration) would direct the agent to use the web search tool (implied by "Bing" context) exclusively for that run. This ensures the agent retrieves information only from the internet, fulfilling the requirement.

B. “type”: “azure_ai-search” : This option is **less suitable** because setting tool_choice to "azure_ai-search" would force the agent to use Azure AI Search, which is configured to retrieve answers from a **private product documentation index**, not the internet. This contradicts the requirement.

C. “auto”: This option is **less suitable** as setting tool_choice to "auto" would likely enable the agent to automatically choose between available tools based on the query or predefined rules. This introduces unpredictability and does not guarantee that the agent will retrieve information **only** from the internet for the specific run.

D. “required”: This option is **less suitable** and possibly **incorrect in context**. Setting tool_choice to "required" does not specify a tool type but rather might imply that a tool is mandatory for the operation. It does not direct the agent to use a specific tool (in this case, the internet search tool), leaving the choice ambiguous and not ensuring the desired behavior.

Correct Answer Justification: Option **A** is correct because it explicitly directs the agent to use the tool associated with internet searches (despite the apparent typo, the intent aligns with the requirement), ensuring deterministic retrieval of information only from the internet for the specific run.

References

1. **Microsoft Azure AI Search Documentation**: https://docs.microsoft.com/en-us/azure/cognitive-search/
2. **Microsoft Foundry (formerly Azure Bot Service) Tools Configuration**: https://docs.microsoft.com/en-us/azure/bot-service/bot-service-channel-connect-azure-foundry?tabs=csharp (Note: Direct "Microsoft Foundry" documentation might be scarce or outdated; the link provided relates to the broader context of tool configuration in similar Microsoft services. For the most current information, always check the latest Microsoft Azure and Foundry documentation.)

Question: 74 CertyIQ

You have a Microsoft Foundry project that contains a customer support agent built on a deployed chat model. The agent responses are validated by using an automated testing system that compares generated answers to stored expected outputs. Identical prompts must return consistent response to prevent automated test failures. You need to reduce response variability, without modifying the prompt or reducing factual accuracy.
What should you do for the model?

A. Increase the max_tokens parameter.
B. Remove stop sequences from the requests.
C. Decrease the temperature parameter.
D. Increase the temperature parameter.

Answer: C

Explanation:

Technical Justification for Correct Answer: C

To address the requirement of reducing response variability in the chat model without modifying the prompt or compromising factual accuracy, let's analyze each option in the context of natural language processing (NLP) and language model behavior:

A. Increase the max_tokens parameter: Increasing max_tokens allows the model to generate longer responses. However, this does not directly impact the variability of responses to identical prompts; it merely allows for more lengthy outputs, which could potentially increase variability if the model explores more diverse endings for the same start. **Not Suitable**.

B. Remove stop sequences from the requests: Stop sequences are used to indicate the end of a response. Removing them could lead to the model generating indefinitely until it hits the max_tokens limit. This change affects response length and form but not the variability of responses to the same prompt. **Not Suitable**.

C. Decrease the temperature parameter: The **temperature** parameter controls the randomness of the model's output. A lower temperature reduces randomness, leading to more deterministic and less varied responses. Decreasing the temperature is directly aimed at reducing the variability of responses to identical inputs without altering the prompt or necessarily impacting factual accuracy, as the model is guided towards more predictable, high-probability outputs. **Correct Answer**.

D. Increase the temperature parameter: Increasing the temperature would have the opposite effect of what is desired. It increases the randomness of the output, leading to more varied responses for the same input, which would exacerbate the issue of test failures due to inconsistency. **Not Suitable**.

Conclusion: Decreasing the temperature parameter (Option C) is the most appropriate action to reduce response variability for identical prompts without modifying the prompt or reducing factual accuracy. This adjustment tunes the model's behavior to produce more consistent outputs, aligning with the requirement for reliable automated testing.

References

1. **Microsoft Azure Cognitive Services - Custom Language Models Documentation**: https://docs.microsoft.com/en-us/azure/cognitive-services/language-customization/
2. **Temperature Parameter Explanation in Context of Language Models**: https://huggingface.co/docs/transformers/generation#temperature (Note: While not Azure-specific, this link provides a clear explanation of the temperature parameter's role in NLP models, which is applicable across platforms including Azure's language model deployments.)

Question: 75 CertyIQ

You are developing prompts for a Micosoft Foundry project that classifies incoming support tickets by category.
You need to improve accuracy by showing the model how correct classifications look, without retaining the model or storing knowledge permanently.
Which prompt engineering approach should you use?

A. Retrieval Augmented Generation (RAG)

B. zero-shot learning

C. chain of thought

D. few-shot learning

Answer: D

Explanation:

Technical Justification for Correct Answer (D. few-shot learning)

To enhance the accuracy of category classification for incoming support tickets in a Microsoft Foundry project without retaining the model or storing knowledge permanently, the most suitable prompt engineering approach is **few-shot learning (D)**. Here's a breakdown of why:

Why D. few-shot learning is the best choice:

Definition Alignment: Few-shot learning involves training a model on a small, limited dataset (typically 1 to 10 examples per class) to perform a specific task, aligning perfectly with the requirement to show the model "how correct classifications look" without permanent knowledge storage.
No Permanent Model Retention: This approach does not necessitate retaining the model post-training, as the goal is to demonstrate immediate classification capability based on the provided few shots.
Accuracy Improvement: By carefully selecting representative examples for the few shots, the model's accuracy on the target task (ticket classification by category) can be significantly improved.

Why other options are less suitable:

A. Retrieval Augmented Generation (RAG)

Requires Stored Knowledge Base: RAG relies on accessing a large, stored knowledge base to generate responses, conflicting with the "without storing knowledge permanently" requirement.
Complexity: More complex to set up compared to few-shot learning, given the need for a robust retrieval system.
B. Zero-shot Learning

No Direct Examples Provided: Zero-shot learning involves no examples of the target class being provided during training, which contradicts the need to "show the model how correct classifications look".
Less Accurate for Specific Tasks: Generally less accurate than few-shot learning for tasks where some examples can be provided.
C. Chain of Thought

Focus on Transparency Over Efficiency for Small Datasets: While useful for understanding model reasoning, it doesn't directly address the efficiency and accuracy needs for a small, non-permanently stored dataset scenario as effectively as few-shot learning.

References

For in-depth information on Few-Shot Learning and its applications: https://learn.microsoft.com/en-us/azure/cognitive-services/custom-question-answering/concept-few-shot-learning
Overview of Prompt Engineering Techniques including those mentioned: https://aka.ms/prompt-engineering-techniques

Question: 78 CertyIQ

You have a web app named App1 that processes user prompts by integrating with a Microsoft Foundry project named Project1. App1 performs the following actions:
•Sends prompts directly to a model by using the Azure OpenAI Responses API
•Invokes the Azure AI Content Safety tool by using a Foundry connection within the same request
You need to configure end-to-end visibility into each step of the request workflow.
What should you do?

A. Enable logging by using the client SDK for Content Safety.
B. Enable logging by using Foundry Local.
C. Enable application tracing in Project1.

D. Route requests through the Azure OpenAI endpoint.

Answer: C

Explanation:

C. Enable application tracing in Project1.

Generative AI Observability: Microsoft Azure AI Foundry features built-in **application tracing** (powered by OpenTelemetry/OpenInference standards).
End-to-End Tracking: Enabling tracing directly inside the project allows you to trace user prompts, model completion outputs, token usage, execution duration, and tool execution (like the Content Safety validation node) as unified workflows within the diagnostics panel.

Question: 79 CertyIQ

HOTSPOT

-

You have a Microsoft Foundry project that contains two agents named PolicyWriter and RskReviewer.
PolicyWriter generates daft updates for customer polices, and RiskReviewer reviews the drafts.
In the visual builder, you need to create a workflow that meets the following requirements:
•Finalizes low-risk updates without manual intervention
•Ensures predictable execution across the agents
•Requires user approval for highs updates
What should you configure? To answer, select the appropriate options in the answer area.
NOTE: Each comet selection is worth one point.

Answer Area

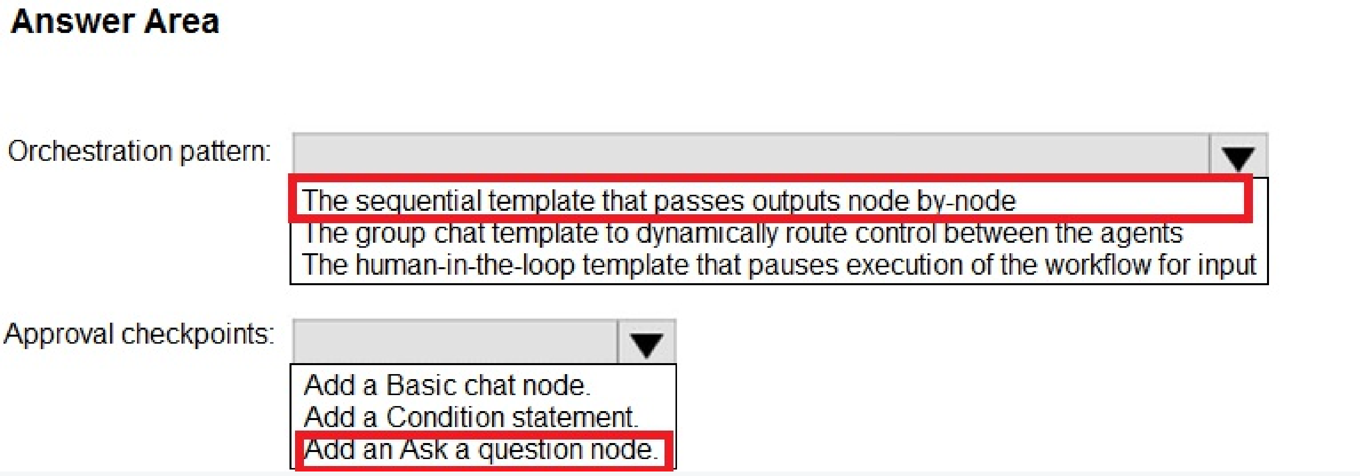
Orchestration pattern:

- The sequential template that passes outputs node by-node
- The group chat template to dynamically route control between the agents
- The human-in-the-loop template that pauses execution of the workflow for input

Approval checkpoints:

- Add a Basic chat node.
- Add a Condition statement.
- Add an Ask a question node.

Answer:



Explanation:

The sequential template that passes outputs node by-node: Choose this if your tasks are linear and must follow a fixed, strict step-by-step order from one agent/node to the next.

Add an Ask a question node.

To create a manual approval checkpoint or gatekeeper step within a conversational workflow, you must pause execution to gather explicit confirmation from a user. This is achieved by adding an interactive **"Ask a question node"** (or a choice/confirm prompt) to capture the human's approval input before continuing down the branch.

Question: 80 **CertyIQ**

You have a Microsoft Foundry project that contains an agent named Agent1.
Agent runs successful, but Foundry Control Plane does NOT display values for error rates, runs, and token usage, and the Traces tab is empty.
You need to ensure that Found Control Plane displays the appropriate values for Agent1.
What should you do?

A. Update Agent1 to a new version.

B. Restart Agent from Foundry Control Plan

C. Assign to a Log Analytics workspace to Agent1.

D. Enable Application Insights for Agent1.

Answer: D

Explanation:

Technical Justification for Correct Answer (D)

Why D (Enable Application Insights for Agent1) is the best option: Enabling Application Insights for Agent1 is the most suitable solution because Application Insights is a monitoring and analytics service that provides detailed metrics and logs, including error rates, run metrics, and usage telemetry (such as token usage). Integrating Agent1 with Application Insights would populate the Foundry Control Plane with the required metrics (error rates, runs, token usage) and also enable trace logging, thus populating the Traces tab. This is a direct solution to the visibility issue described.

Why other options are less suitable:

A. Update Agent1 to a new version: While keeping software up-to-date is crucial, there's no indication that the issue with metrics and trace visibility in Foundry Control Plane is due to a versioning problem with Agent1. This action does not directly address the monitoring and logging visibility issue.

B. Restart Agent from Foundry Control Plane: Restarting the agent might resolve transient issues but does not address the fundamental lack of integration with a monitoring solution that would provide the missing metrics and traces to the Control Plane.

C. Assign to a Log Analytics workspace to Agent1: While Log Analytics is powerful for log analysis, the specific requirement for visibility into error rates, runs, token usage, and traces in the **Foundry Control Plane** suggests a need for a more integrated application monitoring solution like Application Insights, which is more tightly coupled with the metrics and tracing requirements described.

Conclusion: Enabling Application Insights for Agent1 (Option D) is the most direct and effective solution to ensure Foundry Control Plane displays the necessary metrics and traces for Agent1, as it provides the specific types of monitoring and analytics capabilities required to address the identified gaps.

References:

Microsoft Azure - Application Insights Overview
Azure | Microsoft Foundry Documentation - Integrating with Application Insights

Question: 81 CertyIQ

You have a Microsoft Foundry project that contains an agent.
The agent uses Azure Content Understanding in Foundry Too to process vendor onboarding packets. The packs include digital PDFs that contain tables and hyperlinks.
The extracted content is indexed for search and provided to a downstream agent in the Markdown format.
You need to generate a Markdown output that has a layout and a semantic structure optimized for Retrieval Augmented Generation (RAG) workflows.
Which built-in analyzer should you use?

A. prebuilt-documentFieldSchema
B. prebuilt-documentSearch
C. prebuilt-read
D. prebuilt-layout

Answer: B

Explanation:

Technical Justification for Correct Option (B. prebuilt-documentSearch)

The correct option for generating Markdown output with a layout and semantic structure optimized for Retrieval Augmented Generation (RAG) workflows, given the context of processing vendor onboarding packets (PDFs with tables and hyperlinks) using Azure Content Understanding in a Microsoft Foundry project, is **B. prebuilt-documentSearch**. Here's why:

Why B (prebuilt-documentSearch) is the Best Choice:

Optimization for Search and Semantic Structure: prebuilt-documentSearch is specifically designed to extract content in a manner that enhances search capabilities and understands the semantic structure of documents. This aligns perfectly with the requirement for optimizing the Markdown output for RAG workflows, which heavily rely on effective information retrieval and understanding.

Handling of Diverse Content (Tables, Hyperlinks): This analyzer is capable of handling various document elements, including tables and hyperlinks found in the PDF packets, ensuring that the extracted content (and subsequently the Markdown output) retains the necessary structure and links for comprehensive understanding.
Compatibility with Downstream Agents and Markdown Format: The output from prebuilt-documentSearch is well-suited for further processing by downstream agents, as it provides a structured format that can be easily converted into Markdown. This structured approach facilitates the organization of content in a way that supports RAG's augmented generation capabilities.
Why Other Options are Less Suitable:

A. prebuilt-documentFieldSchema:

Primarily focused on extracting specific, predefined fields from documents, which might not cater to the broader semantic understanding required for RAG workflows.
Less adaptable to the varied, unstructured content aspects (like hyperlinks) in the onboarding packets.
C. prebuilt-read:

Designed more for basic reading and text extraction, lacking the specific search and semantic structure optimization needed for RAG.
Might not handle the layout and table extraction as effectively as prebuilt-documentSearch for the stated requirements.
D. prebuilt-layout:

While excellent for understanding and extracting based on the document's layout, it may not offer the same level of semantic understanding and search optimization as prebuilt-documentSearch.
Less focused on the overall content structure beyond layout, potentially leading to less effective RAG workflow integration.
Conclusion

Given the needs of the project — particularly the emphasis on search optimization, semantic structure for RAG workflows, and handling of diverse document content — **B. prebuilt-documentSearch** is the most appropriate analyzer.

References

1. **Azure Cognitive Search - Prebuilt Analyzers**: https://docs.microsoft.com/en-us/azure/search/search-analyzers#prebuilt
2. **Microsoft Azure - Content Understanding in Azure Cognitive Services**: https://azure.microsoft.com/en-us/services/cognitive-services/content-understanding/ (Explore the documentation for the most current Content Understanding capabilities, though note the service's evolution and potential rebranding/refocusing within Azure services)

Question: 82 CertyIQ

You have a Microsoft Foundry project that contains an agent.
The agent uses Azure AI Search for Retrieval Augmented Generation (RAG).
You plan to ingest and index PDF product manuals.
You need to build a solution that supports semantic similarity matching. The solution must ensure that the agent retrieves relevant data when user questions use different wording than the product manuals.
Which indexing approach should you use?

A. vector search
B. semantic ranking

C. suggesters
D. analyzers

Answer: A

Explanation:

Technical Justification for Correct Answer: A. Vector Search

To support semantic similarity matching for the Microsoft Foundry agent utilizing Azure AI Search for Retrieval Augmented Generation (RAG) on ingested PDF product manuals, **Vector Search (A)** is the most appropriate indexing approach. Here's why:

Correct Answer - A. Vector Search:

Semantic Similarity: Vector search uses dense vector representations (embeddings) of text, allowing for the measurement of semantic similarity between queries and documents, even when the wording differs. This is crucial for retrieving relevant data from product manuals based on user questions with varied phrasing.
RAG Compatibility: Given the use of RAG, vector search aligns perfectly as it can efficiently handle the generation and retrieval aspects by understanding the contextual relevance.
PDF Content: Since the content is in PDF format, the semantic understanding provided by vector search can effectively index and retrieve relevant sections of the manuals, regardless of the exact query wording.

Why Other Options are Less Suitable:

B. Semantic Ranking:

While semantic ranking can improve search results by considering the context, it is more about **ranking** relevant documents rather than **indexing** them for semantic similarity. It can complement vector search but doesn't replace its indexing capabilities for this requirement.
C. Suggesters:

Suggesters are useful for auto-completing or suggesting search queries as the user types, based on existing search queries or indexed terms. They do not address the need for semantic similarity matching in the indexed content itself.
D. Analyzers:

Analyzers are crucial for tokenizing, stemming, and normalizing text to improve search accuracy. However, they focus on lexical matching rather than semantic understanding, making them insufficient for the requirement of handling differing wording through semantic similarity.

References

1. **Azure Cognitive Search - Vector Search**: https://docs.microsoft.com/en-us/azure/search/search-vector-search
2. **Understanding RAG in Azure AI Search**: https://docs.microsoft.com/en-us/azure/search/search-howto-ragısıyla

Question: 83 CertyIQ

You have a Microsoft Foundry agent that grounds responses from an Azure Search index that contains the following:
•Searchable text fields for product names and product codes
•A vector field that stores embeddings for product descriptions

You need to ensure that users can query the index by using the following:
•Exact product names or codes
•Natural language descriptions of the products
What should you configure?

A. vector search only
B. hybrid search
C. keyword search only
D. semantic search only

Answer: B

Explanation:

Technical Justification for Correct Answer: B. Hybrid Search

To address the requirement of enabling users to query the Azure Search index using both **exact product names/codes** and **natural language descriptions of products**, configuring **Hybrid Search** is the most suitable approach. Here's why:

Exact Product Names or Codes: This requirement is best met with **keyword search** capabilities, which allow for precise matches. Keyword search is effective for structured data like product names and codes where exact matches are desired.
Natural Language Descriptions of Products: For this, **vector search** (leveraging the pre-stored embeddings for product descriptions) is ideal. Vector search enables semantic matching, allowing users to find relevant products even when using descriptive, non-exact language.

Why Hybrid Search (B) is the Best Choice: Hybrid Search combines the strengths of both keyword and vector search, enabling the index to handle:

Exact matches for product names and codes (keyword search)
Semantic, natural language queries for product descriptions (vector search)

Why Other Options are Less Suitable:

A. Vector Search Only: Would not support exact matches for product names/codes effectively, as it's designed for semantic rather than exact text matching.
C. Keyword Search Only: Fails to leverage the vector field for natural language queries, making product description searches less effective.
D. Semantic Search Only: While similar in concept to vector search for natural language processing, "Semantic Search" in some contexts might imply reliance on external knowledge graphs or more complex AI models not explicitly mentioned in the scenario. It doesn't clearly address the need for exact matches like keyword search does, and in Azure Search, the direct approach for the described vector field use case is more accurately termed as part of a Hybrid Search strategy.

Conclusion: Given the dual nature of the query requirements, **Hybrid Search (B)** is the most comprehensive and technically appropriate configuration, as it effectively combines the precise matching of keyword search with the semantic capabilities of vector search.

References:

1. **Azure Search Hybrid Search Documentation**: https://docs.microsoft.com/en-us/azure/search/search-hybrid-search
2. **Azure Search Vector Search and Embeddings**: https://docs.microsoft.com/en-us/azure/search/search-vector-search-overview

Question: 84 CertyIQ

You have an Azure AI Search indexer that ingests PDF policy manuals.
Client applications must display page-level citations that have bounding polygons for both text and images.
You need to add a single built-in multimodal content extraction skill to the Azure AI Search skillset. The solution must meet the following requirements:
•Provide text and image location metadata.
•Extract tables that span multiple pages.
What should you add?

A. Document Extraction
B. Azure Content Understanding in Foundry Tools
C. GenAI Prompt
D. Document Layout

Answer: B

Explanation:

Technical Justification for Choosing B: Azure Content Understanding in Foundry Tools

For the given scenario, where the requirement is to add a single built-in multimodal content extraction skill to an Azure AI Search skillset that can provide text and image location metadata, extract tables (even those spanning multiple pages), and support page-level citations with bounding polygons for both text and images, **B. Azure Content Understanding in Foundry Tools** is the most suitable option. Here's why:

Multimodal Extraction Capability: Azure Content Understanding in Foundry Tools is designed to handle multimodal data, meaning it can effectively extract and provide metadata for both **text** (including tables across multiple pages) and **images**. This aligns perfectly with the requirement for extracting diverse content types from PDF policy manuals.

Location Metadata and Bounding Polygons: This skill is capable of providing detailed location metadata. For images and text (including tabular content), it can identify and return bounding polygons, which is crucial for client applications to display accurate page-level citations.

Table Extraction Across Pages: Specifically, the capability to extract tables that span multiple pages is a nuanced requirement. Azure Content Understanding in Foundry Tools, with its advanced document understanding capabilities, is better suited for handling such complex table extractions compared to the other options.

Why Other Options Are Less Suitable:

A. Document Extraction: While useful for extracting document content, this option might not provide the same level of granularity in location metadata (especially for images and multi-page tables) as Azure Content Understanding. Its capabilities are more focused on the text content without the advanced layout and image analysis required.

C. GenAI Prompt: GenAI Prompt is more aligned with generating text based on prompts rather than extracting multimodal content with precise location metadata from existing documents. It lacks the specific document layout understanding and extraction capabilities needed for this task.

D. Document Layout: Although Document Layout Analysis can extract text and layout information, the question specifies the need for a skill that also handles **images** with the same level of location precision and extracts tables across multiple pages effectively. Document Layout, while strong in text layout analysis, does not fulfill the multimodal (text + image) requirement as comprehensively as Azure Content Understanding in Foundry Tools.

References

1. **Azure Cognitive Search - Skillset Concepts**: https://docs.microsoft.com/en-us/azure/search/search-concept-skillset
2. **Azure AI Services - Content Understanding**: https://azure.microsoft.com/en-us/services/cognitive-services/content-understanding/ (Note: As of my last update, direct links to "Content Understanding in Foundry Tools" might redirect or require navigation from the Azure AI Services page due to Microsoft's documentation updates. Ensure to explore the latest on Azure Content Understanding capabilities within Azure Search and AI services.)

Question: 85

CertyIQ

You are building an Azure AI Search indexing pipeline named Pipeline1 that ingests invoices stored in Azure Blob Storage. The invoices are stored as scanned images.
You need to enable users to search invoice data across the invoice fields.
Which built-in skill should you add to the skillset of Pipeline1?

A. Text Split
B. Text Translation
C. optical character recognition (OCR)
D. Image Analysis

Answer: C

Explanation:

Technical Justification for Correct Option (C) Optical Character Recognition (OCR)

To enable users to search invoice data across the invoice fields from scanned images stored in Azure Blob Storage, the primary requirement is to extract text from the images. Here's why **Optical Character Recognition (OCR)** is the best choice and why other options are less suitable:

Correct Option: C. Optical Character Recognition (OCR)

Reason for Choice: OCR skills are specifically designed to extract text from images and scanned documents, which aligns perfectly with the need to make invoice fields searchable. Azure's OCR skill can handle various fonts, layouts, and languages found in invoices.
Technical Advantage: By adding an OCR skill to the skillset of Pipeline1, the Azure AI Search indexing pipeline can convert the visual data in scanned invoice images into searchable text, enabling users to query based on the content within the invoices.
Why Other Options are Less Suitable

A. Text Split:

Inappropriateness: Text Split is used to divide large text into smaller segments based on a specified size or delimiter. Since the invoices are images without initially extractable text, this skill does not address the primary requirement.
Use Case Misalignment: Useful after text extraction (e.g., splitting large extracted texts), not before.
B. Text Translation:

Inappropriateness: While useful for multi-language support, the immediate need is to extract text from images, not translate existing text.
Premature Application: Could be considered after text extraction if the invoices are in multiple languages and need translation.
D. Image Analysis:

Partial Relevance but Insufficient for the Goal: Image Analysis can provide metadata about the image (e.g., objects, tags) but does not focus on extracting readable text from documents.
Misalignment with Search Requirement: The primary goal is to search text within invoices, not analyze image contents in a broader sense.
Conclusion

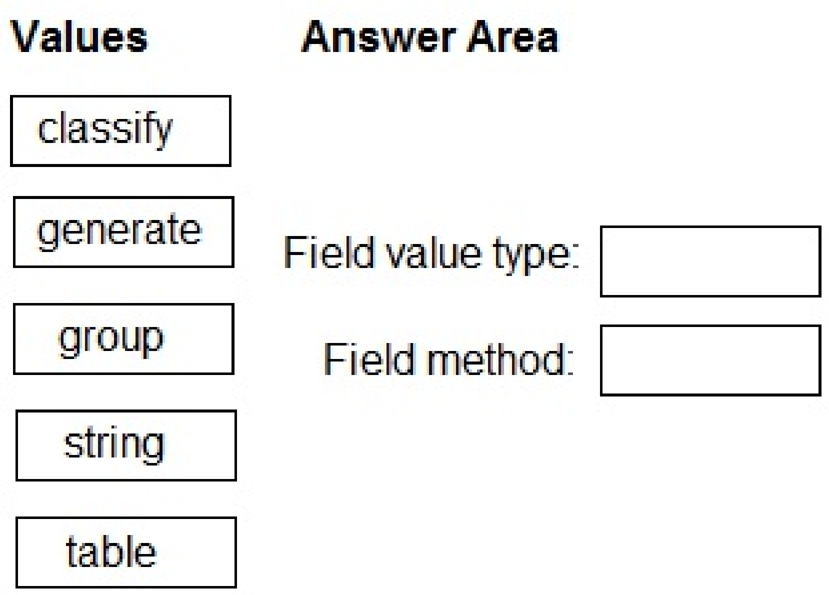
Given the scenario's requirements, **Optical Character Recognition (OCR)** is the most appropriate built-in skill to add to Pipeline1's skillset for enabling search functionality across invoice fields in scanned image invoices.

References

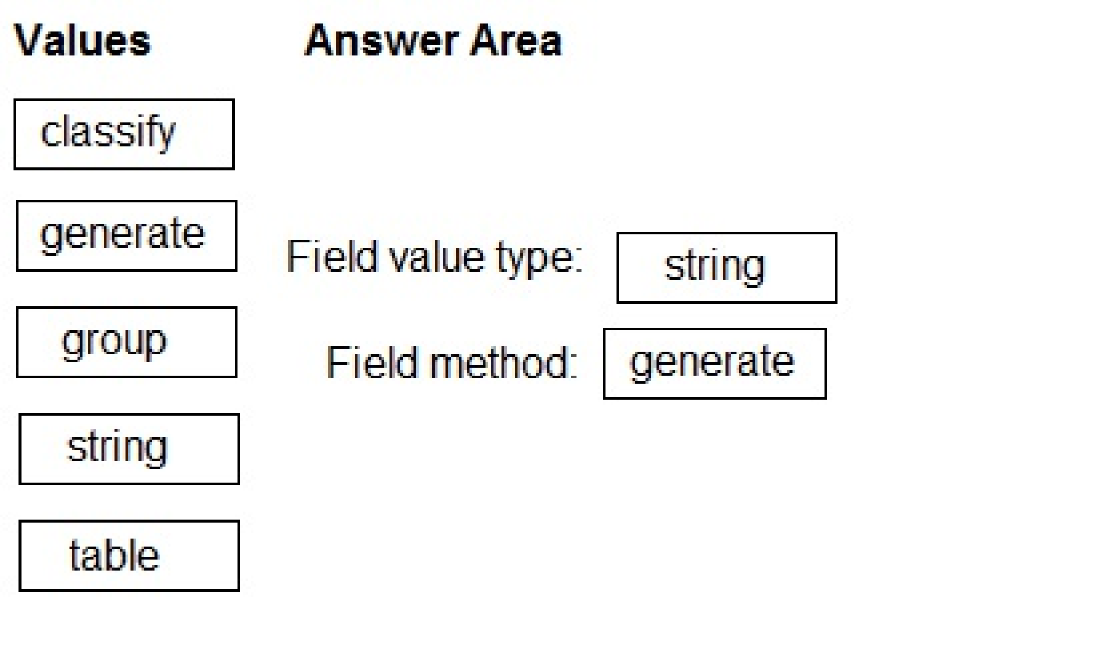
1. **Azure Cognitive Search - OCR Skill**: https://docs.microsoft.com/en-us/azure/cognitive-search/cognitive-search-skill-ocr
2. **Azure Cognitive Search - Skillset Concepts**: https://docs.microsoft.com/en-us/azure/cognitive-search/cognitive-search-concept-skillset

Question: 86 CertyIQ

DRAG DROP
-
You have a Microsoft Foundry project that uses Azure Content Understanding in Foundry Tools to analyze marketing videos.
Video segmentation is enabled.
You need to configure an analyzer to output a generated JSON field that describes the color scheme of each video segment.
How should you configure the analyzer? To answer, drag the appropriate values to the correct targets. Each value may be used once, more than once, or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each correct selection is worth one point.



Answer:



Explanation:

string.

Defines the data format as textual character data.

generate.

Uses an LLM to dynamically create or extract the contents.

Question: 87 CertyIQ

You have an invoice-processing application named App1 that uses Azure Constant Understanding in Foundry Tools.
You are building a new Content Understanding pipeline named Pipeline1 that must meet the following requirements:
•Compare an invoice to its related purchase order
•Validate the voice against static vendor contact documents
•Return a single structured output that includes discrepancy findings
You need to configure Pipeline1 and expose the pipeline as a single analyzer endpoint. What should you configure?

A. a single-file task in standard mode that uses the vendor contract provided as an additional document during analysis.

B. a single-file task in standard mode that uses confidence scores enabled for the extracted fields.

C. a multiple-file task in pro mode that uses the vendor contract files as reference data

D. a multi-file task in standard mode that uses the invoice and purchase order as input to the analyzer

Answer: C

Explanation:

Technical Justification for Correct Answer C

The correct configuration for Pipeline1, given the requirements, is **C. a multiple-file task in pro mode that uses the vendor contract files as reference data**. Here's a detailed breakdown of why C is the best choice and why others are less suitable:

Why C is the Correct Choice

Multiple-File Task: Since the pipeline needs to **compare an invoice to its related purchase order** and **validate the voice against static vendor contact documents**, a multiple-file task is essential to process more than one document type simultaneously.
Pro Mode: Enables advanced features necessary for complex comparisons and validations across different document types, aligning with the requirement for **discrepancy findings** in a structured output.
Vendor Contract Files as Reference Data: Utilizing vendor contracts as reference data in pro mode facilitates the **validation of the invoice against static vendor documents**, meeting the second requirement effectively.
Why Other Options Are Less Suitable

A. Single-file Task in Standard Mode with Vendor Contract as Additional Document

Inadequate for Multi-Document Comparison: Doesn't support comparing invoice and purchase order directly within the task.
Limited Capabilities in Standard Mode: Might not offer the depth of analysis needed for discrepancy findings across different documents.
B. Single-file Task in Standard Mode with Confidence Scores

Misses the Multi-Document Requirement: Fails to address the need to compare and validate across multiple documents (invoice, purchase order, and vendor contract).
Confidence Scores Alone Are Insufficient: While useful, they don't directly solve the comparison and validation requirements.
D. Multi-file Task in Standard Mode with Invoice and Purchase Order as Input

Lacks Pro Mode Capabilities for Advanced Validation: Standard mode may not provide the necessary advanced features for validating against static vendor contracts effectively.
Vendor Contracts Not Utilized as Reference: Ignores the static vendor documents, which are crucial for validation.
References

For deeper understanding of the capabilities and configurations of Azure Cognitive Services' Content Understanding pipelines, especially in the context of multi-file tasks and pro mode features:

1. **Azure Cognitive Search - Content Understanding**: https://learn.microsoft.com/en-us/azure/search/cognitive-search-concept-intro
2. **Configure a Cognitive Search Indexer for Content Understanding**: https://learn.microsoft.com/en-us/azure/search/cognitive-search-howto-content-understanding

Question: 88 CertyIQ

You have a Microsoft Foundry project that generates short promotional product videos.
After several clips are approved, reviewers notice a small watermark in the top-right corner of some videos.
You need to remove the watermark without regenerating the videos.
What should you do?

A. Modify the original prompt to exclude watermarks.
B. Crop the video by using the size parameter.
C. Increase the guidance scale.
D. Apply a mask-based inpainting edit to the affected part of the video.

Answer: D

Explanation:

Technical Justification for Correct Answer: D

The correct approach to remove the watermark without regenerating the videos is **D. Apply a mask-based inpainting edit to the affected part of the video**. Here's why this option is the best fit and why the others are less suitable:

D. Apply a mask-based inpainting edit to the affected part of the video:

Precision: This method allows for precise targeting of the watermark area (top-right corner) without altering the rest of the video.
Non-Destructive to Content: Inpainting, when done correctly, can seamlessly blend the area around the watermark with the rest of the video's content, making the removal nearly undetectable.
No Regeneration Required: Meets the key requirement of not regenerating the videos, as it's an edit applied post-production.

Why Other Options are Less Suitable:

A. Modify the original prompt to exclude watermarks:

Regeneration Required: This would necessitate regenerating the videos, which contradicts the problem's constraint.
Unnecessary Overhaul: If the only issue is the watermark, altering the prompt and regenerating could introduce unintended changes or delays.
B. Crop the video by using the size parameter:

Content Loss: Cropping to remove a watermark in the top-right corner would result in the loss of content around that area, potentially affecting the video's integrity or message.
Aesthetic Implication: Could disrupt the video's compositional balance.
C. Increase the guidance scale:

Irrelevant to Watermark Removal: The guidance scale influences the model's adherence to the prompt but does not directly address or remove existing watermarks in generated content.
Potential for Unwanted Changes: Could alter the video's content in unintended ways, as it influences the generation process rather than editing the output.

References

Microsoft Azure Video Indexer - Advanced Editing Capabilities (See sections on Custom Inpainting for similar concepts)
Azure Cognitive Services - Computer Vision for Image Editing Principles (Although focused on images, the principles of targeted editing like inpainting are relevant)

Question: 89 CertyIQ

You have a web app named App1 that sends requests to a multimodal chat model deployment in a Microsoft Foundry project.
User messages can contain both text and images.
Currently, App1 includes image URL: as plain text inside the message content so the model cannot recognize them as images.
Traces show that the requests contain a single text message instead of a multimodal content array.
You need to send the message as a structured array that includes both the text portion and the image reference to ensure that the model can process the image correctly.
What should you do?

A. Set the user message content array to include items that have type: text and type: image_url.

B. Encode the image to base64 and include the encoded data inside the content string of the user message.

C. Add the image URL to the request metadata section, so the model can resolve the processing issue automatically.

D. Place the image URL inside the System Message and set type to image_url so the model loads the image at initialization.

Answer: A

Explanation:
Technical Justification for Correct Answer A

Why A is the Best Option:

Structured Array for Multimodal Content: Option A suggests formatting the user message content as an array with items explicitly typed as text and image_url. This structured approach aligns with how multimodal models, especially those deployed in Microsoft Foundry projects, are designed to receive and process multiple types of input simultaneously. By clearly defining the type of each content piece, the model can correctly interpret and process both the text and the image referenced by the URL.

Direct Processing Compatibility: Sending the message in this structured format (array with typed items) is compatible with the expected input format for multimodal chat model deployments. This ensures that the model can directly process the image without requiring additional encoding, metadata parsing, or initialization adjustments.

Why Other Options are Less Suitable:

B. Base64 Encoding:

Inefficient for Large Images: Encoding images to base64 significantly increases the request payload size, which can lead to performance issues, especially with large images.
Model Compatibility: While some models can handle base64-encoded images within text fields, this approach does not leverage the model's multimodal input capabilities efficiently and may not be universally supported.

C. Adding Image URL to Request Metadata:

Non-Standard for Model Input: Request metadata is typically used for contextual or operational information, not for model input data. There's no guarantee the model will process metadata as part of the input.
Model Configuration Requirements: Would likely require custom model configuration to look for and process inputs from metadata, which is not a standard or recommended practice for multimodal input handling.

D. Placing Image URL in System Message:

Initialization vs. Runtime Input: System messages are often used for setup or initialization, not for dynamic user input. This approach would not effectively handle images provided at runtime as part of user messages.
Limited Flexibility: Does not provide a scalable solution for handling multiple or variably timed user inputs that include images.
Conclusion

Option A is the most technically sound approach as it leverages the model's capability to process multimodal inputs in a structured and efficient manner, ensuring both text and images are correctly interpreted without unnecessary encoding or reliance on non-standard input locations.

References

Microsoft Azure Cognitive Services: Multimodal Input Handling

Microsoft Foundry Projects: Structured Data Input for AI Models

Question: 90 CertyIQ

You are building a customer support web app named App1 in Microsoft Foundry that uses a GPT realtime model.
App1 must support:
•Live, low-latency voice conversations that use Azure OpenAI
•Streaming audio input from users and playback audio responses
You need to configure a connection method that supports real-time audio streaming in client application and targets approximately 100 ms latency.
Which connection method should you use?

A. RTMP
B. WebRTC
C. SIP
D. WebSocket

Answer: D

Explanation:

Technical Justification for Choosing Connection Method D. WebSocket

For App1, a customer support web app with stringent requirements of live, low-latency voice conversations using Azure OpenAI, streaming audio input, and playback with approximately 100 ms latency, the most suitable connection method is **D. WebSocket**. Here's why:

Low Latency Requirement (~100 ms): WebSocket is particularly well-suited for applications requiring real-time data transfer with minimal latency. It establishes a persistent, bi-directional communication channel between the client and server, enabling immediate push of audio responses from the server to the client, crucial for achieving the desired latency.
Real-Time Audio Streaming: WebSocket supports the continuous, two-way streaming of audio data (both input from the user and output from the Azure OpenAI model) efficiently. This is essential for the live voice conversations App1 aims to facilitate.
Integration with Azure Services: While not exclusive to WebSocket, its native support in most web frameworks and its compatibility with Azure's scalable and real-time capable infrastructure (e.g., Azure SignalR Service for managed WebSocket connections) make it a pragmatic choice for integrating with Azure OpenAI.

Why Other Options Are Less Suitable:

A. RTMP (Real-Time Messaging Protocol): Primarily designed for streaming to a large audience (one-to-many), RTMP is less ideal for low-latency, two-way communications. It also faces challenges with firewall and NAT traversal compared to WebSocket.
B. WebRTC (Web Real-Time Communication): Although WebRTC is designed for real-time communications and supports low latency, it is more commonly used for peer-to-peer connections (e.g., video conferencing) rather than client-server architectures like the one described for App1. Setting up and managing WebRTC for server-centric, scalable audio streaming might be more complex than necessary.
C. SIP (Session Initiation Protocol): SIP is a signaling protocol for setting up, modifying, and terminating real-time communication sessions. It does not handle the media streaming itself and is more suited for managing the session lifecycle in VoIP systems rather than direct, low-latency audio streaming in a web app context.

Conclusion: Given the specific requirements of App1 — low latency, real-time audio streaming, and integration with Azure OpenAI — WebSocket (D) is the most appropriate connection method due to its bi-directional, persistent, and low-latency capabilities, along with its ease of integration into the described architecture.

References

1. **Azure Documentation - Azure SignalR Service**: https://docs.microsoft.com/en-us/azure/azure-signalr/signalr-overview
2. **WebSocket Protocol Overview (W3C)**: https://www.w3.org/TR/websockets/

Question: 91 CertyIQ

You have a Microsoft Foundry project that contains an agent and uses a GitHub repository. The repository contains a YAM file named File1 that defines the evaluation settings of the agent.
You need to create a GitHub Actions workflow that runs the evaluation defined in File1 when a pull request (PR) is opened. How should you configure the workflow?

A. Set project-endpoint to the endpoint of the project.

B. Set evaluation-config to the path of the YAML file.

C. Set model-deployment-name to the deployed model.

D. Set tenant-id to the Microsoft Entra tenant ID

Answer: A

Explanation:

Technical Justification for Correct Option (A)

To configure a GitHub Actions workflow that runs the evaluation defined in File1.yam (note: likely a typo, assuming .yaml) when a pull request (PR) is opened for a Microsoft Foundry project, the correct approach involves specifying the project's endpoint. Here's why option A is the best choice and why others are less suitable:

Correct Option: A. Set project-endpoint to the endpoint of the project.

RATIONALE: Specifying the project-endpoint is crucial because it directs the GitHub Actions workflow to the correct Microsoft Foundry project. This endpoint is unique to your project and is necessary for authentication and authorization within the Azure environment.
FUNCTIONALITY: By setting the project-endpoint, the workflow can authenticate with your Microsoft Foundry project, access the File1.yaml (evaluation settings), and execute the evaluation as defined when a PR is opened.
RELEVANCE TO TRIGGER CONDITION: The project-endpoint setting is fundamental to triggering the correct evaluation workflow in response to a PR event, as it ensures the workflow operates on the intended project resources.

Why Other Options Are Less Suitable:

B. Set evaluation-config to the path of the YAML file.
INSUFFICIENT FOR TRIGGER SETUP: While specifying the evaluation-config path is necessary for identifying the evaluation settings, **it does not authenticate the workflow with the project**. This setting would be used in conjunction with, not instead of, the project-endpoint.
LACK OF PROJECT CONTEXT: Without the project endpoint, the workflow cannot determine which project's evaluation to run.

C. Set model-deployment-name to the deployed model.

IRRELEVANT TO EVALUATION TRIGGER SETUP: This setting pertains to the deployment aspect of a model, not the triggering of an evaluation based on a PR event. The evaluation might indirectly relate to a model, but this setting doesn't address the immediate need of running the evaluation defined in File1.yaml upon a PR.

TOO NARROW IN SCOPE: Focusing on the model deployment name overlooks the broader project context needed for the workflow trigger.

D. Set tenant-id to the Microsoft Entra tenant ID.

TOO BROAD IN SCOPE, INSUFFICIENT FOR PROJECT IDENTIFICATION: While the tenant-id is crucial for Azure authentication at a tenant level, **it does not specify which project within the tenant** should be targeted for the evaluation trigger.

SECURITY AND PERMISSION CONCERNS: Alone, it doesn't ensure the workflow has the necessary permissions for the specific project, as permissions can be more granular at the project level.

Configuration Summary for Correct Workflow Setup:

To correctly configure the GitHub Actions workflow:

MUST: Set project-endpoint to the endpoint of the project (Option A).

SHOULD (but not for the question's specific task): Additionally, in the workflow steps, specify evaluation-config to point to File1.yaml to ensure the correct evaluation settings are used.

References

[1] **Microsoft Documentation - Azure DevOps & GitHub Integration**: https://docs.microsoft.com/en-us/azure/devops/integrate/overview?view=azure-devops

[2] **GitHub Actions Workflow Syntax for Azure Services**: https://docs.github.com/en/actions/deploying-to-microsoft-azure/deploying-to-azure-app-service

Question: 92 CertyIQ

HOTSPOT

-

You have a Microsoft Foundry project that contains an agent.

The agent uses a stored access key to retrieve secrets from an Azure key vault, which violates a keyless-credentials requirement.

You need to ensure that the agent can retrieve the secrets. The solution must follow the principle of least privilege.

What should you configure? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Managed identity scope:
- Enable a system-assigned managed identity at the Foundry level.
- Enable a system-assigned managed identity at the project level.
- Create a service principal and store the principal's client secret.

Key Vault authorization method:
- Add an API key to application settings.
- Add a Key Vault access policy for the secrets.
- Assign the Key Vault Secrets User role to the managed identity.

Answer:

Answer Area

Managed identity scope:
- Enable a system-assigned managed identity at the Foundry level.
- Enable a system-assigned managed identity at the project level.
- Create a service principal and store the principal's client secret.

Key Vault authorization method:
- Add an API key to application settings.
- Add a Key Vault access policy for the secrets.
- Assign the Key Vault Secrets User role to the managed identity.

Explanation:

Managed identity scope:

Enable a system-assigned managed identity at the Foundry level.

In Azure AI Foundry, core dependencies (such as Azure Key Vault, Azure Storage, and Azure Container Registry) are connected and governed primarily at the hub/Foundry workspace level. Enabling a system-assigned managed identity at the Foundry level ensures that the entire collaborative environment can securely manage credentials, connections, and metadata using a single identity baseline.

Key Vault authorization method:

Assign the Key Vault Secrets User role to the managed identity.

Azure strongly recommends using Azure role-based access control (Azure RBAC) instead of legacy vault access policies for modern cloud resources. Assigning the **Key Vault Secrets User** built-in role to the managed identity enforces the principle of least privilege, allowing the platform to securely retrieve secrets without granting excessive administrative permissions.

Question: 93

CertyIQ

HOTSPOT

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You have a Microsoft Foundry project that contains a customer support application.
You create an evaluation named Run1 that has the following configurations:
•Includes risk and safety metrics
•Includes the protected material evaluation
•Includes harmful content metrics that use a medium severity threshold
You create an evaluation named Run2 that has the following configurations:
•Includes risk and safety metrics
•Includes the protected material evaluation
•Includes harmful content metrics that use a high severity threshold
You run both evaluations against a dataset named DB1 and receive the following results:
•Content harm defect rate of Run1: 12%
•Content harm defect rate of Run2: 4%
•Protected material evaluation of Run1: 6%
•Protected material evaluation of Run1: 6%
You start a fine-tuning job by using DB1. The job fails during automatic RAI checks for multiple content harm types.
You discover that the content filtering configuration is set to high severity.
For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
Changing the content filtering configuration to low severity will resolve the fine-tuning job issues.	○	○
The difference between the 12% and 4% content harm defect rate is consistent with the different severity thresholds used in Run1 and Run2.	○	○
The identical 6% protected material evaluation values across Run1 and Run2 indicate that this metric is unaffected by the change in the severity threshold.	○	○

Answer:

Answer Area

Statements	Yes	No
Changing the content filtering configuration to low severity will resolve the fine-tuning job issues.	○	$\boxed{○}$
The difference between the 12% and 4% content harm defect rate is consistent with the different severity thresholds used in Run1 and Run2.	$\boxed{○}$	○
The identical 6% protected material evaluation values across Run1 and Run2 indicate that this metric is unaffected by the change in the severity threshold.	$\boxed{○}$	○

Explanation:

Statement 1: Changing the content filtering configuration to low severity will resolve the fine-tuning job issues.
Answer: **No**
Explanation: Fine-tuning jobs can fail if the training dataset triggers Azure OpenAI's content safety filters. Setting a filter to a "low" severity threshold makes it **more restrictive** (blocking content even with low levels of sensitive terms), which would increase data rejections rather than resolving the job failure.
Statement 2: The difference between the 12% and 4% content harm defect rate is consistent with the different severity thresholds used in Run1 and Run2.
Answer: **Yes**
Explanation: Content harm metrics (such as violence, sexual content, or hate speech) evaluate responses against specific severity tier thresholds. Changing these thresholds directly alters the criteria for what is flagged as a "defect," which naturally accounts for a shift in defect rates (e.g., from 12% down to 4%).
Statement 3: The identical 6% protected material evaluation values across Run1 and Run2 indicate that this metric is unaffected by the change in the severity threshold.
Answer: **Yes**
Explanation: "Protected material" checks function as a binary text-matching detector for copyrighted or proprietary text/code. Because it operates independently of the category-based **content harm severity tiers**, changing the harm severity threshold has zero impact on this percentage.

Question: 94 **CertyIQ**

DRAG DROP

-

You have a Microsoft Foundry project that contains a multi-agent solution. The agents use tool calling to query internal systems.

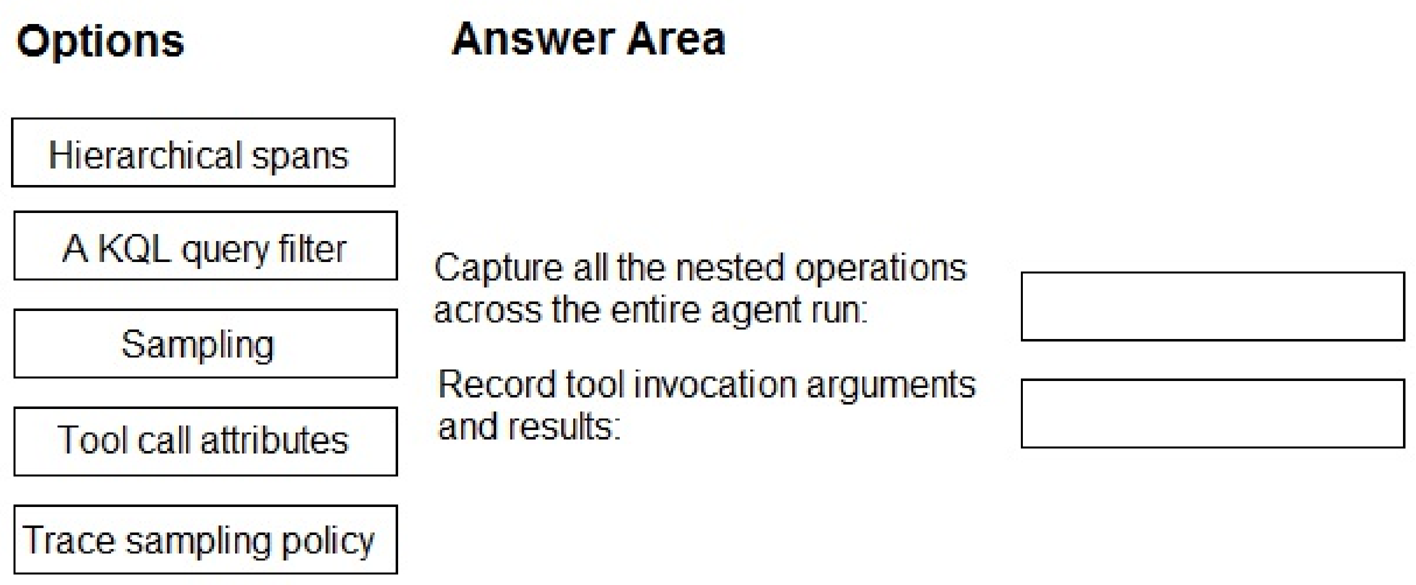
You need to implement responsible AI auditing to meet the following requirements:

•Capture all the nested operations across the entire agent run.

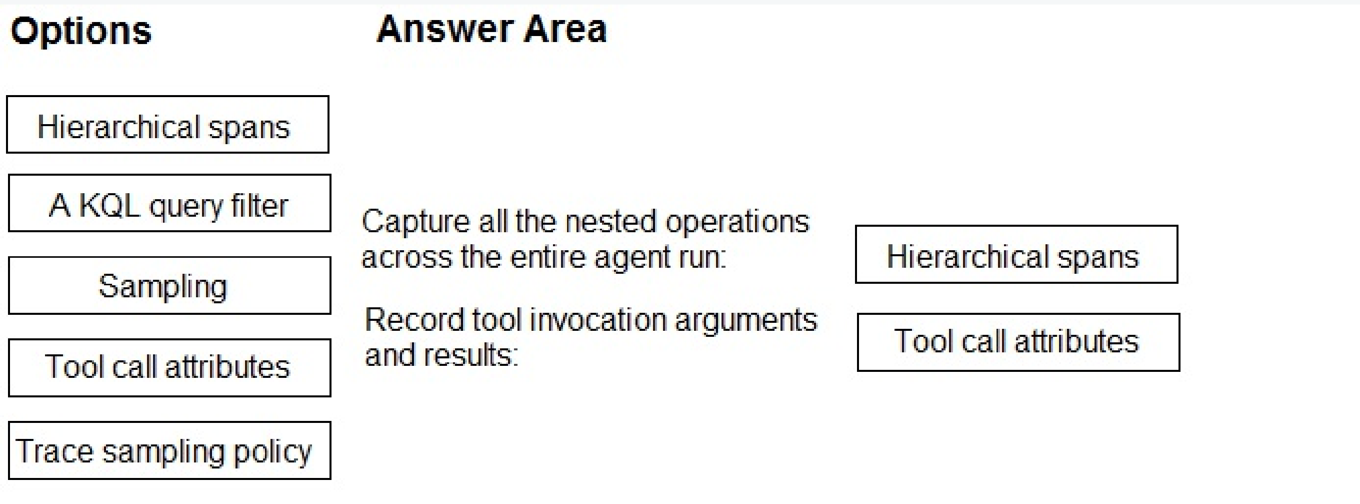
•Record tool invocation arguments and retuned results as metadata.

What should you use for each requirement? To answer, drag the appropriate options to the correct targets Each option may be used once, more than once, or not at all. You may need o dag the split bar between panes or scroll to view content.

NOTE: Each correct selection is worth one point.



Answer:



Explanation:

Hierarchical spans

In distributed tracing frameworks, an agent execution sequence involving nested steps (such as orchestration flows calling an agent, which then runs a planner or invokes tools) is modeled using parent-child tree elements known as **hierarchical spans**.

Tool call attributes

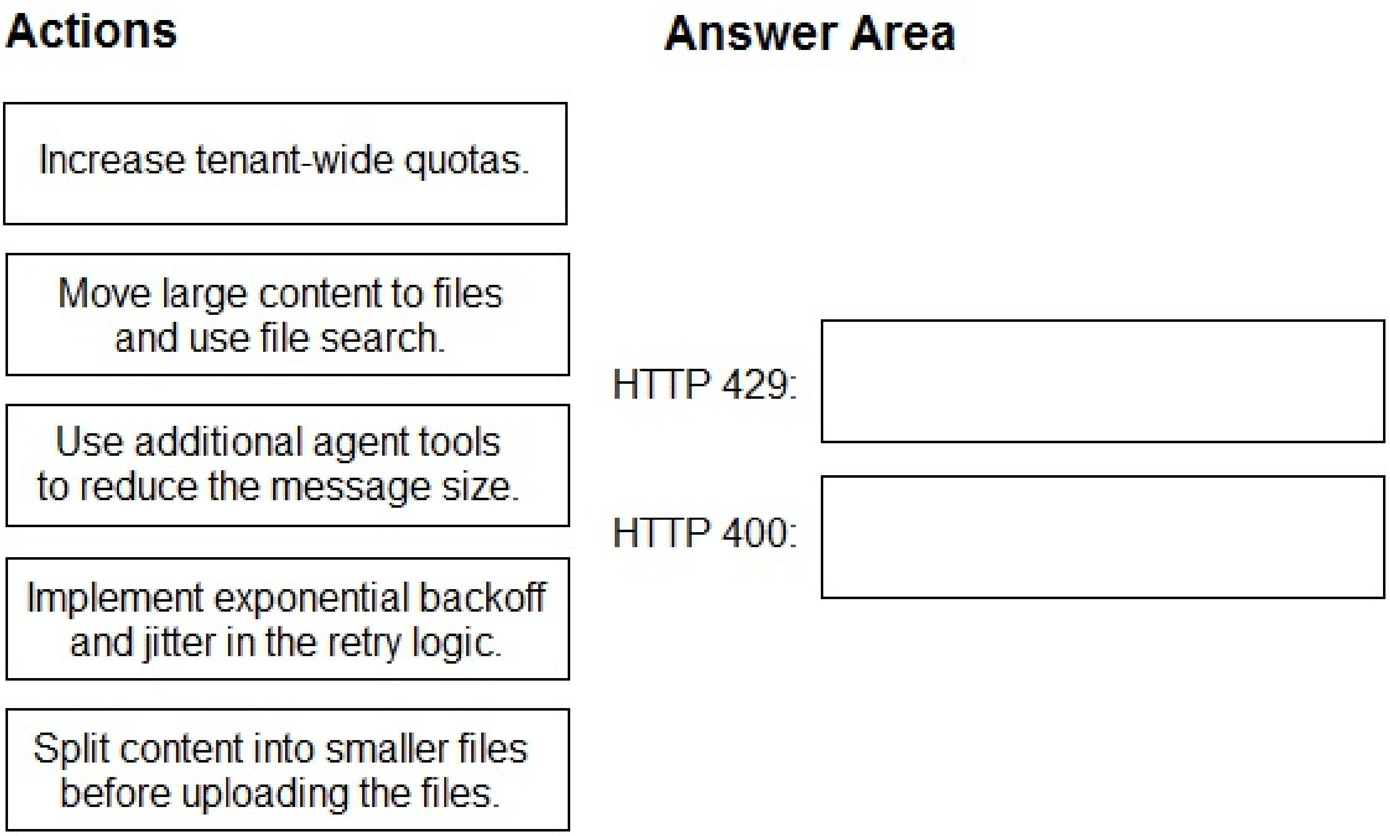
Specific operational parameters, inputs, outputs, and return metadata associated with an individual execution block are recorded structurally as key-value pairs called **attributes** (specifically **tool call attributes** in LLM tracing specifications).

Question: 95 CertyIQ

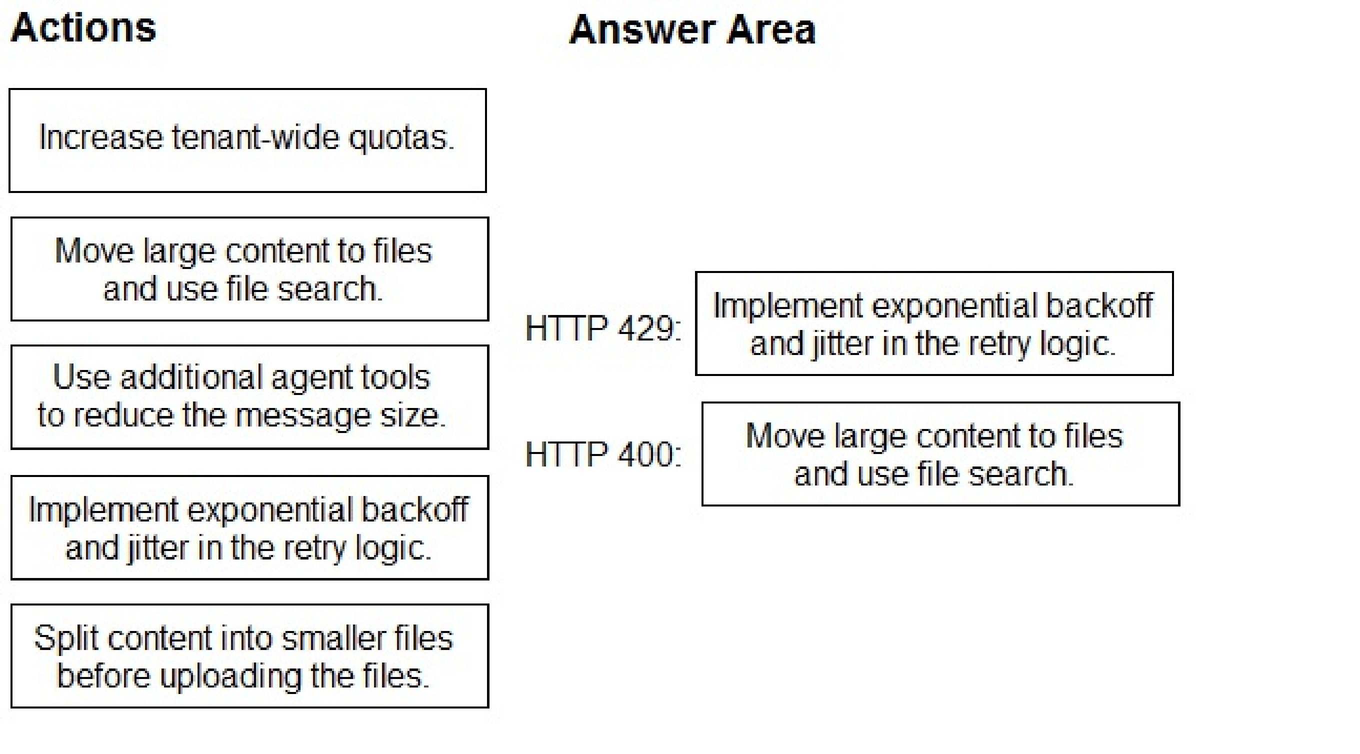
DRAG DROP

-

You have a Microsoft Foundry project that contains an agent. The agent uses threads and file uploads and calls an Azure OpenAI model deployment.
During load testing, calls intermittently fall and return an HTTP 429 rate limit exceeded error. Some user uploads fail and generate an HTTP 400 file size exceeded error.
You need to mitigate the errors and reduce call failures. The solution must remain within the service and model limits.
What should you do to resolve each error? To answer, drag the appropriate actions to the correct errors. Each action may be used once, more than once or not at all. You may need to drag the split bar between panes or scroll to view content.
NOTE: Each comet selection is worth one point.



Answer:



Explanation:

Implement exponential backoff and jitter in the retry logic.

An HTTP 429 status code indicates that you have exceeded your assigned Requests Per Minute (RPM) or Tokens Per Minute (TPM) limit. The standard programming practice to gracefully handle rate limiting is to retry the request using a randomized, expanding time window (backoff with jitter).

Move large content to files and use file search.

In the AI Agents API, sending massive blobs of raw text directly inside a prompt message will trigger an HTTP 400 error if it exceeds the maximum input payload or token context window. Moving that context into reference documents and using the built-in **file search tool** keeps the active API payload small and compliant.

Question: 96 CertyIQ

You have a Microsoft Foundry project.
You need to deploy a model from the model catalog to support a search solution for internal policy documents. The model must generate vector representations of the text in the documents and of user queries.
Which type of model should you use?

A. an embedding model

B. an image generation model

C. a large language model (LLM)

D. a small language model (SLM)

Answer: A

Explanation:

Technical Justification for Correct Answer: A. an embedding model

The correct answer for deploying a model to generate vector representations of text in documents and user queries for a search solution is **A. an embedding model**. Here's why:

Embedding Models are specifically designed to convert text (or other data types) into dense vector representations (embeddings) in a high-dimensional space, where semantically similar texts are closer together. This property is crucial for effective search solutions, as it enables the comparison of similarity between user queries and document texts at a granular level.
Why A is Best:

Directly addresses the requirement for generating vector representations of text.
Optimized for semantic similarity, which is essential for accurate search results.
Typically more lightweight and efficient for this specific task compared to larger language models.

Why Other Options are Less Suitable:

B. an Image Generation Model:

Irrelevant to Text Search: Designed for image creation, not text vectorization or search.
Misses the Requirement: Cannot generate text vector representations for search purposes.
C. a Large Language Model (LLM):

Overkill for the Task: While capable of text understanding, LLMs are computationally intensive and not optimized solely for generating vector representations for search.
Indirect Approach: Would require additional processing to extract relevant vector representations, adding unnecessary complexity.
D. a Small Language Model (SLM):

Similar to LLM but Smaller: Shares the indirect approach issue with LLMs, albeit with reduced computational requirements.
May Lack Sufficient Contextual Understanding: Depending on the model's training, it might not capture the nuanced similarities between texts as effectively as an embedding model.

References

[1] Microsoft Learn - Embedding Models for Text Search: https://learn.microsoft.com/en-us/azure/cognitive-search/concept-embedding-models
[2] Azure AI Documentation - Choosing the Right Model for Text Analysis: https://docs.microsoft.com/en-us/azure/machine-learning/how-to-choose-algorithms-text-analysis#embedding-models

Question: 97

CertyIQ

You have a Microsoft Foundry project.
You need to deploy a model from the model catalog to support real-time inference. The solution must meet the following requirements:
•Use key-based authentication.
•Support real-time REST API access.
•NOT consume the vCPU quota of the virtual machines in the Azure subscription.
Which type of deployment should you use?

A. serverless API
B. batch
C. self-hosted container
D. standard

Answer: D

Explanation:

Technical Justification for Choosing Deployment Type

Given the requirements for deploying a model from the Microsoft Foundery model catalog for real-time inference, the best deployment type is **D. standard**. Here's why each option aligns or fails to align with the specified requirements:

D. Standard

Use key-based authentication: Standard deployments support key-based authentication for securing REST API access, aligning with this requirement.
Support real-time REST API access: Designed for real-time inference, standard deployments provide the necessary infrastructure for immediate API responses.
NOT consume the vCPU quota of the virtual machines in the Azure subscription: Since standard deployments are managed and run on dedicated resources provisioned by Azure (not leveraging the user's existing VM quotas), this requirement is met.

Why Other Options are Less Suitable

A. Serverless API
Issue: While serverless options like Azure Functions can support real-time API access and key-based auth, they **may consume vCPU quota** indirectly through the underlying compute resources managed by Azure, though this is more about cost optimization than quota consumption. The primary concern here is the potential for quota impact if not properly managed, making it less straightforward for the "NOT consume vCPU quota" requirement.
Note: Serverless might still be a good fit in many scenarios but doesn't perfectly align with the "vCPU quota" requirement as directly as Standard does.
B. Batch
Issue: Batch processing is **not suited for real-time inference** as it's designed for scheduled or triggered bulk processing, failing the real-time REST API requirement.
C. Self-hosted Container
Issue:
vCPU Quota: **Will consume the vCPU quota** of the virtual machines in the Azure subscription since you manage the infrastructure.
Management Overhead: Implies more management responsibility for the user, which isn't necessarily an issue but doesn't provide an advantage over "Standard" in this context.

ConclusionGiven the constraints, **D. Standard** is the most appropriate choice because it directly addresses all requirements without the ambiguities or mismatches present in the other options.

References

1. **Azure Machine Learning - Deploy Model**: https://docs.microsoft.com/en-us/azure/machine-learning/how-to-deploy-and-where?tabs=python
2. **Azure Resource Quotas and Limits**: https://docs.microsoft.com/en-us/azure/azure-resource-manager/management/azure-subscription-service-limits#virtual-machines-limits

Question: 98 CertyIQ

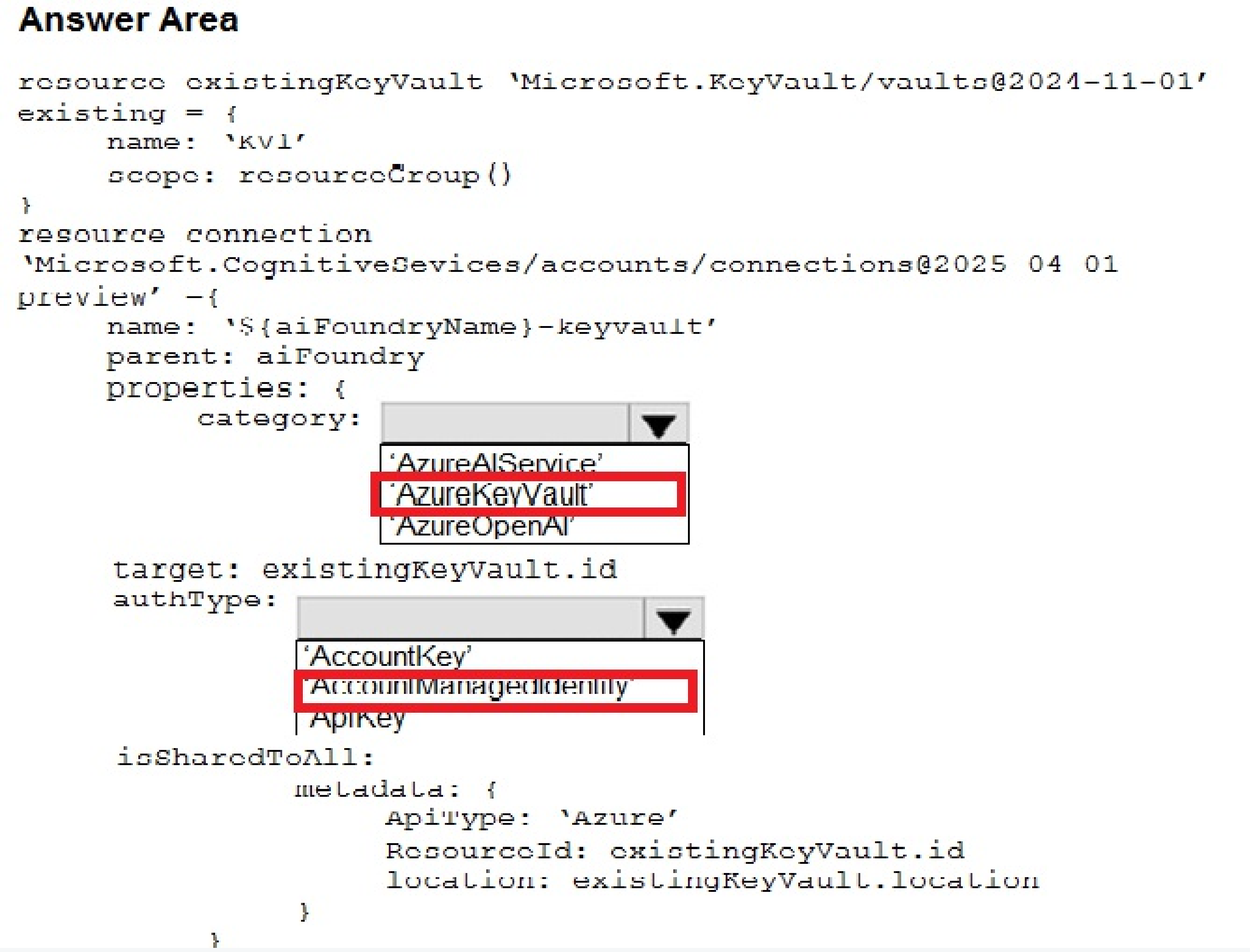
HOTSPOT
-
You plan to create a Microsoft Foundry project named Project1 that will contain an agent and use an Azure key

vault named KV1.
You need to configure a connection from Project1 to KV1.
How should you complete the Bicep code? To answer, select the appropriate options in the answer area?
NOTE: Each correct selection is worth one point.

Answer Area

```
resource existingKeyVault 'Microsoft.KeyVault/vaults@2024-11-01'
existing = {
     name: 'KV1'
     scope: resourceGroup()
}
resource connection
'Microsoft.CognitiveSevices/accounts/connections@2025-04-01-
preview' ={
     name: '${aiFoundryName}-keyvault'
     parent: aiFoundry
     properties: {
          category: [dropdown]
                    'AzureAIService'
                    'AzureKeyVault'
                    'AzureOpenAI'
      target: existingKeyVault.id
      authType: [dropdown]
                'AccountKey'
                'AccountManagedIdentity'
                'ApiKey'
      isSharedToAll:
                metadata: {
                     ApiType: 'Azure'
                     ResourceId: existingKeyVault.id
                     location: existingKeyVault.location
                }
          }
      }
```

Answer:



Explanation:

AzureKeyVault'

The Bicep block targets existingKeyVault.id and establishes a secure link to an Azure Key Vault workspace backend. In the Azure AI Hub/Foundry connection schema, the category property must match the resource class being integrated, which is 'AzureKeyVault'.

AccountManagedIdentity'

Azure Key Vault relies on identity-based authentication mechanisms (such as Azure RBAC or vault access policies) via system-assigned or user-assigned identities. It does not utilize symmetric account keys or API keys for direct workspace connections, meaning 'AccountManagedIdentity' is the structurally valid authorization protocol.

Question: 99 CertyIQ

HOTSPOT

-

You have a Microsoft Foundry project.
You need to create a customer support agent that meets the following requirements:
•Grounds responses only in company policy documents stored in curated repositories
•Retains customer preferences across separate chat sessions
How should you configure the agent? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

Knowledge grounding:

- Configure revival from approved data sources.
- Upload the policy documents direct to the agent.
- Embed the policy documents directly into the agent instructions.

Memory:

- Use orchestration-managed session context.
- Enable agent memory that uses persistent storage.
- Retain user preferences in the state of the client application

Answer:

Answer Area

Knowledge grounding:

- Configure revival from approved data sources.
- Upload the policy documents direct to the agent.
- Embed the policy documents directly into the agent instructions.

Memory:

- Use orchestration-managed session context.
- Enable agent memory that uses persistent storage.
- Retain user preferences in the state of the client application

Explanation:

Configure retrieval from approved data sources.

In enterprise AI applications, **Knowledge Grounding** is best achieved via Retrieval-Augmented Generation (RAG) architecture by connecting your model or agent to an enterprise search index or specific storage accounts (approved data sources). This prevents hallucination without risking token context limitations or model instruction degradation that occurs when trying to upload or embed massive raw policy documents directly into the agent's prompt instructions.

Enable agent memory that uses persistent storage.

To implement true agentic long-term memory that preserves user history, preferences, and state context across multiple independent conversational sessions, you should configure **persistent storage-backed agent memory** (such as Azure AI Projects' MemorySearchTool mapped to a persistent store resource).

Question: 100 CertyIQ

You have a Microsoft Foundry project that contains an agent. The agent generates summaries from retrieved policy documents.
You need to improve response completeness. The solution must be implemented in the logic of the application code before responses are returned.
What should you do?

A. Add a retry evaluation before the responses are returned.

B. Decrease the value of the temperature parameter.

C. Increase the value of the presence_penalty parameter

D. Replace the model with a smaller deployment.

Answer: A

Explanation:

Technical Justification for Correct Answer: A

To enhance response completeness for the Microsoft Foundry agent that generates summaries from retrieved policy documents, the most effective solution among the provided options is to **Add a retry evaluation before the responses are returned (Option A)**. Here's why:

Option A: Add a retry evaluation before the responses are returned is the best choice because it directly addresses the issue of response completeness by allowing the system to re-attempt the generation of a summary if the initial response does not meet a predefined threshold of completeness. This can be implemented in the application logic to catch and retry incomplete responses before they are returned to the user, thus improving overall completeness.

Option B: Decrease the value of the temperature parameter would actually have the opposite effect of what's intended for improving completeness. The temperature parameter in AI models controls randomness and creativity; lowering it increases the model's tendency to produce the most probable (but not necessarily more complete) responses, potentially leading to less diverse and possibly less complete outputs if the model is overly constrained.

Option C: Increase the value of the presence_penalty parameter is relevant in the context of controlling repetition in responses. While useful for readability, increasing this parameter does not directly contribute to ensuring the completeness of the response regarding the coverage of the policy document's content.

Option D: Replace the model with a smaller deployment would likely worsen response completeness. Smaller models typically have less capacity to understand and generate comprehensive text, leading to potentially more incomplete summaries compared to larger, more capable models.

Why A is Best:

Direct Addressal: A directly addresses the need for more complete responses by allowing for retries.
Implementation Feasibility: Can be implemented in the application logic without altering the model's configuration or size.
Targeted Improvement: Specifically aims at improving completeness without unnecessarily affecting other aspects of response quality.

References

For deeper understanding of the concepts mentioned:

1. **Microsoft Azure Documentation - Retry Policies in Azure**: https://learn.microsoft.com/en-

us/azure/architecture/patterns/retry-with-exponential-backoff
2. **Understanding AI Model Parameters (Temperature, etc.)**: https://huggingface.co/blog/how-to-fine-tune (Refer to the sections on model parameters for insight into how temperature and penalty parameters function)

Question: 101 CertyIQ

HOTSPOT

-

You develop a test method to verify the results retrieved from a call to the Azure Vision in Foundry Tools API. The call is used to analyze the existence of company logos in images. The call returns a collection of brands named brands.

You have the following code segment:

```
for brand in image_analysis.brands:
    if brand.confidence >= 0.75:
        print(f"\nLogo of {brand.name} between
{brand.rectangle.x}, {brand.rectangle.y} and {brand.rectangle.w},
{brand.rectangle.h}")
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.

NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
The code will display the name of each detected brand with a confidence equal to or higher than 75 percent.	○	○
The code will display coordinates for the top-left corner of the rectangle that contains the brand logo of the displayed brands.	○	○
The code will display coordinates for the bottom-right corner of the rectangle that contains the brand logo of the displayed brands.	○	○

Answer:

Answer Area

Statements	Yes	No
The code will display the name of each detected brand with a confidence equal to or higher than 75 percent.	○	○
The code will display coordinates for the top-left corner of the rectangle that contains the brand logo of the displayed brands.	○	○
The code will display coordinates for the bottom-right corner of the rectangle that contains the brand logo of the displayed brands.	○	○

Explanation:

Statement 1 (Yes): The code filters results using a conditional check (brand.Confidence >= 0.75), meaning only brands meeting or exceeding 75% confidence are displayed.

Statement 2 (Yes): The bounding box properties X and Y provided by the Azure Computer Vision API directly represent the **top-left corner** horizontal and vertical coordinates of the bounding box.

Statement 3 (No): The API provides width (W) and height (H) rather than the coordinates of the bottom-right corner. To obtain the bottom-right coordinates, one would have to calculate them via X + W and Y + H.

Question: 102 CertyIQ

You have an Azure subscription.
You plan to build an app that will use the Azure AI DALL-E model.
You need to deploy the model.
What should you use?

A. the Azure SDK for Python and PowerShell cmdlets.
B. the Azure SDK for JavaScript and Azure Machine Learning Studio.
C. Microsoft Foundry and the Azure Command Line Interface (CLI)
D. the Azure portal and Microsoft Graph API

Answer: C

Explanation:

Justification for Correct Answer: C

The correct option for deploying the Azure AI DALL-E model is **C. Microsoft Foundry and the Azure Command Line Interface (CLI)**. Here's why:

Microsoft Foundry is specifically designed for building, deploying, and managing AI and machine learning models at scale on Azure. It provides a streamlined workflow for model deployment, which aligns perfectly with the requirement of deploying the DALL-E model. Foundry handles the complexities of model serving, scaling, and monitoring, making it the most suitable choice for this task.

Azure Command Line Interface (CLI) complements Foundry by offering a flexible, scriptable way to manage and automate the deployment process across Azure services. It's particularly useful for configuring the environment, setting up resources (e.g., compute targets), and integrating the deployed model with other Azure services, all of which are crucial steps in deploying a model like DALL-E.

Why Other Options are Less Suitable:

A. The Azure SDK for Python and PowerShell cmdlets: While powerful for managing Azure resources and possible for deployment, this combination is more oriented towards general Azure resource management and custom scripting, not specifically optimized for the streamlined deployment of AI models like DALL-E. Python SDK could be used in conjunction with other services for model training or custom integration but lacks the direct, model-deployment focused capabilities of Microsoft Foundry.

B. The Azure SDK for JavaScript and Azure Machine Learning Studio: Azure Machine Learning Studio is indeed useful for the lifecycle of machine learning models, including deployment. However, for specifically leveraging the DALL-E model (which implies a need for a more specialized, possibly pre-configured environment for immediate deployment), Microsoft Foundry offers a more direct path. The JavaScript SDK, meanwhile, is more suited for web application integrations rather than the deployment process itself.

D. The Azure portal and Microsoft Graph API: The Azure portal provides a GUI for deploying models through certain Azure services but may not offer the same level of automation, scalability, and model-specific deployment features as Microsoft Foundry for AI models. Microsoft Graph API is primarily focused on accessing Microsoft 365 data and services, not AI model deployment on Azure.

References

Microsoft Foundry Documentation
Azure Command Line Interface (CLI) Documentation

Question: 103 CertyIQ

HOTSPOT
-
You have an Azure subscription.
You need to create a new resource that will generate fictional stores in response to user prompts. The solution must ensure that the resource uses a customer-managed key to protect data.
How should you complete the script? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
az cognitiveservices account create -n myresource -g
myResourceGroup --kind [          ▼] --sku S -l WestEurope
```

- AIServices
- LanguageAuthoring
- OpenAI

```
[          ▼] '{
```

- --api-properties
- --assign-identity
- --encryption

```
"keySource": "Microsoft.KeyVault",
"keyVaultProperties": {
     "keyName": "KeyName",
     "keyVersion": "secretVersion",
     "keyVaultUri": "https://issue23056kv.vault.azure.net/"
     }
   }'
```

Answer:

Answer Area

```
az cognitiveservices account create -n myresource -g
myResourceGroup --kind [          ▼] --sku S -l WestEurope
```

- AIServices
- LanguageAuthoring
- **OpenAI**

```
[          ▼] '{
```

- --api-properties
- --assign-identity
- **--encryption**

```
"keySource": "Microsoft.KeyVault",
"keyVaultProperties": {
     "keyName": "KeyName",
     "keyVersion": "secretVersion",
     "keyVaultUri": "https://issue23056kv.vault.azure.net/"
     }
   }'
```

Explanation:

1. **OpenAI**: The question specifically configures an OpenAI model resource container. Choosing AIServices or LanguageAuthoring would create incorrect resource kinds.
2. **--encryption**: This parameter accepts the JSON-formatted key configuration string specifying keySource as Microsoft.KeyVault and the parameters mapping out the target encryption key.

Question: 104 CertyIQ

HOTSPOT

-

You have a Python application collects customer comments before posting them to a public forum.
You need to send a text comment to Azure AI Content Safety and return the self-harm severity from the response.
How should you complete the code? To answer, select the appropriate options in the answer area.
NOTE: Each correct selection is worth one point.

Answer Area

```
def get_self_harm_severity (comment: str) -> int:
     key = os.environ["CONTENT_SAFETY_KEY"]
     endpoint = os.environ
["CONTENT_SAFETY_ENDPOINT"]
     client = ContentSafetyClient(endpoint,
AzureKeyCredential(key))
     request = [dropdown]
```

- AnalyzeTextOptions(categories=comment)
- AnalyzeTextOptions(text=[comment])
- AnalyzeTextOptions(text=comment)
- TextCategory.SELF_HARM(comment)

```
     response = [dropdown]
```

- client.analyze_image(request)
- client.analyze_text(request)
- client.moderate_text(request)
- client.path("/text:analyze").post(request)

```
     result = next(
             item for item in response.categories_analysis
             if item.category == TextCategory.SELF_HARM

        )
        return result.severity
```

Answer:

Answer Area

```
def get_self_harm_severity (comment: str) -> int:
      key = os.environ["CONTENT_SAFETY_KEY"]
      endpoint = os.environ
["CONTENT_SAFETY_ENDPOINT"]
      client = ContentSafetyClient(endpoint,
AzureKeyCredential(key))
      request = 
```

- AnalyzeTextOptions(categories=comment)
- AnalyzeTextOptions(text=[comment])
- AnalyzeTextOptions(text=comment)
- TextCategory.SELF_HARM(comment)

```
      response = 
```

- client.analyze_image(request)
- client.analyze_text(request)
- client.moderate_text(request)
- client.path("/text:analyze").post(request)

```
      result = next(
            item for item in response.categories_analysis
            if item.category == TextCategory.SELF_HARM

        )
        return result.severity
```

Explanation:

1. First Drop-Down Selection (request =)

Correct Value: AnalyzeTextOptions(text=comment)
Explanation: The AnalyzeTextOptions class takes a named parameter text which expects a single string value containing the input text content to be evaluated.

2. Second Drop-Down Selection (response =)

Correct Value: client.analyze_text(request)
Explanation: The synchronous method to submit a text moderation analysis request via a configured ContentSafetyClient instance is explicitly named analyze_text.

Question: 105 **CertyIQ**

You have a custom named entity recognition (NER) project in Azure Language in Foundry Tools for support tickets.
The schema for the project contains an entity type named ContactInfo.
In tagged training files, ContactInfo is used for phone numbers, email addresses, and social media handles.
Model evaluation shows low precision for ContactInfo, including false positives in which nearby text is extracted as ContactInfo.
You need to improve the precision of the project.
What should you do before retraining the model?

A. Lower the confidence threshold for ContactInfo.

B. Trigger an auto-labeling job.

C. Add more support tickets as training data and label more ContactInfo entities.

D. Replace ContactInfo by using Phone, Email, and SocialMedia entities. Relabel every matching span.

Answer: D

Explanation:

Technical Justification for Correct Answer D

Correct Answer: D. Replace ContactInfo by using Phone, Email, and SocialMedia entities. Relabel every matching span.

Why D is the Best Option: Replacing the broad ContactInfo entity with more specific entities (Phone, Email, and SocialMedia) directly addresses the issue of low precision and false positives. By differentiating between these types, the model can learn more distinct features for each, reducing the likelihood of incorrectly extracting nearby text as ContactInfo. Relabeling ensures the training data accuracy, which is crucial for improving model precision.

Why Other Options are Less Suitable:

A. Lower the confidence threshold for ContactInfo: Decreasing the confidence threshold would actually **increase** the number of false positives, as the model would be more likely to predict ContactInfo for less confident matches. This contradicts the goal of improving precision.
B. Trigger an auto-labeling job: While auto-labeling can save time, without first addressing the schema's ambiguity (by splitting ContactInfo), auto-labeling would likely perpetuate the existing inaccuracies at scale, making subsequent retraining less effective.
C. Add more support tickets as training data and label more ContactInfo entities: Although increasing the size and diversity of the training dataset can generally improve model performance, the underlying issue of ContactInfo's broad definition would remain. This approach might slightly improve precision due to the law of large numbers but would not fundamentally solve the precision problem without redefining the entity for clearer differentiation.

Key Technical Points for Improvement:

Entity Granularity: More granular entities improve model precision by reducing ambiguity.
Data Quality Over Quantity (Initially): Ensuring the labeled data's accuracy is more critical than increasing its volume when faced with a precision issue stemming from entity design.
Retraining After Data Refinement: Retraining after refining the entity schema and relabeling ensures the model learns from accurately defined data, maximizing the likelihood of improved precision.

References

Microsoft Azure Documentation: Understand entities in Azure Language Understanding
Microsoft Azure Blog: Best Practices for Named Entity Recognition (NER) in Azure Language Service

Question: 106 CertyIQ

You are building a text-to-speech solution that uses Azure Speech in Foundry Tools to read instructions from the script in a text file.
You discover that the solution often pronounces technical terms incorrectly.
You need to prevent the incorrect pronunciations. The solution must minimize development effort.
What should you do?

A. From Speech Studio, train a custom neural voice
B. Use Speech Synthesis Markup Language (SSML) to specify phonemes.
C. Use Speech Synthesis Markup Language (SSML) to apply say as rules.
D. Use Speech Synthesis Markup Language (SSML) to adjust the prosody of the voice.
E. From Azure OpenAI use the Whisper model.

Answer: B

Explanation:

Technical Justification for Correct Answer: B

Correct Answer: B. Use Speech Synthesis Markup Language (SSML) to specify phonemes.

Why B is the Best Option: Specifying phonemes using SSML directly addresses the issue of incorrect pronunciations of technical terms by providing explicit pronunciation guidance for the text-to-speech engine. This approach minimizes development effort as it doesn't require training new models (like in A) or leveraging entirely different services (like in E). Instead, it involves adding specific markup to the text, which is a relatively lightweight development task. SSML is natively supported by Azure Speech Services, ensuring seamless integration.

Why Other Options are Less Suitable:

A. Train a custom neural voice: While effective for broad, consistent voice customization or for voices not currently offered, training a custom neural voice is resource-intensive (both in terms of development effort and potentially cost) for a problem as specific as correcting the pronunciation of certain technical terms. This option does not minimize development effort as required.
C. Use SSML to apply say-as rules: Say-as rules in SSML are useful for indicating the interpretation of certain words or phrases (e.g., dates, times, abbreviations) but may not offer the fine-grained control over pronunciation that specifying phonemes does for technical terms.
D. Use SSML to adjust the prosody of the voice: Adjusting prosody (the rhythm, stress, and intonation of speech) enhances the natural flow or emotional content of synthesized speech but does not correct the pronunciation of specific words.
E. From Azure OpenAI use the Whisper model: The Whisper model is primarily a speech-to-text model, not designed for text-to-speech tasks or correcting pronunciation issues in synthesized speech. This option is completely off-target for the problem at hand.

Conclusion: Given the need to minimize development effort while correcting the pronunciation of technical terms in a text-to-speech solution using Azure Speech, specifying phonemes with SSML is the most direct, efficient, and effective approach.

References:

Microsoft Documentation: Customize speech synthesis with SSML
Microsoft Documentation: Phonemes in SSML for Azure Speech Services

Question: 107 CertyIQ

HOTSPOT
-
You have a Python application that redacts sensitive information before sending prompt text to a language model. The application has the following code:

```
sample_text = "Contact John Doe at 312-555-1234 or
john.doe@contoso.com. His SSN is 859-98-0987."

def redact_for_model(input_text):
    payload = {
        "kind": "PiiEntityRecognition",
        "parameters": {
            "modelVersion": "latest",
            "piiCategories": ["Person", "PhoneNumber"],
            "redactionPolicies": [
                {
                    "policyKind": "entityMask"
                }
            ]
        },
        "analysisInput": {
            "documents": [
            {
                "id": "1",
                "language": "en",
                "text": input_text
                }
            ]
        }
    }
…
    return {
        "text for model": doc.get("redactedText", ""),
        "audit": [
            (entity["text"], entity["category"], entity
["confidenceScore"])
            for entity in doc["entities"]
        ]
    }
result = redact_for_model(sample_text)
```

For each of the following statements, select Yes if the statement is true. Otherwise, select No.
NOTE: Each correct selection is worth one point.

Answer Area

Statements	Yes	No
For `sample_text, audit` will include entity records for Contact and SSN.	○	○
For `sample_text, text_for_model` will include john.doe@contoso.com and 859-98-0987.	○	○
For `sample_text, text_for_model` will contain entity type masks for John Doe and 312-555-1234.	○	○

Answer:

Answer Area

Statements	Yes	No
For `sample_text, audit` will include entity records for Contact and SSN.	○	○
For `sample_text, text_for_model` will include john.doe@contoso.com and 859-98-0987.	○	○
For `sample_text, text_for_model` will contain entity type masks for John Doe and 312-555-1234.	○	○

Explanation:

1. **Statement 1 (No)**: The Azure PII recognition service uses highly specific, predefined entity categories (such as Person, Email, PhoneNumber, and SSN). There is no overarching single category called Contact.
2. **Statement 2 (No)**: The purpose of creating a sanitized string (like text_for_model) through PII redaction is to remove or mask sensitive information. Consequently, raw values like the email address (john.doe@contoso.com) and the SSN (859-98-0987) will be stripped out and will not appear in the final text.
3. **Statement 3 (Yes)**: When executing PII redaction with an entity-masking configuration, the sensitive elements (such as the name "John Doe" and the telephone number "312-55-1234") are replaced by their corresponding type labels (e.g., [Person] and [PhoneNumber]).

Question: 108 CertyIQ

DRAG DROP

-

You are developing an application that will detect faulty components produced on a factory production line. The components are specific to your business.
You need to use the Azure Custom Vision API to help detect common faults.
Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.

Actions

- Initialize the training dataset.
- Train the classifier model.
- Create a project.
- Upload and tag images.
- Train the object detection model.

Answer Area

1
2
3

Answer:

Actions

- Initialize the training dataset.
- Train the classifier model.
- Create a project.
- Upload and tag images.
- Train the object detection model.

Answer Area

1 Create a project.
2 Upload and tag images.
3 Train the classifier model.

Explanation:

Step 1: Create a project.

Before any other actions can be taken, you must configure a new project container within the Custom Vision portal and choose the project type (e.g., Image Classification).

Step 2: Upload and tag images.

You need to provide the dataset for training by uploading the required set of target images and assigning labels (tags) to define the classes.

Step 3: Train the classifier model.

Once your images are tagged, you initiate model training using either Fast Training or Advanced Training to construct your custom classifier.

Question: 109 **CertyIQ**

You are developing a new sales system that will process user-generated video and text from a public-facing website.
You plan to notify users that their data has been processed by the sales system.
Which responsible AI principle does this help meet?

A. fairness
B. transparency
C. inclusiveness
D. reliability and safety

Answer: B

Explanation:

Technical Justification for Correct Answer: B. Transparency

Notifying users that their data has been processed by the sales system directly aligns with the principle of **transparency** in responsible AI practices. Transparency involves clearly communicating to users how their data is collected, used, and processed. By informing users about the processing of their video and text data, the system ensures that individuals are aware of what happens to their input, fostering trust and allowing them to make informed decisions about their data sharing.

Why Other Options are Less Suitable:

A. Fairness: While fairness ensures that AI systems do not discriminate against certain groups, notifying users about data processing does not directly address fairness in decision-making or outcomes. Fairness would be more relevant if the question discussed ensuring equitable treatment across different user groups in the sales system's processing.
C. Inclusiveness: Inclusiveness pertains to designing systems that are accessible and beneficial to diverse populations. Notifying users about data processing is a universal practice and does not specifically enhance or address the inclusivity of the system for diverse user groups.
D. Reliability and Safety: Reliability and safety focus on the system's performance, accuracy, and potential risks. While crucial, notifying users about data processing is more about ethical communication than ensuring the system's operational reliability or safety aspects.

References

[1] Microsoft - Responsible AI Principles: https://www.microsoft.com/en-us/trust-center/ai/responsible-ai/principles
[2] Azure - Transparency in AI and Machine Learning: https://azure.microsoft.com/en-gb/resources/transparent-ai-and-machine-learning/

Question: 110 **CertyIQ**

You are designing a content management system.
You need to ensure that the reading experience is optimized for users who have reduced comprehension and

learning differences, such as dyslexia. The solution must minimize development effort.
Which Azure service should you include in the solution?

A. Azure Document Intelligence in Foundry Tools
B. Azure Language in Foundry Tools
C. Azure AI Immersive Reader
D. Azure Translator in Foundry Tools

Answer: C

Explanation:

Technical Justification for Correct Answer: C (Azure AI Immersive Reader)

Why Azure AI Immersive Reader (C) is the Best Option:

Optimized Reading Experience for Dyslexia and Learning Differences: Azure AI Immersive Reader is specifically designed to improve the reading experience for users with dyslexia, learning differences, and second-language learners. It offers features like font size adjustment, line spacing control, reading aloud with natural speech, translation, and highlighting of important words or phrases, directly addressing the requirement.
Minimized Development Effort: Immersive Reader provides a pre-built, configurable widget that can be easily integrated into web applications with minimal code, thus minimizing development effort. This aligns perfectly with the need to reduce development overhead.

Why Other Options are Less Suitable:

A. Azure Document Intelligence in Foundry Tools:

Purpose: Primarily designed for extracting insights and structured data from unstructured documents, not for enhancing reading experiences.
Development Effort: Would require significant customization to meet the reading experience requirements, increasing development effort.
B. Azure Language in Foundry Tools:

Purpose: Focuses on advanced natural language understanding and generation capabilities, not specifically on reading accessibility features.
Development Effort: Integrating the necessary reading experience features would demand substantial development work.
D. Azure Translator in Foundry Tools:

Purpose: Mainly for translating text in real-time, with some reading aloud capabilities but lacks the comprehensive accessibility features for dyslexia and learning differences.
Development Effort/Feature Alignment: While easier to integrate than A or B for basic reading aloud, it doesn't fully address the optimized reading experience for the specified user group without additional development.

References

Azure AI Immersive Reader Documentation
Comparing Azure Cognitive Services for Text and Language

Question: 111 CertyIQ

You have an Azure subscription that contains an Azure App Service app named App1.
You provision a Microsoft Foundry Service resource named CSAccount1.
You need to configure App1 to access CSAccount1. The solution must minimize administrative effort.
What should you use to configure App1?

A. the endpoint URI and subscription key

B. the endpoint URI and an OAuth token

C. the endpoint URI and a shared access signature (SAS) token

D. a system assigned managed identity and an X.509 certificate

Answer: A

Explanation:

Justification for Correct Answer (A) and Analysis of Other Options

To configure App1 (an Azure App Service app) to access CSAccount1 (a Microsoft Foundry Service resource) with minimal administrative effort, the most suitable option is:

A. the endpoint URI and subscription key

Why A is the Best Choice:

Minimal Administrative Effort: Using the endpoint URI along with a subscription key requires the least setup and administrative overhead compared to other methods. It involves simply obtaining the URI and a key from the Azure portal, which is a straightforward process.
Direct Access: For many Azure services, including accessing APIs or resources directly, using a subscription key (when available and appropriate for the service) is the most direct method, aligning with the requirement for minimal effort.
Appropriateness for Foundry Service: Given the context of a Foundry Service, which often involves data or service access, using a subscription key is a common and supported method for authentication, making it the most fitting choice here.

Why Other Options are Less Suitable:

B. the endpoint URI and an OAuth token
Increased Complexity: Implementing OAuth requires setting up an application registration, configuring consent scopes (if necessary), and handling token refreshes, significantly increasing administrative effort.
Not Necessarily Required for Resource Access: Unless App1 is acting on behalf of a user or needs delegated permissions, OAuth's complexity isn't justified for direct service-to-service communication in this scenario.
C. the endpoint URI and a shared access signature (SAS) token
SAS for Resource-Specific Access: SAS tokens are typically used for granting temporary, restricted access to specific Azure resources (e.g., Blob Storage), not generally for service endpoints like Foundry Service.
Additional Setup: Generating and managing SAS tokens adds administrative overhead without clear benefits over a subscription key for this use case.
D. a system assigned managed identity and an X.509 certificate
Maximum Administrative Effort: This option involves the most setup, including enabling managed identity, configuring IAM roles for CSAccount1, and managing X.509 certificates, which is contrary to the requirement for minimal effort.
Overkill for Direct Service Access: Managed identities with X.509 certificates are more suited for scenarios requiring secure, credential-less access across services with fine-grained permissions, not the straightforward access described.

References

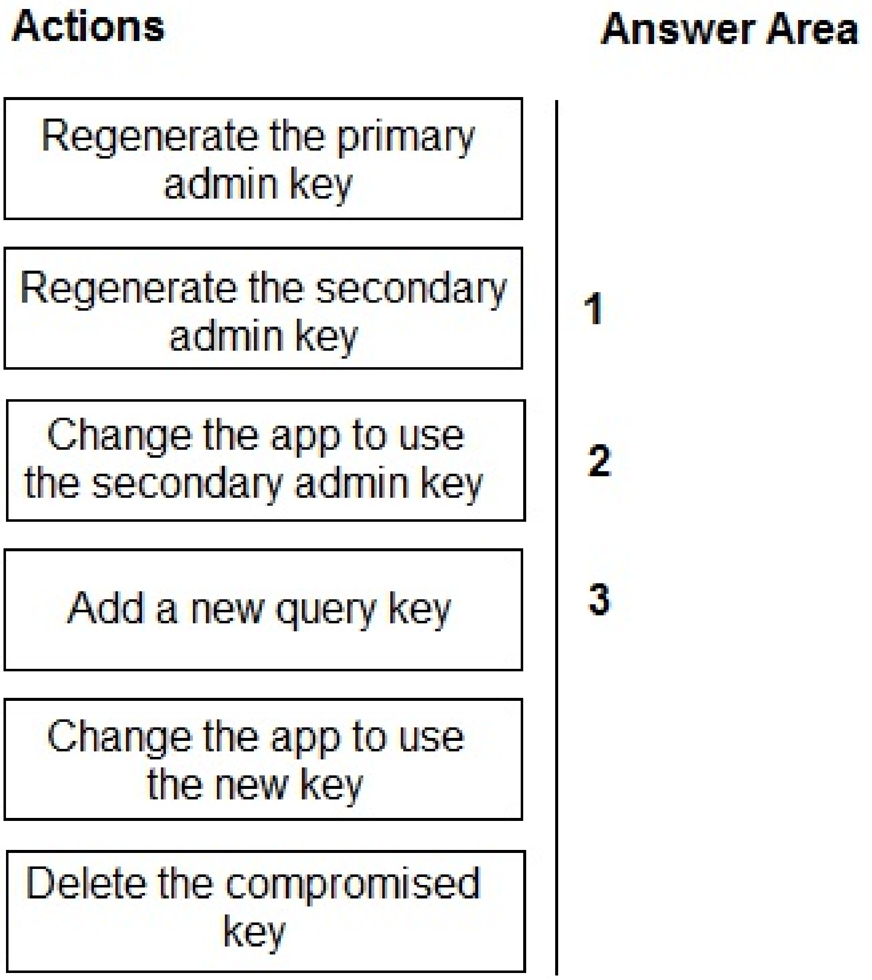
Microsoft Azure: Authenticate with Azure services using the subscription key
Microsoft Azure: Microsoft Foundry Service Overview

Question: 112 CertyIQ

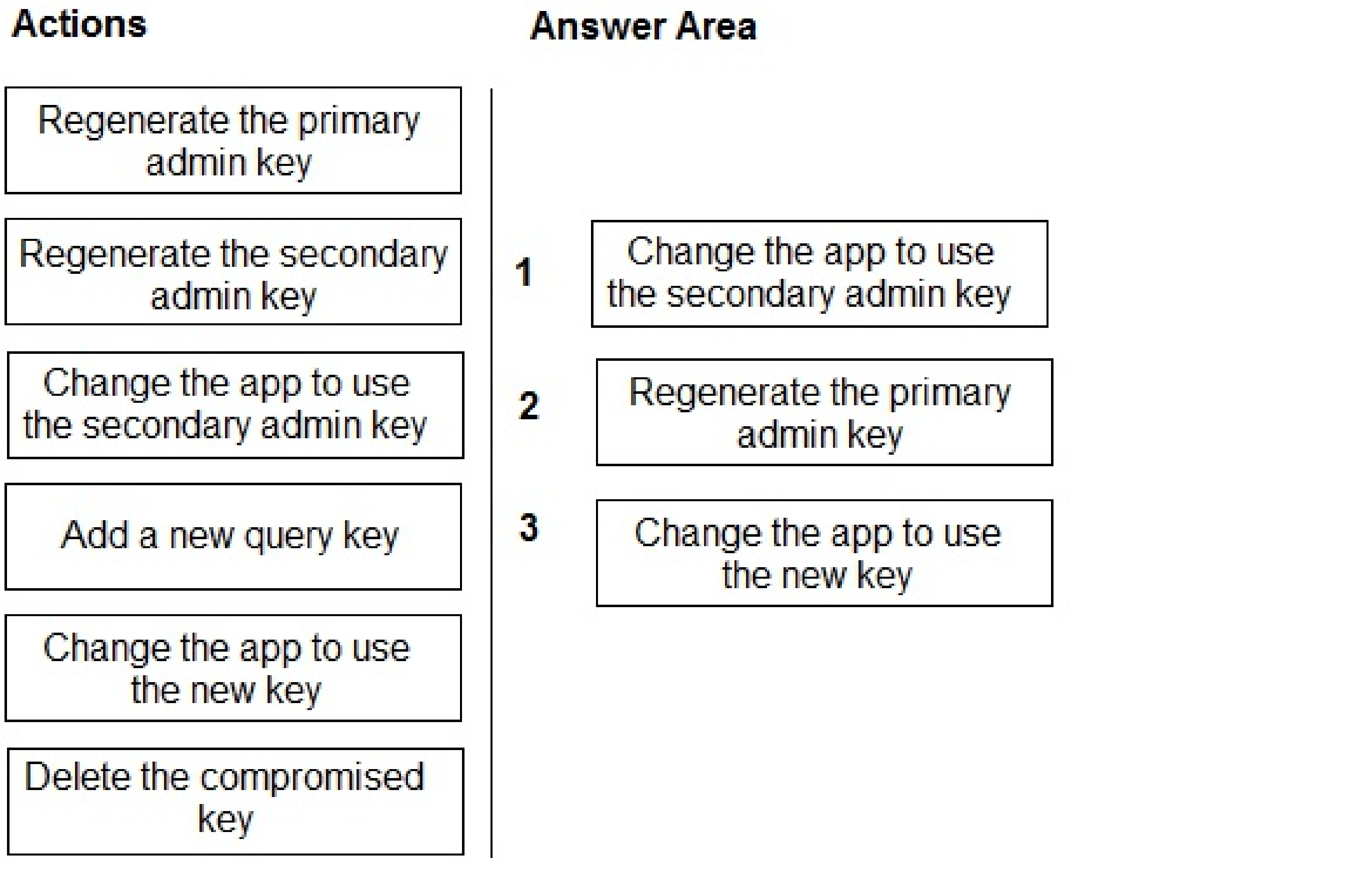
DRAG DROP

-

You have a web app that uses Azure AI Search.
When reviewing activity you see greater than expected search query volumes. You suspect that the query key is compromised.
You need to prevent unauthorized access to the search endpoint and ensure that users only have read only access to the documents collection. The solution must minimize app downtime.
Which three actions should you perform in sequence? To answer, move the appropriate actions from the list of actions to the answer area and arrange them in the correct order.



Answer:



Explanation:

Step 1: Change the app to use the secondary admin key

Before changing or invalidating the compromised primary key, update your application code or configuration settings to use the valid secondary admin key. This ensures the application continues to run without experiencing connection downtime.

Step 2: Regenerate the primary admin key

Now that your application is safely routing requests using the secondary key, you can safely regenerate the compromised primary admin key. This immediately revokes unauthorized access using the old, leaked key.

Step 3: Change the app to use the new key

Once the new primary key is generated, update your application configurations once more to point back to this new, uncompromised primary credential.

Question: 113 CertyIQ

HOTSPOT

-

You need to create a new resource that will be used to perform sentiment analysis and optical character recognition (OCR). The solution must meet the following requirements:

•Use a single key and endpoint to access multiple services.

•Consolidate billing for future services that you might use.

•Support the use of Azure Vision in Foundry Tools in the future.

How should you complete the HTTP request to create the new resource? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.

Answer Area

PATCH
POST
PUT

```
https://management.azure.com/subscriptions/
xxxxxxxx-xxxx-xxxx-xxxx-
xxxxxxxxxxxx/resourceGroups/RG1
```

```
/providers/Microsoft.CognitiveService/accounts/CS1?api-
version=2021-04-30
{
     "location": "West US",
     "kind": "
```

CognitiveServices
ComputerVision
TextAnalytics

```
",
     "sku": {
               "name": "S0"
          },
          "properties": {},
          "identity": {
               "type": "SystemAssigned"
          }
        }
```

Answer:

Answer Area

PATCH
POST
PUT

```
https://management.azure.com/subscriptions/
xxxxxxxx-xxxx-xxxx-xxxx-
xxxxxxxxxxxx/resourceGroups/RG1
```

CognitiveServices
ComputerVision
TextAnalytics

```
/providers/Microsoft.CognitiveService/accounts/CS1?api-
version=2021-04-30
{
     "location": "West US",
     "kind": "                ",
     "sku": {
          "name": "S0"
     },
     "properties": {},
     "identity": {
          "type": "SystemAssigned"
     }
}
```

Explanation:

1. HTTP Method Selection

Correct Value: **PUT**
Explanation: Under Azure Resource Manager REST API guidelines, creating or performing an idempotent replacement of a specific named cloud resource (such as /accounts/CS1) requires using the **PUT** verb. POST is typically used for action endpoints or non-idempotent creations without a predefined resource name, and PATCH is strictly for partial delta updates on an already existing resource.

2. Resource Kind Selection

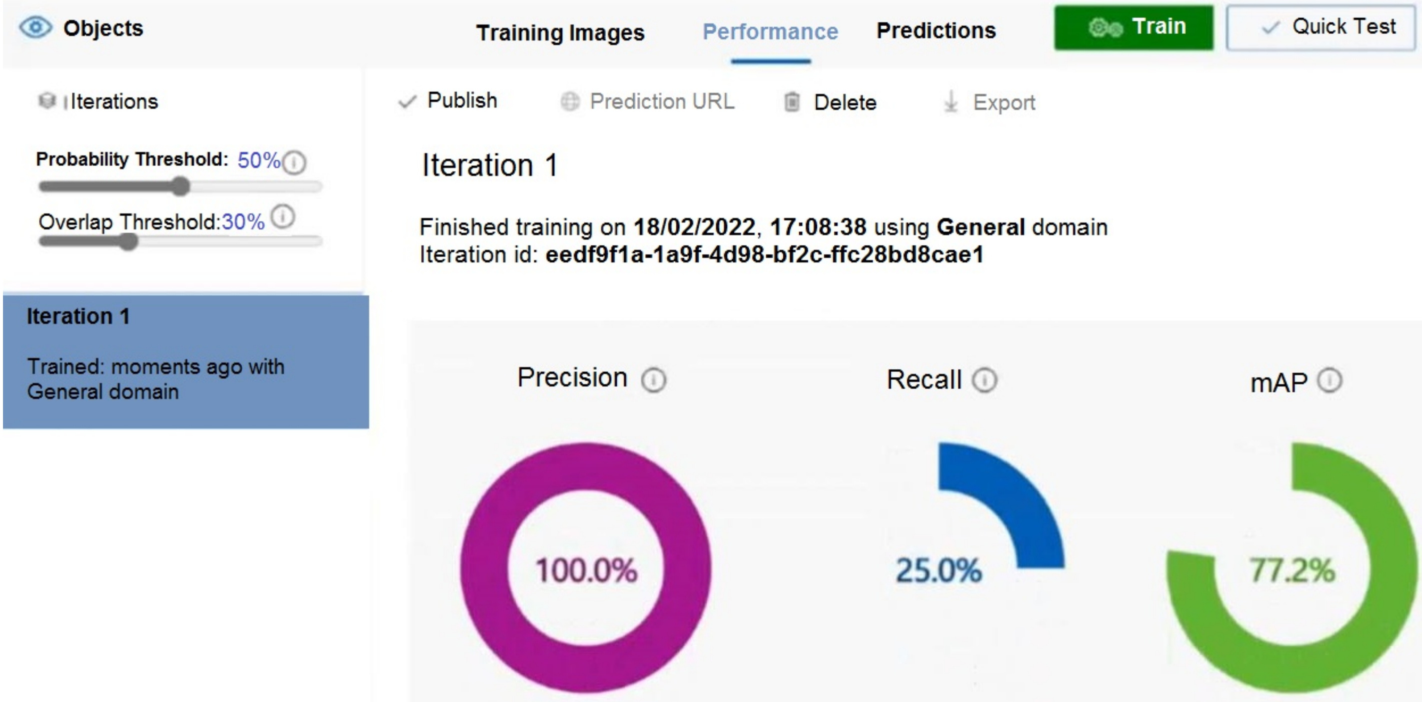
Correct Value: **CognitiveServices**
Explanation: The target provider namespace in the URI is explicitly defined as Microsoft.CognitiveServices/accounts. For an all-in-one multi-service Azure AI resource type within this namespace, the exact case-sensitive kind value required by the schema is **CognitiveServices**.

Question: 114 CertyIQ

HOTSPOT

-

You are building a model to detect objects in images.
The performance of the model based on training data is shown in the following exhibit.



Use the drop-down menus to select the answer choice that completes each statement based on the information presented in the graphic.
NOTE: Each correct selection is worth one point.

Answer Area

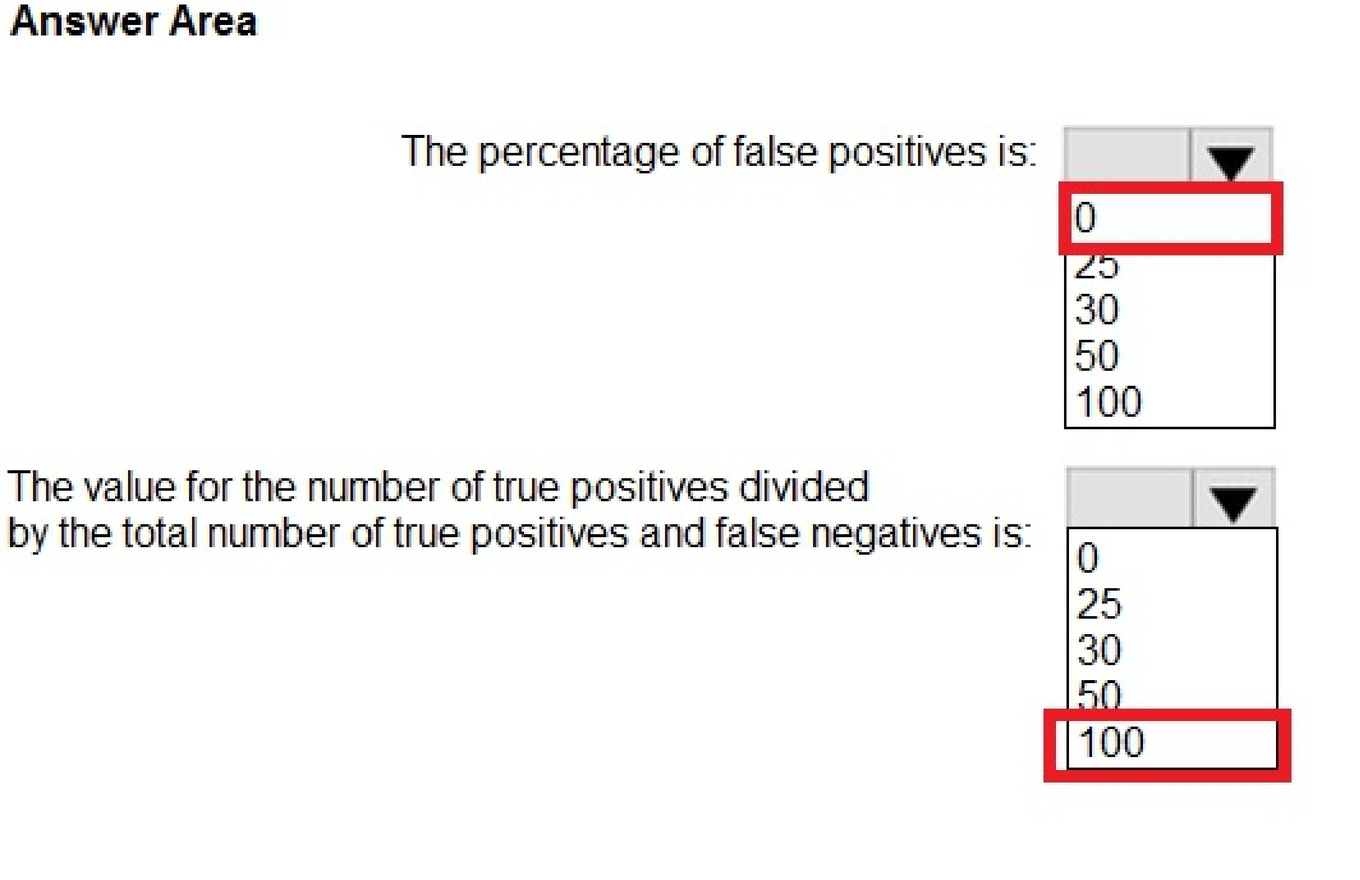
The percentage of false positives is:

- 0
- 25
- 30
- 50
- 100

The value for the number of true positives divided by the total number of true positives and false negatives is:

- 0
- 25
- 30
- 50
- 100

Answer:



Explanation:

Percentage of False Positives (0):

When a model achieves a Precision rate of 100%, it means every single object predicted by the model was correct. Consequently, there are no incorrect positive predictions, resulting in a **0% false positive** rate.

True Positives ÷ (True Positives + False Negatives) (100):

This formula calculates **Recall**, which measures the model's ability to find all the actual positive objects in the dataset. In this specific question variant, the training summary dashboard displays a Recall value of **100%**.

Question: 115 **CertyIQ**

You are developing an application that will use Azure AI Search for internal documents.
You need to implement document-level filtering for Azure AI Search.
Which three actions should you include in the solution? Each correct answer presents part of the solution.
NOTE: Each correct selection is worth one point.

A. Add allowed groups to each index entry

B. Create one index per group.

C. Send access tokens from Microsoft Entra ID, with the search request.

D. Retrieve all the groups.

E. Retrieve the group memberships of the user

F. Supply the groups as a filter for the search requests

Answer: AEF

Explanation:

Technical Justification for Correct Option (AEF) and Analysis of Alternatives

Correct Option: AEF

A. Add allowed groups to each index entry: This action is crucial for document-level filtering as it enables the association of specific groups with each document (index entry) in Azure AI Search. By embedding group permissions at the index level, the search service can efficiently filter results based on user access rights. This approach ensures that during query execution, the search service can quickly determine which documents a user is authorized to see, making the filtering process efficient.
E. Retrieve the group memberships of the user: Before executing the search query, fetching the user's group memberships is essential. This step ensures that the application knows which groups the current user belongs to, allowing for the accurate application of filters based on the user's entitlements. This dynamic retrieval ensures flexibility in handling changing group memberships without altering the search index.
F. Supply the groups as a filter for the search requests: By passing the user's group memberships as a filter with the search request, Azure AI Search can effectively filter the results to only include documents (index entries) that the user has access to, based on the group associations defined in the index. This real-time filtering ensures that search results are personalized to the user's permissions.

Why Other Options are Less Suitable:

B. Create one index per group: While this approach could theoretically work for filtering, it is highly inefficient for management and scalability, especially with a large number of groups or frequently changing group memberships. Maintaining multiple indexes increases operational overhead and complicates search queries that might need to span across groups.
C. Send access tokens from Microsoft Entra ID, with the search request: While sending access tokens might seem like a viable security measure, Azure AI Search's filtering capabilities are more effectively utilized with explicit group filters (as in option F) rather than relying solely on external token validation for each query. Tokens are better used for authentication, not for the fine-grained filtering that document-level access requires.
D. Retrieve all the groups: This action is unnecessary for the implementation of document-level filtering for a specific user's search request. Retrieving all groups would be inefficient and irrelevant to determining the access rights of the current user, especially in environments with a large number of groups not related to the user.

References

For detailed information on implementing security and access control in Azure AI Search, visit:
https://learn.microsoft.com/en-us/azure/search/search-security-rbac
For guidance on filters in Azure AI Search queries, refer to: https://learn.microsoft.com/en-us/azure/search/query-lucene-filter

Question: 116

CertyIQ

You have an Azure AI Search indexer that ingest PDF policy manuals.
Client applications must display page-level citations that have bounding polygons for both text and images.
You need to add a single built-in multimodal content extraction skill to the Azure AI Search skillset. The solution must meet the following requirements:
•Provide text and image location metadata.
•Extract tables that span multiple pages.
What should you add?

A. Document Layout
B. Document Extraction
C. Azure Content Understanding

D. GenAI Prompt

Answer: C

Explanation:

Technical Justification for Choosing C. Azure Content Understanding

To fulfill the requirements of extracting page-level citations with bounding polygons for both text and images, and additionally extracting tables that span multiple pages from PDF policy manuals in an Azure AI Search indexer, **Azure Content Understanding (C)** is the most suitable built-in multimodal content extraction skill to add to the search skillset. Here's why:

Requirement 1: Provide text and image location metadata with bounding polygons

Azure Content Understanding (C): Capable of extracting both text and images with their respective locations, including bounding box coordinates. This meets the need for precise page-level citations.
Insufficiency of Other Options:

A. Document Layout: While it can extract text and layout information, it's primarily focused on the document's structural layout rather than providing detailed metadata for images or complex bounding polygons for both text and images.
B. Document Extraction: Excels at extracting specific data fields but doesn't inherently provide the location metadata with bounding polygons for images and text as required.
D. GenAI Prompt: Not designed for extracting existing content with metadata but rather for generating text based on prompts, making it unsuitable for this requirement.

Requirement 2: Extract tables that span multiple pages

Azure Content Understanding (C): Advanced in handling complex document structures, including the capability to identify and extract tables even when they span across multiple pages, thanks to its understanding of document context.
Insufficiency of Other Options:

A. Document Layout & B. Document Extraction: While they can extract tables, their capability to handle tables spanning multiple pages is limited compared to Azure Content Understanding's contextual understanding.
D. GenAI Prompt: Again, not applicable for extraction tasks, especially complex ones like multi-page table extraction.

Why C. Azure Content Understanding is the Best Choice:

Multimodal Extraction: Handles both text and image extraction with location metadata.
Advanced Table Extraction: Capable of extracting tables across multiple pages.
Alignment with Requirements: Directly meets both specified requirements without needing additional skills, keeping the solution streamlined.

References

Azure Cognitive Search Skills Overview - Microsoft Docs
Azure Content Understanding Skill - Microsoft Docs

Question: 117 CertyIQ

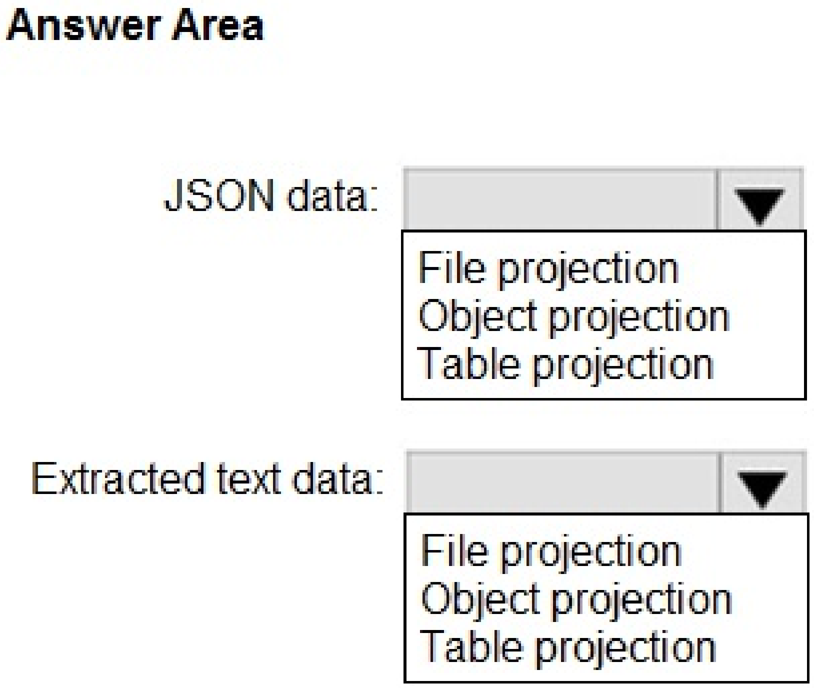
HOTSPOT

-

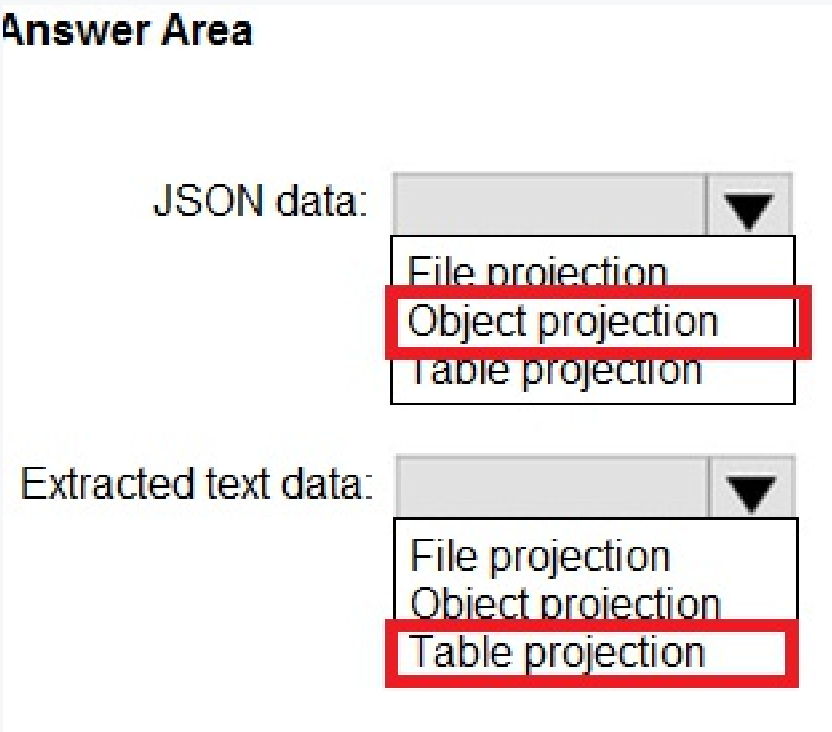
You are creating an enrichment pipeline that will use Azure AI Search. The knowledge store contains unstructured JSON data and the text from scanned PDF documents.

Which projection type should you use for each data type? To answer, select the appropriate options in the answer area.

NOTE: Each correct selection is worth one point.



Answer:



Explanation:

1. **Object projection**: This type is used when you need a full, hierarchical **JSON representation** of your enrichment tree data structures, making it the perfect fit for unstructured JSON content.
2. **Table projection**: When text is extracted from documents (like scanned PDFs or articles), it is usually broken down into tabular structures (such as paragraphs, key phrases, or localized lines of text) that map natively into rows and columns within Azure Table Storage.

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